Industrial Capacity as a Military Deterrent*

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Abstract

Military outcomes depend not only on initial military strength but also on the capacity to expand it during war.We present a model in which industrial capacity can have a strategic role as a military deterrent. We study how a government can sustain peace through military deterrence at the lowest cost. We show that when wars are short, peacetime procurement of military equipment suffices because wartime production does not affect military outcomes. When wars are long, however, efficient deterrence also requires subsidies to industrial capacity because private investors do not internalize its strategic value as a military deterrent. Such industrial policy should favor technologies that allow the defense industrial base to scale output efficiently during war. We develop an Integrated Deterrence–Economy Model to assess the welfare gains and identify priority sectors. Industrial policy can raise welfare by reducing the procurement expenditure required to sustain a given level of deterrence. Industrial priorities include munitions, guided missiles, and propulsion units, and the model shows how these priorities vary with the type and expected duration of war.

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1 Introduction v7

The economic study of optimal defense policy—the allocation of goods, capital and labor to strengthen national defense—occupies a relatively small place in contemporary economic research. This relative neglect is surprising: national defense is among the state's most fundamental responsibilities and commands substantial public resources. Although early work thought about the distinctively economic foundations of national security such as the role of the defense industrial base, theoretical research on these issues has largely lain dormant since at least the Cold War. Therefore, economic theory provides little guidance on how policy should support the economic foundations of national security.

The need for such guidance is growing. Governments are not just responding to heightened security concerns by increasing their defense budgets but also pursuing policies to strengthen their defense industrial bases. The U.S. National Defense Industrial Strategy, for example, calls for investment in additional capacity and in the ability to expand production rapidly U.S. Department of Defense (2023). But the theoretical argument for such policies is unclear. Is it efficient for governments to support industrial capacity directly, or should they simply procure military equipment? What externality would justify direct support, and how does the case for intervention depend on the type of war a government expects to fight? And if support is warranted, how should it be targeted across sectors and technologies?

The usual neoclassical model of public-good provision does not provide the guidance needed to address these questions. It treats public-good provision as a production problem: the government procures military equipment and other inputs to provide military services, which enter household preferences. When embedded in a dynamic general equilibrium model with investment, this approach implies that, in the absence of other distortions, optimal procurement resolves the public-good problem and induces efficient investment in the industrial base.

We therefore develop a framework that embeds a strategic model of public-good provision in a dynamic general equilibrium model with investment. We model national security as peace sustained through military deterrence. In peacetime, the military thus provides a strategic service by deterring war. The key idea is that this role extends to the industrial base: beyond supplying military equipment, industrial capacity can itself act as a military deterrent. We begin with a simple baseline model to show how this strategic role depends on the expected duration of war and when it creates a national security externality that justifies direct peacetime support for the industrial base. We then use a more general model to study how support should be targeted across sectors.

Baseline Model Our model of military deterrence places industrial capacity at the center. Consider a country that faces an adversary who contemplates starting a war to challenge the status quo through military force. Such a war can either be long or short, and for simplicity we assume both parties know which it will be. During a war, territory changes hands depending on the relative strength of the militaries. Military strength includes both initial strength and any increases throughout the war. To sustain the peace, the country needs to make war sufficiently costly for its adversary. It can do so during peace by investing in both the size of its armed forces and the size of the defense industrial base.

The key point is that industrial capacity can act as a form of deterrence. Deterrence does not just depend on military equipment a country *has* at the start of the war but also on what it *can produce* over the duration of a longer war. An adversary understands that a country with larger industrial capacity will leverage it to expand military capacity throughout the war. In a way, the peacetime defense budget determines the army a country starts with; industrial capacity determines what it can end up with. Because it affects an adversary's calculus of aggression, industrial capacity is not just a supplier to the military but can directly contribute to deterrence.

In this framework, we ask how a government can sustain peace through deterrence at the lowest possible cost. We study two types of peacetime policy: *procurement* and *industrial* policy. Procurement consists of purchases of military equipment such as ammunition and tanks. Industrial policy influences the inputs firms use to produce that equipment. Since private firms choose investment in the defense industrial base, we focus on subsidies through which the government can encourage capital accumulation across sectors.[1]

Economists have traditionally distinguished sharply between these two types of policy. The classical policy prescription to solve the underprovision of public goods is to replace private provision with public provision—through procurement of military equipment in our case. Economists have often been skeptical of additional interventions such as industrial policies beyond such procurement policies. The seminal production efficiency result of Diamond and Mirrlees (1971) justifies this skepticism. It shows that industrial policy, as defined above, is indeed not needed for efficient public good provision in standard neoclassical models in which public goods are provided by the government by combining goods with capital or labor.

Throughout the paper, we emphasize the dynamic aspect of war. We have two reasons. First, as we will see, it matters for the argument for industrial policy. Second, assessments of the conflicts countries are likely to face are changing. Much of the eco-

[1] In practice, procurement contracts can combine what we call procurement and industrial policy by paying producers both for equipment and to retain additional production capacity.

nomic research on defense was conducted during the Cold War, when some influential analyses placed greater emphasis on forces available at the outset than on the capacity to expand production during war. For example, in *The Economics of Defense in the Nuclear Age* (1960), Hitch and McKean argued that technological advances had made wars requiring prolonged industrial mobilization relatively unlikely. In their assessment, the principal threats had shifted toward thermonuclear exchange and limited conflicts, placing greater weight on forces available at the outset. Capacity to expand wartime production therefore played a much smaller role in their account than it had during the world wars.[2]

The war in Ukraine has brought renewed attention to the possibility of prolonged conventional conflict, including conflicts involving a nuclear power. Such wars can quickly exhaust munitions stocks and destroy the equipment accumulated before the war. Wartime production during such conflicts can dwarf prewar inventories. Before World War II, the U.S. Army was the eighteenth largest, yet American industry subsequently produced roughly two-thirds of Allied military equipment (Herman, 2012). Industrial capacity—and the ability to expand it—therefore becomes central to military success. Reflecting renewed recognition of its role in deterrence, William LaPlante, while serving as Under Secretary of Defense for Acquisition and Sustainment, put it succinctly: "production itself is deterrence."[3] This renewed emphasis helps explain the widespread policy interest in expanding the defense industrial base and making it more scalable.[4]

Results Our first result provides a useful baseline: when wars are short, efficient deterrence only requires procurement policy. Wars end before wartime production can affect military outcomes, and so the defense industrial base has no direct role in deterrence. The

[2]This emphasis on forces available at the outset was especially pronounced in analyses of nuclear war. In a passage citing Hitch and McKean, Rowen (1960) argued that a nuclear war would be decided before the year or more required for mobilization, making forces maintained in peacetime much more important. NATO's 1957 strategic concept likewise prioritized existing forces, anticipating that nuclear destruction would severely restrict subsequent mobilization (NATO Military Committee, 1957). The emphasis also appeared in later Army doctrine: the 1976 edition of *FM 100-5: Operations* warned that a future war could be decided by initial combat, contrasting this possibility with America's historical reliance on resources mobilized after hostilities began (Department of the Army, 1976).

[3]LaPlante made this remark when discussing lessons from Ukraine (Clark, 2023). Similarly, NATO Secretary General Jens Stoltenberg argued that Ukraine's war of attrition had exposed insufficient Western industrial capacity to sustain such conflicts, prompting efforts to expand weapons and ammunition production (Stoltenberg, 2024).

[4]The U.S. National Defense Industrial Strategy calls for incentives to build spare capacity and develop flexible production (U.S. Department of Defense, 2023). The United Kingdom's Defence Industrial Strategy calls for maintaining flexible production capacity and assessing scalability in major procurements (Ministry of Defence, 2025). The European Defence Industrial Strategy proposes funding spare industrial capacity to enable rapid expansion during crises (European Commission and High Representative of the Union for Foreign Affairs and Security Policy, 2024). NATO's Updated Defence Production Action Plan likewise calls for industrial planning that addresses long-term scalability and surge requirements (NATO, 2025).

defense industrial base acts as a producer of military equipment, but this by itself is not a reason to intervene because market prices lead to the efficient investment in capacity. The short war view is an interesting benchmark because thinking about defense as a neoclassical public goods provision problem, while ignoring its strategic nature, essentially leads to the right economic policy recommendation.

When wars are long, this is no longer the case. Our second result shows that industrial policy will also be needed for efficient deterrence. When wars are long, industrial capacity acts as a military deterrent, but the private sector does not internalize this strategic benefit. This national security externality therefore generates underinvestment in the defense industrial base and hence justifies industrial policy that increases industrial capacity. Without industrial policy, firms choose capacity to meet relatively stable peacetime procurement. But a production structure that serves this demand at low cost need not be well suited to expanding output during war.

Our third result shows how this mismatch affects the composition of investment: industrial policy should encourage technologies that make the defense industrial base more scalable. We extend the baseline to include two production technologies that differ in their scalability—the extent to which they can expand output at low costs in the short run. By tilting the composition of investment towards the more scalable technology, industrial policy increases the capacity to expand wartime production. By changing the composition of technologies, industrial policy can raise deterrence even if the aggregate capital stock in the defense industrial base remains fixed.

The interaction between the two key model ingredients—deterrence as a public good and war duration—is central to these results. The former introduces a national security externality that can overturn the production-efficiency conclusion of Diamond and Mirrlees (1971) and create a role for industrial policy; the latter determines whether industrial capacity actually contributes to deterrence. Together, these ingredients provide a framework for understanding why economists working under the Cold War view of short wars were justified in paying relatively little attention to industrial policy for the defense industrial base. The same framework explains why the growing emphasis on prolonged wars provides a rationale for renewed support for industrial capacity and the accompanying emphasis on scalability.

Quantitative exercise In the final part of our paper, we study how industrial policy should be targeted. When wars are long, policymakers must decide which industries to support alongside military procurement. These decisions depend on both the structure of the economy and how war is expected to unfold. We therefore develop an Integrated

Defence Economics Model (IDEM) to study how industrial capacity contributes to deterrence and study how policy should be targeted.

IDEM combines a dynamic general equilibrium economic block with a military block. The economic block captures military producers' reliance on other industries, both for intermediate inputs and for the capital goods needed to expand production during war. The military block combines different types of equipment into military power, allowing their relative importance to vary across conflicts. War duration is uncertain and depends on the military balance: a sufficiently strong aggressor may end a war quickly, while a prolonged conflict gives industrial capacity time to shape the outcome. We calibrate the model using U.S. production and investment networks and historical evidence on war durations and military buildups.

We find that industrial policy raises welfare relative to procurement policy alone. It raises steady-state consumption by reducing military expenditure and the aggregate capital stock, holding deterrence fixed. Optimal support is concentrated in a relatively narrow set of industries, providing quantitative support for the view that the national security argument for industrial policy applies to only a limited set of industries. Priority sectors include guided missiles, propulsion units, and ammunition. Missiles and ammunition can be stockpiled but are expended in combat, so demand for them can rise especially sharply during war. Our exercise also suggests that industrial policy should depend on the conflict being deterred. A land war in Europe places greater emphasis on tank production, while a war over Taiwan favors aircraft and shipbuilding. When wars are long, economic policymaking therefore cannot be separated from military analysis. We conclude by discussing how IDEM could support closer integration between the two.

Related Literature The framework developed in this paper brings together two literatures that study different aspects of war or military buildups: their economic consequences and the strategic decisions that generate them. The theoretical argument for industrial policy follows from the interaction between the strategic nature of public-good provision and the fact that it is often the private sector that invests in industrial capacity. We draw particular inspiration from Hitch and McKean (1960), which offers the clearest account of the interplay between the defense industrial base, military capacity, deterrence, and the nature of warfare. Our contribution is to provide a formal framework of these interactions, characterize the resulting rationale for industrial policy, and operationalize it through IDEM.

The neoclassical literature in macroeconomics and public finance uses general equilibrium models to study investment, production, and government policy in response to war

or military buildups. It typically treats these events as exogenous shocks and abstracts from the strategic decisions behind them. Empirical work in macroeconomics has used military spending to estimate fiscal multipliers (Ramey, 2011; Barro and Redlick, 2011; Nakamura and Steinsson, 2014).[5] Recent work studies military buildups in their own right (see Ilzetzki (2025) for a review), examining their effects on military-goods prices Antonova et al. (2025) and their scale during major wars Marzian and Trebesch (2026). We draw on this evidence in our quantitative exercise.

The strategic literature in political economy and international relations uses game-theoretic models to study war or military buildups as endogenous outcomes of countries' assessments of the costs and benefits of conflict. It typically abstracts from private-sector investment decisions and from features of the economy such as input–output networks. Building on Schelling (1960, 1966), it recognizes that because war is costly, national security policy is often about preventing conflict through deterrence. Work in this tradition includes Mearsheimer (1983); Baliga et al. (2020); Baliga and Sjöström (2020); Luo and Wolitzky (2025); Herrera et al. (2022); Di Lonardo and Tyson (2022, 2026).

By providing a rationale for supporting the defense industrial base, our paper also contributes to the industrial policy literature (reviewed in Juhász et al. (2024)). This literature has studied the empirical effects of defense-related policies, including military procurement, public R&D, and direct investment in manufacturing capacity Lichtenberg (1988); Gross and Sampat (2023); Ilzetzki (2024); Moretti et al. (2025); Garin and Rothbaum (2025), but formal arguments for prioritizing the defense industrial base in peacetime remain less developed.[6] Thompson (1979) provides a rationale in a model in which wartime price controls generate underinvestment during peace. Kooi (2025) formalizes and quantifies a strategic rationale based on the benefits of lowering the domestic cost of war. Lower costs can strengthen bargaining power or, in a deterrence model, make a country's commitment to fight if the status quo is challenged more credible. Our baseline takes this commitment as given, so this motive is absent. Section 3.5 extends our model to incorporate this additional strategic rationale and clarify how it is distinct from the one developed here.

[5]There is also a related public finance literature that initially focused on war finance Pigou (1921); Keynes (1940) but now more commonly studies fiscal policy over the business cycle Barro (1979); Lucas and Stokey (1983); Bhandari et al. (2017).

[6]Standard rationales for industrial policy, such as knowledge spillovers and credit constraints, can in principle justify support for the defense industrial base. To justify prioritizing defense over other sectors, however, these rationales would need to be quantitatively more important in defense. This is not obvious: knowledge spillovers can also pose national security risks, while defense firms are often large and established, potentially limiting their exposure to credit constraints. An argument grounded in the distinctive nature of public-good provision in national defense therefore has particular appeal.

Finally, our paper relates to the nascent geoeconomics literature (see Mohr and Trebesch (2025) for a recent review). This literature also studies the interaction between economic structure and geopolitics, with much recent work emphasizing trade and financial relationships as sources of geopolitical power rather than military production. Our model shares structural features with work combining national-level bargaining and private-sector investment in irreversible capital McLaren (1997); Kooi (2025); Becko and O'Connor (2025). Kooi (2025) highlights similarities between the policy implications of bargaining in the shadow of trade and military conflict. Other work examines the links between military security and international economic relationships Alekseev and Lin (2025); Mayer et al. (2025); Adão and Kooi (2026). Further contributions to geoeconomics include Clayton et al. (2024, 2026) and Liu and Yang (2025).

2 Model

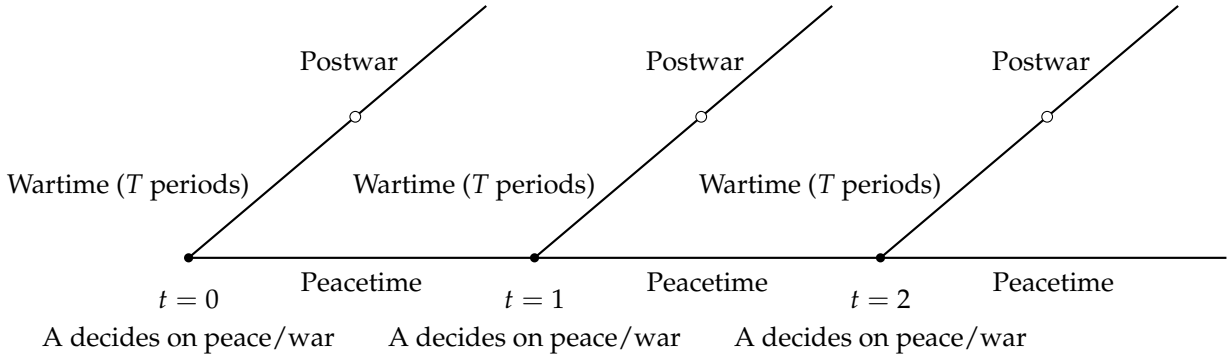
Overview There are two countries, Sovereign and Adversary, indexed by $i \in \{S, A\}$, each inhabited by a representative agent. Time is discrete and infinite, with periods denoted $t = 0, 1, \ldots$.

The countries compete over a divisible public good, which we interpret as territory. Both countries value territory, but each unit can belong to only one of them. Its allocation remains at the status quo during peace and can change only through war. At the end of a war, territory is reallocated according to the countries' relative military strength over the course of the conflict. This strength reflects both their initial military stocks and the additional military equipment accumulated during the war. The resulting territorial allocation remains fixed thereafter.

The economy begins in peace. At the beginning of each period, Adversary decides whether to maintain the peace or start a war. A war lasts one or two periods, $T \in \{1, 2\}$, which we refer to as short and long wars, respectively. Its duration is exogenous and known to both countries. The key difference is that during long wars a country can produce additional military equipment through its defense industrial base but during short wars they cannot. War can occur only once: after it ends, economic activity continues but no further war is possible.

Figure 1 illustrates this timing. We study how Sovereign can use military procurement and industrial policies that increase the capital stock of the defense industrial base to sustain the peace, so the economy follows the horizontal peacetime path in the equilibria we consider. The wartime and postwar branches describe what would happen if Adversary attacked. These possible continuations are central to the analysis because they determine Adversary's willingness to break the peace and challenge the status quo through military force.

Figure 1: Timing of peace, war, and the postwar period.



2.1 Economic Block

We begin by describing the economic block of the model, which is common to both countries. Households supply labor and own firms, while the government procures military goods and can subsidize capital in the defense industrial base. We denote the geopolitical state by $z_t \in \{P, C\}$, where $z_t = P$ if no war has begun and $z_t = C$ once a war has begun, including the postwar period. In what follows, all equilibrium variables are, in general, functions of the history $z^t = (z_0, z_1, \ldots, z_t)$, but whenever no confusion is possible we leave this dependence implicit.

Preferences The representative agent in each country has preferences given by

$$U^i = \sum_{t=0}^{\infty} \beta^t \left(G_t^i + c_t^i \right), \tag{1}$$

where $\beta \in (0, 1)$ is the discount factor and $\{c_t^i\}_{t=0}^{\infty}$ is the sequence of consumption. Here G_t^i is the public good introduced above, which is not chosen by households but is instead determined by geopolitics.

Capital Accumulation There are two capital stocks in this economy. There is a stock of military capital, M_t^i, which represents the strength of the armed forces and depreciates at rate δ_M. There is also capital in the defense industrial base, K_t^i, which serves as an input into the production of military capital and depreciates at rate δ_K. Let m_t^i denote government purchases of new military equipment and i_t^i investment in industrial capital during period t. The two capital stocks therefore evolve according to

$$M_{t+1}^i = M_t^i(1 - \delta_M) + m_t^i \tag{2}$$

$$K_{t+1}^i = (1 - \delta_K)K_t^i + i_t^i. \tag{3}$$

Both stocks are predetermined at the beginning of period t. Purchases and investment during period t augment the stocks available in period $t + 1$.

Production There are two production technologies in this economy. Output of the numeraire good is denoted by $y_{n,t}^i$ and is produced through a linear technology that uses labor $l_{n,t}^i$ as input. The defense sector produces military goods, denoted by $y_{m,t}^i$, which are used to accumulate military capital. Military goods are produced using a Cobb–Douglas technology with $\alpha \in (0, 1)$ that takes K_t^i and labor, $l_{m,t}^i$, as inputs. We normalize produc-

tivity in military goods production to one. The two technologies are given by

$$y_{n,t}^i = l_{n,t}^i \tag{4}$$

$$y_{m,t}^i = \left(K_t^i\right)^\alpha \left(l_{m,t}^i\right)^{1-\alpha}. \tag{5}$$

Market Clearing The numeraire good can be used for consumption or investment in the defense industrial base. The military good is used only to accumulate military capital. The economy is endowed with $\bar{l}^i$ units of labor, which are allocated between the numeraire sector and the defense sector. The three market-clearing conditions are therefore given by

$$c_t^i + i_t^i = y_{n,t}^i \tag{6}$$

$$m_t^i = y_{m,t}^i \tag{7}$$

$$l_{n,t}^i + l_{m,t}^i = \bar{l}^i. \tag{8}$$

Competitive equilibrium Households choose consumption to maximize utility subject to their budget constraint, taking as given the price of the numeraire good $p_{n,t}^i$, the wage w_t^i, firm profits Π_t^i, and the lump-sum tax T_t^i levied by the government. They have access to a risk-free bond with return r_t^i between periods t and $t+1$, which is available in zero net supply. Let $R_t^i = \prod_{u=0}^{t-1}(1+r_u^i)$ denote the cumulative gross return from date 0 to date t, with $R_0^i = 1$. The household's problem is

$$\max_{\{c_t^i\}_{t=0}^\infty} \left\{ \sum_{t=0}^\infty \beta^t c_t^i \,\middle|\, \sum_{t=0}^\infty \frac{1}{R_t^i}\left(p_{n,t}^i c_t^i - w_t^i \bar{l}^i - \Pi_t^i + T_t^i\right) = 0 \right\}. \tag{9}$$

The producers of the numeraire good solve a static problem in which they maximize profits by hiring labor:

$$\max_{y_{n,t}^i, l_{n,t}^i} \left\{ p_{n,t}^i y_{n,t}^i - w_t^i l_{n,t}^i \,\middle|\, (4),\ l_{n,t}^i \geq 0 \right\}. \tag{10}$$

The defense industrial base maximizes profits by choosing investment and labor to produce military goods. It takes as given the price of military goods, $p_{m,t}^i$, and the subsidy on its capital stock, s_t^i, which the government uses to affect investment decisions. The subsidy is a payment of s_t^i per unit of installed capital in period t. Given its initial capital

stock K_0^i, its optimization problem is therefore

$$\max_{\{y_{m,t}^i, l_{m,t}^i, i_t^i, K_{t+1}^i\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \frac{1}{R_t^i} \left(p_{m,t}^i y_{m,t}^i + s_t^i K_t^i - w_t^i l_{m,t}^i - p_{n,t}^i i_t^i \right) \middle| (3),\ (5),\ l_{m,t}^i \geq 0,\ i_t^i \geq 0 \right\}. \quad (11)$$

The government raises revenue to fund the subsidies it pays to the defense industrial base and to procure military goods. It uses a lump-sum tax to balance its budget period by period and starts with no initial debt. Its budget constraint is therefore given by

$$T_t^i = p_{m,t}^i m_t^i + s_t^i K_t^i. \quad (12)$$

Definition 1 (Competitive Equilibrium)**.** *Taking as given the sequence of geopolitical states $\{z_t\}_{t=0}^{\infty}$, the sequence of government purchases and subsidies $\{m_t^i, s_t^i\}_{t=0}^{\infty}$, and the initial capital stocks K_0^i and M_0^i, a competitive equilibrium in country i is a sequence of prices $\{r_t^i, p_{n,t}^i, p_{m,t}^i, w_t^i\}_{t=0}^{\infty}$, quantities $\{c_t^i, y_{n,t}^i, y_{m,t}^i, i_t^i, K_t^i, M_t^i, l_{n,t}^i, l_{m,t}^i\}_{t=0}^{\infty}$, and taxes and profits $\{T_t^i, \Pi_t^i\}_{t=0}^{\infty}$ such that households and firms solve their optimization problems, markets clear, and the government budget constraint holds in every period.*

2.2 Deterrence Block

We now describe the deterrence block. We use t to index calendar time and write peacetime quantities without a regime superscript. When considering a war that begins at date t, we index its continuation by $\tau = 0, 1, \ldots$, where $\tau = 0$ is the first period of war. For a generic variable X, we write $X_\tau^{i,C}$ for its value τ periods after the outbreak, including the postwar period. Since capital stocks are predetermined when Adversary decides whether to attack, the initial stocks in this continuation satisfy $K_0^{i,C} = K_t^i$ and $M_0^{i,C} = M_t^i$.

The allocation of the public good Countries are endowed with an initial allocation of the public good, $\bar{G}^i$. During peacetime, the allocation of the public good is fixed at its initial level. This implies that

$$G_t^i = \bar{G}^i. \quad (13)$$

The allocation of the public good can only be changed through war. At the end of the war, the public good is reallocated based on the parties' relative military capacity throughout the war. The duration of the war, $T \in \{1, 2\}$, is exogenous and known to both parties. The technology is linear, and the parameter $\chi > 0$ captures the effectiveness of military power in obtaining territory. The new allocation applies from the final period of war onward and

remains fixed thereafter:

$$G_{\tau}^{i,C} = \begin{cases} \bar{G}^i, & \tau < T-1, \\ \bar{G}^i + \chi \sum_{u=0}^{T-1} \left(M_u^{i,C} - M_u^{-i,C} \right), & \tau \geq T-1, \end{cases} \tag{14}$$

where $-i$ denotes the other country.

Adversary's Policy Rules Adversary's government is assumed to follow simple policy rules that keep its military and industrial capital stocks at their initial levels, denoted by $\bar{M}^A$ and $\bar{K}^A$, respectively. It replaces depreciated military equipment and industrial capital in every period, during peace, war, and the postwar period.

The reason for restricting attention to such simple rules is that the focus is on the strategic interaction between Sovereign's military procurement and industrial policies, on the one hand, and Adversary's aggression decision, on the other. Allowing for richer behavioral responses by Adversary would introduce additional strategic interactions. Section 3.5 explains how to incorporate these and discusses their implications for policy.

Adversary's Challenge Decision At the beginning of any peacetime period, Adversary can challenge the status-quo allocation of territory by starting a war. Let $W_t^{A,P}(\mathbf{x}_t)$ denote its discounted welfare from maintaining peace at date t, and let $W_t^{A,C}(\mathbf{x}_t)$ denote its discounted welfare from starting a war at that date. Both values include consumption and public-good flows from date t onward. Here $\mathbf{x}_t = \{G_t^i, K_t^i, M_t^i\}_{i \in \{S,A\}}$ denotes the vector of state variables at the time of the decision. Choosing peace at date t leaves open the possibility of starting a war at a later date; the superscript P therefore refers to the current decision, rather than a commitment to permanent peace.

Let η_t^A denote Adversary's additional political payoff from going to war. Adversary starts a war whenever

$$W_t^{A,C}(\mathbf{x}_t) + \eta_t^A > W_t^{A,P}(\mathbf{x}_t). \tag{15}$$

A higher η_t^A therefore makes Adversary more willing to attack. When indifferent, it maintains the peace.

2.3 Deterrence Equilibrium

We consider equilibria in which Adversary maintains the peace at every point in time. We define deterrence as

$$\mathcal{D}_t(\mathbf{x}_t) \equiv W_t^{A,P}(\mathbf{x}_t) - W_t^{A,C}(\mathbf{x}_t). \tag{16}$$

This measures Adversary's welfare advantage from maintaining peace, before including the additional political payoff from attacking. Adversary is deterred when this advantage is at least as large as η_t^A. We therefore define a deterrence equilibrium as follows.

Definition 2 (Deterrence equilibrium)**.** *A deterrence equilibrium is a deterministic sequence of quantities, prices, subsidies, taxes, and profits that (i) is consistent with the competitive equilibrium and (ii) sustains the peace by satisfying the deterrence constraint*

$$\mathcal{D}_t(\mathbf{x}_t) \geq \eta_t^A, \quad \forall t. \tag{17}$$

There may be many allocations that satisfy these equilibrium conditions.[7] Specifically, there may be several feasible combinations of military and defense-industrial capital that satisfy the deterrence constraint (17) and can be implemented in a competitive equilibrium. Section 3 studies which policies Sovereign needs in order to attain the deterrence equilibrium that maximizes domestic welfare. First, however, it is useful to characterize the deterrence constraint more explicitly.

2.4 Simplifying the Deterrence Constraint

Our definition of equilibrium is expressed in terms of $\mathcal{D}_t(\mathbf{x}_t)$. As will become clear in the next section, the characterization of optimal subsidies applies to general deterrence functions. However, our comparative statics use the microfoundations of $\mathcal{D}_t(\mathbf{x}_t)$. In this section, we show how our assumptions simplify this function.

Adversary's welfare during peace We begin by deriving an expression for Adversary's welfare during peace. Because Adversary maintains its military and industrial capital stocks at $\bar{M}^A$ and $\bar{K}^A$, its consumption is constant. It equals the total labor endowment minus the labor used to produce replacement military equipment and industrial capital:

$$c_t^A = \bar{c}^A \equiv \bar{l}^A - \delta_K \bar{K}^A - \left(\frac{\delta_M \bar{M}^A}{(\bar{K}^A)^\alpha}\right)^{\frac{1}{1-\alpha}}. \tag{18}$$

[7] We assume that a feasible deterrence equilibrium exists even in the absence of industrial policy.

In a deterrence equilibrium, Adversary expects peace to continue at every future date. Its peacetime welfare is therefore

$$W_t^{A,P} = \frac{\bar{G}^A + \bar{c}^A}{1-\beta}. \tag{19}$$

Sovereign's wartime problem Adversary's welfare if it chooses war depends in part on how much military equipment Sovereign produces during the conflict. We therefore begin by studying Sovereign's optimal policy during war.

Consider a war beginning at date t. Sovereign enters the conflict with capital stocks K_t^S and M_t^S and the status-quo territorial allocation $\bar{G}^S$. Substituting the conflict technology (14) into the household objective function yields the following policy problem:

$$\max_{\substack{\{c_\tau^{S,C}, y_{n,\tau}^{S,C}, y_{m,\tau}^{S,C}, l_{n,\tau}^{S,C}, l_{m,\tau}^{S,C}, \\ i_\tau^{S,C}, m_\tau^{S,C}, K_{\tau+1}^{S,C}, M_{\tau+1}^{S,C}\}_{\tau=0}^{\infty}}} \left\{ \sum_{\tau=0}^{\infty} \beta^\tau \left(c_\tau^{S,C} + \bar{G}^S \right) + \frac{\chi \beta^{T-1}}{1-\beta} \sum_{\tau=0}^{T-1} \left(M_\tau^{S,C} - \bar{M}^A \right) \,\middle|\, (2)\text{–}(8) \right\}. \tag{20}$$

The constraints are evaluated along the conflict continuation, with initial stocks inherited from date t and all allocations subject to non-negativity constraints. Since $\bar{G}^S$ and $\bar{M}^A$ enter the objective additively and separably, Sovereign's optimal military stocks depend only on its inherited capital stocks, holding parameters fixed. We denote this response by $M_\tau^{S,C}(K_t^S, M_t^S)$.

Adversary's welfare during war Having characterized Sovereign's military stocks as a function of its inherited capital stocks, we can now characterize Adversary's welfare during war. Adversary maintains the same consumption during peace and conflict, so $c_\tau^{A,C} = \bar{c}^A$. Substituting Sovereign's military stocks into the conflict technology (14), Adversary's welfare if it attacks at date t is

$$W_t^{A,C} = \frac{\bar{G}^A + \bar{c}^A}{1-\beta} + \frac{\chi \beta^{T-1}}{1-\beta} \sum_{\tau=0}^{T-1} \left(\bar{M}^A - M_\tau^{S,C}(K_t^S, M_t^S) \right). \tag{21}$$

The Simplified Deterrence Constraint We now combine (19) and (21) to simplify the deterrence constraint. The common consumption and status-quo territorial terms cancel. Since the capital stocks available when war begins are predetermined, the wartime response is evaluated at Sovereign's peacetime stocks K_t^S and M_t^S. Keeping Adversary's fixed military stock $\bar{M}^A$ implicit as a parameter, we can therefore write deterrence as

$$\mathcal{D}_t(K_t^S, M_t^S) = \frac{\chi \beta^{T-1}}{1-\beta} \sum_{\tau=0}^{T-1} \left(M_\tau^{S,C}(K_t^S, M_t^S) - \bar{M}^A \right). \tag{22}$$

Deterrence is increasing in the amount of military power deployed throughout the war.[8] The deterrence constraint used going forward is therefore

$$\mathcal{D}_t(K_t^S, M_t^S) \geq \eta_t^A. \tag{23}$$

2.5 Discussion

Before turning to optimal economic policy, we briefly comment on some of the modelling assumptions.

The Asymmetries of Deterrence In our model, there is a clear asymmetry in the security problem for the two countries. Adversary is the challenger to the status quo, whereas Sovereign is its defender. We view this as a useful simplification that allows us to develop the strategic argument for industrial policy. But such asymmetries are also quite natural from a historical perspective. When the balance of geopolitical power changes—for example, because the economies of rising powers grow faster than those of existing powers (Kennedy, 1987)—existing powers find themselves trying to preserve a status quo that favors them for historical reasons. The assymetry thus arises endogenously from changes in the balance of power. We view this as a useful description of the security problem facing Western countries seeking to deter challenges to the geopolitical status quo by China and Russia.

Policy instruments We study a specific form of industrial policy in the form of subsidies to the capital stock. In these types of models, there are in general many policies the government could use to implement its desired allocation of capital, including investment subsidies or direct government control of the defense industrial base. The subsidies are therefore best interpreted as wedges that clarify when the government should deviate from a competitive equilibrium without industrial policy rather than as a prescription for the use of a particular policy instrument.

Deterrence and beliefs We have emphasized that the duration of war matters for deterrence. In the baseline model, countries have rational expectations in the sense that they agree on T, and their beliefs coincide with the true duration of war. But because

[8]The model can be extended to allow Adversary's stocks to follow exogenous paths that differ between peace and conflict. These paths would change the level of $\mathcal{D}_t$ but leave its marginal effects with respect to K_t^S and M_t^S unchanged. Such shifts in the deterrence constraint can be represented through changes in η_t^A.

war never occurs in a deterrence equilibrium, what matters for deterrence is how Adversary expects a war to unfold if it attacks. Sovereign must therefore take into account Adversary's beliefs about the duration of war, which may not always coincide with its true duration. For example, Mearsheimer (1983) argues that deterrence is more likely to fail when an adversary believes that a war will be short—that its troops will be home by Christmas. Misperceptions can also go in the opposite direction: leaders may expect wars to be long even when they turn out to be short.[9] The argument for industrial policy does not require Adversary's beliefs about war duration to be correct. What matters is that Adversary assigns some positive probability to a war lasting long enough for Sovereign's wartime production to affect the outcome. Of course, when Adversary believes that wars are likely to be short, it places less weight on Sovereign's ability to produce additional equipment, weakening the deterrent value of industrial capital.

3 Optimal Economic Policy to Sustain the Peace

In this section, we study how the government should use procurement and industrial policy to maximize domestic welfare while sustaining peace through deterrence. After stating the peacetime planning problem, we explain when and why capital in the defense industrial base contributes to deterrence. We then characterize the optimal policy and explain why, in the absence of policy, the market and the planner solve different cost-minimization problems. Finally, we show why optimal industrial policy can encourage investment in technologies that allow military production to scale during war.

3.1 Sovereign's Peacetime Planning Problem

The central question is how Sovereign should choose investment in military capital and in the defense industrial base in order to attain the deterrence equilibrium that maximizes domestic welfare. Given initial capital stocks K_0^S and M_0^S, we therefore focus on the fol-

[9] For example, Hitler was preparing during the fall of 1939 for a long, drawn-out war against France, yet the campaign ended after only six weeks. See chapter 10 in Tooze (2007).

lowing planning problem:

$$\max_{\{c_t^S, y_{n,t}^S, y_{m,t}^S, l_{n,t}^S, l_{m,t}^S, i_t^S, m_t^S, K_{t+1}^S, M_{t+1}^S\}_{t=0}^{\infty}} \quad \sum_{t=0}^{\infty} \beta^t \left(\bar{G}^S + c_t^S \right)$$

s.t. (i) Technology and market-clearing constraints (2)–(8), for all t,

(ii) Deterrence constraint (23), for all t,

(iii) Non-negativity constraints, for all t.

The planner chooses only peacetime allocations. In the deterrence equilibria we study, wars are off the equilibrium path, and the planner is not allowed to commit to off-path allocations. If war begins, Sovereign solves the wartime problem in Section 2.4 given its inherited capital stocks. This response is already incorporated into the deterrence constraint.

Note that this is a primal planning problem. The planner chooses the entire sequence of peacetime allocations directly. Household and firm optimization constraints are not imposed explicitly. Instead, we later verify that the allocations solving this problem can be decentralized as a competitive equilibrium using procurement and capital subsidies.

3.2 The Optimal Deterrence Decision

In this section we study the optimal investment decisions and explain when and why capital in the defense industrial base contributes to deterrence.

Optimal investment At an interior peacetime allocation, the optimality conditions for investment in military capital and defense capital are, respectively,

$$\lambda_{m,t}^S = \beta \left(\gamma_{t+1}^S \frac{\partial \mathcal{D}_{t+1}}{\partial M_{t+1}^S} + (1 - \delta_M) \lambda_{m,t+1}^S \right) \tag{24}$$

$$1 = \beta \left(\gamma_{t+1}^S \frac{\partial \mathcal{D}_{t+1}}{\partial K_{t+1}^S} + (1 - \delta_K) + \lambda_{m,t+1}^S \alpha \left(K_{t+1}^S \right)^{\alpha - 1} \left(l_{m,t+1}^S \right)^{1-\alpha} \right) \tag{25}$$

where γ_t^S is the multiplier on (23) and $\lambda_{m,t}^S$ is the multiplier on (2), both at date t.

Equation (24) characterizes optimal investment in the stock of military capital. Its value is determined by two forces: its deterrent value next period, given by the shadow value of deterrence γ_{t+1}^S multiplied by its marginal contribution to deterrence, and the value of the equipment that remains after depreciation. Equation (25) characterizes optimal investment in the defense industrial base. The planner values defense capital both

because it contributes directly to deterrence and because it raises military-goods output, as captured by the final term. As we show below, the productive role of defense capital during peace is reflected in market prices, whereas its role as a deterrent is not.

The deterrent value of military and industrial capacity Before turning to the analysis of optimal policy, it is useful to clarify the determinants of the deterrent value of capital.

Proposition 1 (Capital and Deterrence)**.** *Suppose wars are short ($T = 1$). Then the marginal deterrent values of military and industrial capacity at a possible attack date t are*

$$\frac{\partial \mathcal{D}_t}{\partial M_t^S} = \frac{\chi}{1-\beta} > 0, \qquad \frac{\partial \mathcal{D}_t}{\partial K_t^S} = 0.$$

Suppose instead that wars are long ($T = 2$) and some labor remains in numeraire production in the first period of war. Then the deterrent values of military and industrial capacity are

$$\frac{\partial \mathcal{D}_t}{\partial M_t^S} = \frac{\chi\beta}{1-\beta}\Big(1 + (1-\delta_M)\Big) > 0, \qquad \frac{\partial \mathcal{D}_t}{\partial K_t^S} = \left(\frac{\chi\beta}{1-\beta}\right)^{\frac{1}{\alpha}} (1-\alpha)^{\frac{1-\alpha}{\alpha}} > 0. \tag{26}$$

Suppose instead that wars are long ($T = 2$) and the labor constraint binds strictly in the first period of war, so all labor is allocated to military production. Then the deterrent values of military and industrial capacity are

$$\frac{\partial \mathcal{D}_t}{\partial M_t^S} = \frac{\chi\beta}{1-\beta}\Big(1 + (1-\delta_M)\Big) > 0, \qquad \frac{\partial \mathcal{D}_t}{\partial K_t^S} = \frac{\chi\beta}{1-\beta}\alpha \left(K_t^S\right)^{\alpha-1} \left(\bar{l}^S\right)^{1-\alpha} > 0. \tag{27}$$

Proof. Found in Appendix A.2. □

The proposition makes three points. First, military capacity deters in both short and long wars. Second, holding the initial military stock fixed, industrial capacity acts as a deterrent only when wars are long. The reason is that, in this model, wartime production can affect battlefield outcomes only if there is time to build before the war ends. Industrial capacity can still contribute to building the peacetime military stock before a short war. Third, the deterrent value of the defense industrial base depends on whether wartime production is limited by incentives or feasibility. We discuss these cases next.

Incentive Compatibility versus Feasibility Why does capital contribute to deterrence? In this model, it does so during long wars because, holding initial military stocks fixed, an increase in K raises wartime military production $m^{S,C}(K)$ and thereby reduces Adver-

sary's gain from attacking. There are two reasons why capital can raise wartime production, illustrated in Figure 2.

First, capital can relax a *feasibility constraint*, as shown in panel (a). Sovereign would like to expand military production further, but all available labor is already allocated to it. Production is therefore limited by the capacity of the defense industrial base. Raising capital from K_L to K_H allows the same labor endowment to produce more military equipment, shifting the capacity constraint outward and strengthening deterrence.

Second, capital can relax a *wartime incentive compatibility constraint* arising from limited commitment. During peace, Sovereign would like to commit to allocating all available labor to military production if war begins. The prospect of a larger wartime buildup strengthens deterrence, while the costs of carrying it out are never incurred in a deterrence equilibrium. Once war begins, however, Sovereign must weigh the military benefits of production against the loss of consumption. It may therefore prefer to retain some labor in numeraire production rather than carry out the promised buildup. This difference between what Sovereign would like to promise during peace and what it would choose during war is the time-inconsistency problem.

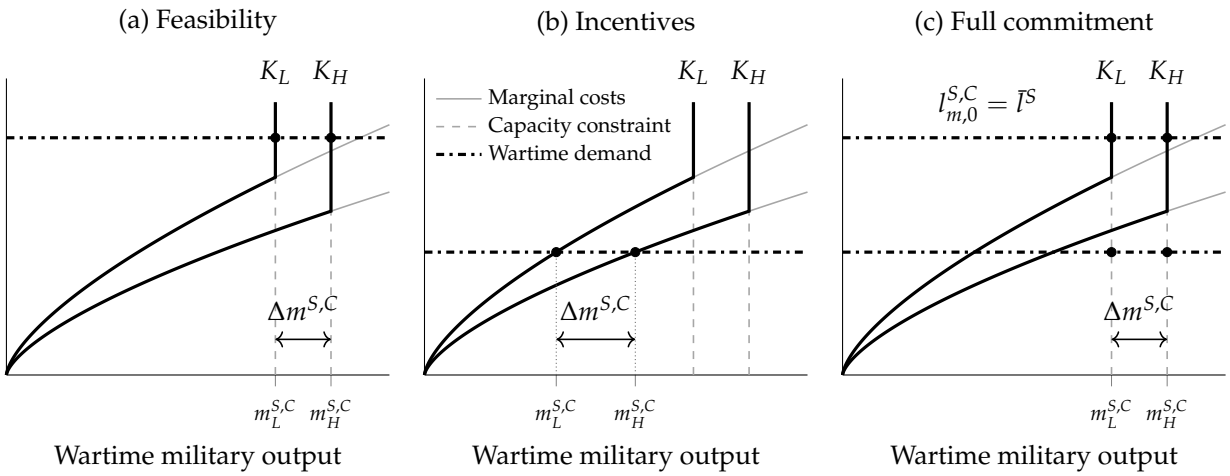
Panel (b) illustrates this case: wartime demand intersects the upward-sloping part of the supply curve, so some labor remains in numeraire production. Here, an increase in capital raises wartime military output through two effects. First, it raises output for a given allocation of labor, just as in the feasibility-constrained case. Second, it raises the marginal product of military labor, inducing Sovereign to reallocate additional labor toward military production. This latter effect brings the labor allocation closer to the full-commitment solution in which all available labor is mobilized. It is this crowding-in of military labor that we refer to as relaxing the wartime incentive compatibility constraint.

Finally, panel (c) shows that capital continues to deter even when full commitment removes the wartime incentive compatibility constraint and the associated time-inconsistency problem. Sovereign promises full labor mobilization at either demand level, raising wartime production relative to panel (b) but leaving it unchanged relative to panel (a). With labor already fully mobilized, additional capital cannot crowd in more military labor. It nevertheless raises the output that the same labor endowment can produce. Capital therefore continues to strengthen deterrence by relaxing the feasibility constraint.

3.3 Optimal industrial policy

In this section we first provide a sufficient statistic for optimal industrial policies. We then explain why industrial policy can be needed to efficiently deter long but not short wars.

Figure 2: Industrial capital and wartime military production



Note: All panels use the same supply curves; $m_L^{S,C}$ and $m_H^{S,C}$ denote wartime output at $K_L < K_H$. Under full commitment, Sovereign promises full labor mobilization at either demand level.

Lastly, we clarify the market failure by explaining why procurement policy alone does not minimize the cost of deterrence during long wars.

Optimal Subsidies We begin by studying the firm's decision to invest in capital in the defense industrial base. At an interior allocation, its first-order condition for optimal investment can be expressed as

$$1 = \beta\Big((1-\delta_K) + p^S_{m,t+1}\alpha(K^S_{t+1})^{\alpha-1}(l^S_{m,t+1})^{1-\alpha} + s^S_{t+1}\Big). \tag{28}$$

It shows that, in the absence of industrial policy, the firm invests because it increases its output, which it gets to sell at market prices. It does not consider the effect of its investment on the deterrence constraint. Each competitive firm takes prices and government procurement as given. Slackening the deterrence constraint reduces the resources needed to sustain deterrence and benefits all households, but the firm receives no payment for this contribution. At the allocation being implemented, $p^S_{m,t} = \lambda^S_{m,t}$, so comparison with (25) gives the following result.

Proposition 2. *An optimal deterrence equilibrium can be implemented by choosing procurement optimally and setting subsidies equal to*

$$s^S_t = \gamma^S_t \frac{\partial \mathcal{D}_t}{\partial K^S_t}. \tag{29}$$

Proof. Delegated to Appendix A.3. □

The expression shows that industrial policy is used because capital has a strategic value as a deterrent. The subsidy equals its marginal contribution to deterrence multiplied by the shadow value of deterrence. The other reason for a firm to value capital – because it uses it to produce output – is valued appropriately because it receives the market price for that output and hence does not appear in the expression.

We now turn to a key comparative static. The next proposition shows how optimal industrial policy depends on the length of the war.

Proposition 3. *Suppose wars are short ($T = 1$). Then the planner does not use industrial policy, $s_t^S = 0$. Suppose instead wars are long ($T = 2$) and the deterrence constraint binds, $\gamma_t^S > 0$. Then optimal subsidies are positive, $s_t^S > 0$, regardless of whether it is the incentive compatibility or feasibility constraint that binds.*

Proof: Immediate from the above propositions.

The proposition shows that whether there is a role for industrial policy is endogenous to the nature of the war. When wars are short, the defense industrial base does not add to deterrence holding the initial military stock fixed, and so industrial policy is not needed. When wars are long and the deterrence constraint has a positive shadow value, it does, and hence industrial policy is needed.[10]

On the difference between the Market and the Planner Lastly, we explain why the competitive equilibrium can be inefficient in the absence of industrial policy. We show that the planner minimizes the cost of deterrence while the market minimizes the cost of military equipment. These two problems happen to coincide when wars are short but not when wars are long.

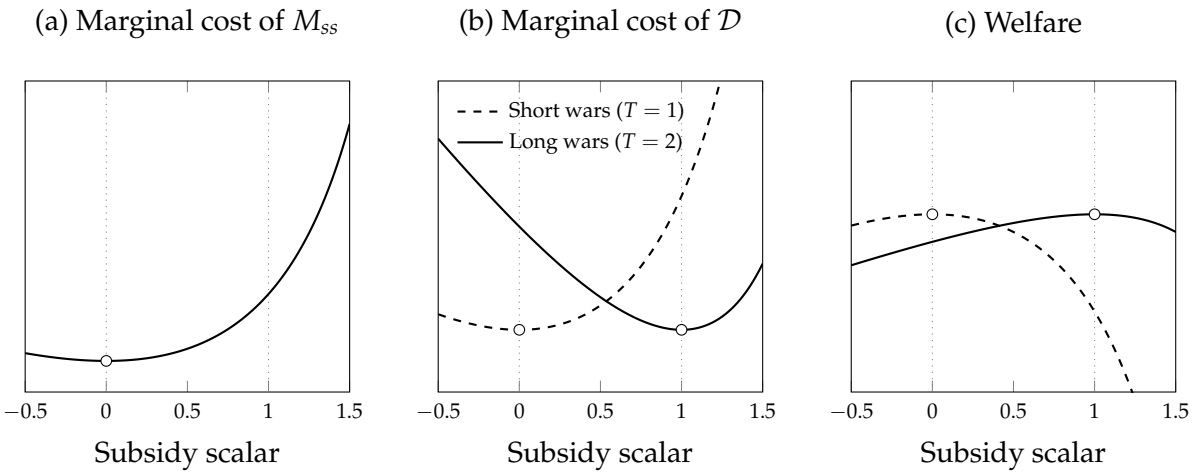
We illustrate this point by studying the steady state of the model when $\beta \to 1$.[11] This case is convenient because the steady-state problems of both the planner and the market are effectively static. The solution of the optimal static problem then coincides with the limiting steady-state allocation of the dynamic model.[12] We then obtain the following result.

[10]The deterrence constraint may not always bind. The reason it may not is that an economy could start with very high M_0^S and K_0^S. In an optimal steady state with positive military capital, the deterrence constraint always binds because otherwise a country could lower the steady-state military capital stock and enjoy higher consumption.

[11]To ensure that the value of military equipment during conflict does not explode as $\beta \to 1$, the marginal benefit of military equipment is rescaled accordingly, i.e., $\chi = (1-\beta)\hat{\chi}$ for some $\hat{\chi} > 0$.

[12]The logic is analogous to the neoclassical growth model, where the Ramsey capital stock coincides with the consumption-maximizing capital stock when $\beta \to 1$.

Figure 3: Deterrence equilibria and optimal subsidies



Note: The subsidy scalar multiplies the planner's optimal subsidy for long wars: 1 is the optimum and 0 implies no subsidies. Panel (a) plots the steady-state marginal cost of M_{ss}. Panels (b) and (c) plot the steady-state marginal cost of $\mathcal{D}$ and welfare for short and long wars. Cost curves are normalized at their own minima and welfare curves at their own maxima.

Proposition 4. *Consider a steady state of the Ramsey problem when $\beta \to 1$ with a positive shadow value of deterrence. The input choices in the defense industrial base of the competitive equilibrium in the absence of industrial policy minimize the cost of producing the steady-state level of military equipment. The planner instead minimizes the cost of providing deterrence. These problems coincide when wars are short but not when they are long. This implies that conditional on government procurement there is too little capital and hence too much labor used in military production when wars are long.*

The intuition is best explained graphically. Figure 3(a) plots the steady-state marginal cost of maintaining a larger military stock. This cost is minimized when the government does not use industrial policy to distort the private sector's choice of inputs. Panel (b) instead plots the steady-state marginal cost of providing additional deterrence. For short wars, this cost is also minimized without industrial policy. For long wars, however, it is minimized with positive subsidies, because industrial capacity itself contributes to deterrence. Panel (c) shows the corresponding welfare implications: welfare is maximized at the subsidy level that minimizes the cost of providing deterrence.

3.4 Encouraging the capacity to scale

As policymakers revisit the role of industrial policy in the defense industrial base, they have emphasized investments that allow production to scale during war. We now study

the sense in which policy should encourage this capacity to scale. In the baseline model, only a single technology is used in the defense industrial base, so there is no sense in which policy can explicitly encourage "scalability". We therefore introduce a second technology to capture the idea that some production technologies allow output to expand more readily than others. Our notion of scalability builds on Murphy and Topel (2013), who think of a technology as scalable if it has a high output elasticity with respect to a flexible input. We augment this idea by also modelling the elasticity of supply of that input. For a technology to be scalable in practice, output must respond strongly to flexible inputs, and those inputs must be available to the defense sector without sharply increasing their cost.

The baseline model is therefore extended by allowing military goods to be produced using two technologies $g \in \{1, 2\}$. Each technology has a Cobb–Douglas production function with constant returns to scale, $m_g = K_g^{\alpha_g} l_g^{1-\alpha_g}$, and total output of the military good is $m = \sum_g m_g$. One can think of the two technologies as representing different ways to produce the same equipment, or as different types of equipment that serve a similar function on the battlefield. Labor is supplied separately to each technology according to $w_g(l_g) = l_g^{1/\varepsilon_{\ell,g}}$, where $\varepsilon_{\ell,g}$ is the sectoral labor supply elasticity. The upward-sloping labor supply curves make the marginal cost of production rise as each technology expands, allowing both technologies to be used despite constant returns in production.[13]

We begin by establishing that the key results from the previous sections generalize.

Proposition 5. *When there are two technologies, optimal investment subsidies are given by*

$$s_{K,g,t+1}^{S,P} = \gamma_{t+1}^{S,P} \frac{\partial \mathcal{D}_{t+1}}{\partial K_{g,t+1}^{S,P}}. \tag{30}$$

If the deterrence constraint has a positive shadow value, the planner chooses strictly positive investment subsidies for each active technology in long wars but does not use them in short wars.

Proof. Found in Appendix A.4. □

Next we ask how investment subsidies depend on the capacity to scale production. In this extension, this capacity depends on both the output elasticity of labor and the labor

[13] We assume $0 < \alpha_g < 1$ and $0 < \varepsilon_{\ell,g} < \infty$. The labor supply curves are the same during peace and war. We interpret $w_g(l_g)$ as the marginal opportunity cost of labor in units of the numeraire. The total opportunity cost is therefore $C_g(l_g) = l_g^{1+1/\varepsilon_{\ell,g}}/(1 + 1/\varepsilon_{\ell,g})$, and the resource constraint becomes $c + \sum_g i_g + \sum_g C_g(l_g) = \bar{l}$. We focus on allocations with positive numeraire consumption during peace and in the first period of conflict, and retain a common capital depreciation rate δ_K.

supply elasticity. Holding capital fixed, the short-run supply elasticity of technology g is

$$\varepsilon_g \equiv \left.\frac{\partial \log m_g}{\partial \log p_m}\right|_{K_g} = \frac{1-\alpha_g}{\alpha_g + 1/\varepsilon_{\ell,g}}. \tag{31}$$

A higher output elasticity of labor, $1-\alpha_g$, or a higher labor supply elasticity, $\varepsilon_{\ell,g}$, therefore makes production more scalable in the short run. The supply elasticity is constant, so the same expression applies during peace and war.

Proposition 6. *Consider optimal subsidies in the steady state when wars are long, both technologies are active, and the deterrence constraint has a positive shadow value. Suppose that military production rises during conflict relative to the peacetime steady state and that technology 1 is more scalable, $\varepsilon_1 > \varepsilon_2$. In this case the planner subsidizes technology 1 relative to technology 2:*

$$s^{S,P}_{K,1,t+1} > s^{S,P}_{K,2,t+1}. \tag{32}$$

Proof. Found in Appendix A.5. □

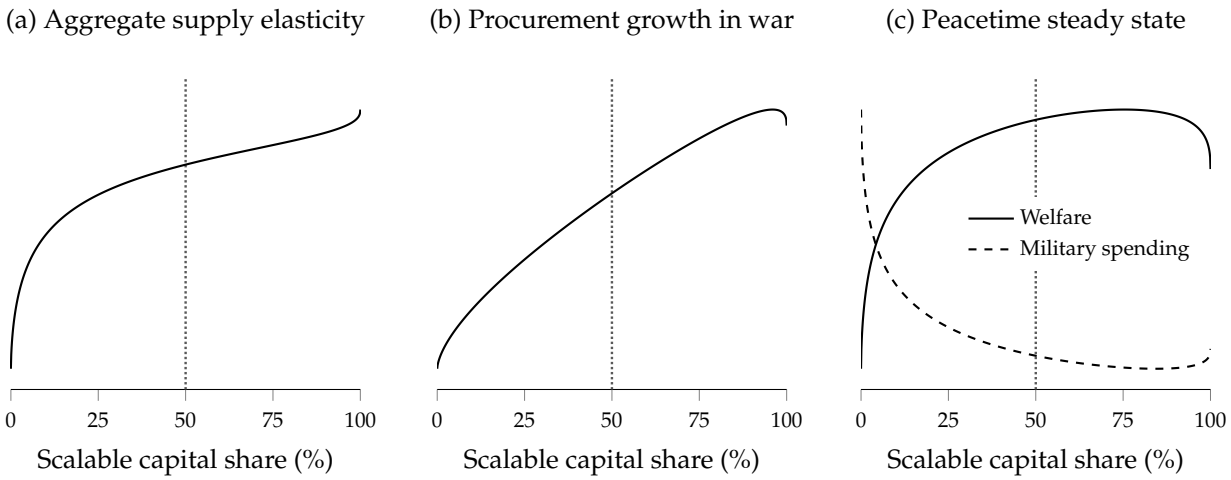
The optimal subsidies combine two effects: they expand the defense industrial base and change its composition. Having illustrated the first effect above, we use Figure 4 to illustrate the second. It plots a numerical example where in the no-subsidy steady state, both technologies have the same amount of capital. We then vary the share allocated to the more scalable technology, holding total industrial capital and deterrence fixed. Panel (a) shows that aggregate short-run supply elasticity rises as more capital is allocated to the scalable technology. Panel (b) illustrates how greater scalability can support a larger wartime expansion in procurement.[14] Panel (c) shows that changing the composition of capital alone can raise peacetime steady-state welfare and lower military expenditure: a greater capacity to expand wartime production reduces the peacetime buildup needed to sustain deterrence.

3.5 Extensions

In this section, we discuss a number of extensions to the baseline model. We provide an overview here and delegate details to Appendix **??**.

[14]The increase in procurement growth reverses at very high shares of the scalable technology. This reflects two opposing effects of changing the composition of capital. Below the no-subsidy benchmark, increasing the scalable technology's capital share makes supply more elastic and lowers the cost of producing the benchmark peacetime quantity. Above the benchmark, further increases make supply more elastic but raise that cost. Initially, the scalability effect dominates. At very high shares, however, the cost effect dominates: the marginal cost of producing a given wartime quantity rises and wartime output falls. This explains why procurement growth rises monotonically below the benchmark but not above it.

Figure 4: Scalability and the composition of industrial capital



Note: The dotted line marks the no-subsidy benchmark. Supply elasticity is evaluated at wartime output; procurement growth is relative to peacetime output. In panel (c), welfare is measured by consumption-equivalent gains and military spending by $p_m^P m^P$, on separate vertical scales.

Traditional Rational Deterrence Theory Traditional rational deterrence theory has long emphasized the problem of credibility: a country may threaten to fight but prefer to back down once challenged. Our baseline model abstracts from this problem by assuming that Sovereign always fights when Adversary challenges the status quo. But there are historical examples in which countries have not responded to such challenges with war. For example, France did not respond to Germany's reoccupation of the Rhineland with military force. Similarly, Ukraine in 2014 did not attempt to retake Crimea.

To study how this credibility problem affects optimal industrial policy, we study an extended model in which we explicitly model the possibility of backing down. A country that wants to deter needs to commit to using military force when the status quo is challenged. This implies that countries also need to satisfy a limited commitment constraint. The cost of war needs to be sufficiently low for a country to be able to commit to going to war. This constraint generates another argument for industrial policy. Countries choose capital stocks in order to lower the cost of war. The optimal industrial policy that implements these capital stocks subsidizes capital that is valuable during war. The expression is identical to the one found in Kooi (2025). That paper develops many of its implications in an open-economy general-equilibrium model. It shows that this argument for industrial policy can also support investment in scalable technologies. Because many of the implications are already developed there, we do not develop them again here.

Equilibrium wars In our baseline model, we study a deterrence equilibrium in which war never occurs. We now add shocks to Adversary's willingness to attack, so that it sometimes chooses war despite Sovereign's deterrence efforts. We study how industrial and procurement policy can attain a generalized deterrence equilibrium—a given sequence of war probabilities—most efficiently. The deterrence constraint now depends on capital and military stocks in all future periods. The intuition is that Adversary compares attacking today with waiting and retaining the option to attack later. Anticipated future stocks therefore affect its current incentives to attack. For example, stronger future defenses can make waiting less attractive and encourage an earlier attack. The basic principle therefore carries over, but the optimal investment subsidy at date t now reflects how the anticipated capital stock at that date affects attack incentives both at date t and at earlier dates.

Endogenous War Duration Our baseline model assumes that the duration of war is exogenous. We therefore study an extended model in which the probability of reaching the second period depends on the initial military balance, assuming that a larger Sovereign military stock makes a long war more likely. We show that this does not change the optimal subsidy formula in Proposition 2. But it does change the shape of the deterrence function in two ways. First, industrial capital contributes to deterrence only in long wars, so its deterrent value is weighted by the probability of reaching the second period. Second, there is now an additional effect of initial military equipment, namely that it changes the probability that the war lasts into the second period. If long wars are at least as favorable for deterrence as short wars, then in the steady state Sovereign chooses a larger military stock relative to capital in the defense industrial base, and therefore chooses lower subsidies than in the case where wars are always long.

Strategic Interactions In our baseline model, Adversary chooses whether to attack, but its military stock is fixed. We extend the model to allow for a more general form of strategic interaction in which Adversary adjusts its military stock in response to Sovereign's peacetime choices. Specifically, Adversary's military stock is allowed to respond separately to Sovereign's military stock and to Sovereign's capital in the defense industrial base. In the steady state, when wars are long, a stronger response to Sovereign's military stock makes that stock less effective at providing deterrence, so Sovereign relies more on the defense industrial base and optimal capital subsidies rise. A stronger response to Sovereign's defense-industrial capital makes that capital less effective at providing deterrence, so optimal capital subsidies fall.

4 Quantitative model

In our baseline model, a single sector represents the defense industrial base. In practice, the defense industrial base and the civilian sector consist of multiple industries, all relying on other industries' output to produce and invest. To study how industrial policy should be targeted, we develop an Integrated Defence Economics Model (IDEM). It combines an economic block describing how the economy responds to military buildups and industrial policy with a military block describing how wars unfold. Together, these blocks determine how capital investment contributes to deterrence.

We present the model below and study optimal policy in the next section. To simplify the exposition, we describe Sovereign's economy, suppressing the country index and dependence on the history of geopolitical states.

4.1 Economic Block

Let $\mathcal{G}$ denote the set of N sectors. All aggregators below have constant returns to scale.

Preferences The representative household consumes sectoral goods $c_{g,t}$, supplies labor $\bar{L}_{g,t}$, and values the public good G_t. The discount factor is β and consumption weights are ϵ_g. Following Vom Lehn and Winberry (2022), τ is the elasticity of labor supply across sectors and ξ is the Frisch elasticity of the aggregate labor index. Preferences are

$$\sum_{t=0}^{\infty} \beta^t \Big(\log C_t - \frac{\tilde{L}_t^{1+1/\xi}}{1+1/\xi} + G_t \Big), \quad C_t = \prod_{g \in \mathcal{G}} c_{g,t}^{\epsilon_g}, \quad \tilde{L}_t = \Big[\sum_{g \in \mathcal{G}} \bar{L}_{g,t}^{\frac{\tau+1}{\tau}} \Big]^{\frac{\tau}{\tau+1}}.$$

Production Sector g uses capital $K_{g,t}$, labor $L_{g,t}$, and intermediate goods $x_{g,r,t}$ purchased from sectors r. Let $V_{g,t}$ denote value added, $X_{g,t}$ the intermediate composite, and $Y_{g,t}$ gross output. The α's are sector-specific technology weights. The elasticities σ_{VA}, σ_X, and σ_Y govern substitution between capital and labor, across intermediate goods, and between value added and intermediates, respectively, and are common across sectors:

$$V_{g,t} = \Big[\alpha_{K,g} K_{g,t}^{\frac{\sigma_{VA}-1}{\sigma_{VA}}} + \alpha_{L,g} L_{g,t}^{\frac{\sigma_{VA}-1}{\sigma_{VA}}} \Big]^{\frac{\sigma_{VA}}{\sigma_{VA}-1}}, \qquad X_{g,t} = \Big[\sum_{r \in \mathcal{G}} \alpha_{g,r} x_{g,r,t}^{\frac{\sigma_X-1}{\sigma_X}} \Big]^{\frac{\sigma_X}{\sigma_X-1}},$$

$$Y_{g,t} = \Big[\alpha_{V,g} V_{g,t}^{\frac{\sigma_Y-1}{\sigma_Y}} + \alpha_{X,g} X_{g,t}^{\frac{\sigma_Y-1}{\sigma_Y}} \Big]^{\frac{\sigma_Y}{\sigma_Y-1}}.$$

Capital Accumulation Sector-specific capital depreciates at rate δ_g. Investment is irreversible, $I_{g,t} \geq 0$, and adjustment costs governed by $\varphi > 0$ reduce effective investment,

following Vom Lehn and Winberry (2022). The weights $\vartheta_{g,r}$ describe the composition of sector g's investment purchases $i_{g,r,t}$:

$$K_{g,t+1} = (1-\delta_g)K_{g,t} + I_{g,t} - \frac{\varphi}{2}\Big(\frac{I_{g,t}}{K_{g,t}} - \delta_g\Big)^2 K_{g,t}, \qquad I_{g,t} = \prod_{r\in\mathcal{G}} i_{g,r,t}^{\vartheta_{g,r}}, \qquad g \in \mathcal{G}.$$

Military Power The government accumulates J military stocks $M_{j,t}$, indexed by $j \in \mathcal{J}$. Suppliers are partitioned into sets $\mathcal{G}_j \subset \mathcal{G}$; weights $\psi_{j,g}$ aggregate purchases $m_{j,g,t}$ into non-negative procurement $m_{j,t}$. Depreciation $\delta_{j,t}$ equals δ_j in peace and after war, and $\delta_j^w \geq \delta_j$ during war to reflect combat losses. The weights ϕ_j determine each stock's contribution to military power M_t:[15]

$$m_{j,t} = \prod_{g\in\mathcal{G}_j} m_{j,g,t}^{\psi_{j,g}}, \qquad M_{j,t+1} = (1-\delta_{j,t})M_{j,t} + m_{j,t}, \qquad M_t = \prod_{j\in\mathcal{J}} \Big(\frac{M_{j,t}}{\phi_j}\Big)^{\phi_j}.$$

Market Clearing Sectoral output meets consumption, intermediate, investment, and military demand. At sectoral wage $w_{g,t}$, labor demand $L_{g,t}$ equals supply $\bar{L}_{g,t}$:

$$Y_{g,t} = c_{g,t} + \sum_{b\in\mathcal{G}} x_{b,g,t} + \sum_{b\in\mathcal{G}} i_{b,g,t} + \sum_{j\in\mathcal{J}} m_{j,g,t}, \qquad L_{g,t} = \bar{L}_{g,t}, \qquad g \in \mathcal{G}.$$

Competitive Equilibrium The competitive equilibrium is analogous to that in Section **??**. The key change is that subsidies apply at rate $s_{g,t}$ to capital's replacement value $P^I_{g,t}K_{g,t}$, where $P^I_{g,t}$ is the price of sector g's investment basket. The baseline per-unit subsidy corresponds to $s_{g,t}P^I_{g,t}$. Let $R_t = \prod_{v=0}^{t-1}(1+r_v)$, with $R_0 = 1$, denote the cumulative gross return. With goods prices $p_{g,t}$ and wages $w_{g,t}$, the producer's objective now reads

$$\sum_{t=0}^{\infty} \frac{1}{R_t}\Big(p_{g,t}Y_{g,t} + s_{g,t}P^I_{g,t}K_{g,t} - w_{g,t}L_{g,t} - \sum_{r\in\mathcal{G}} p_{r,t}\big(x_{g,r,t} + i_{g,r,t}\big)\Big).$$

Appendix C.1 defines equilibrium and derives the optimality conditions.

4.2 Deterrence Block

Timing and war duration Military outcomes in the initial period determine whether a war is short or long. If Sovereign avoids a quick defeat, the conflict becomes an "industrial war of attrition," whose duration follows an exogenous distribution. Let π_T denote

[15] We collect military stocks in $\mathbf{M}_t$ and sectoral capital stocks in $\mathbf{K}_t$.

baseline probabilities of wars lasting $T \in \{1, \ldots, \bar{T}\}$ periods. The initial balance of power scales the probability of a long war by $a(M_t)$: a stronger Adversary is more likely to defeat Sovereign quickly. With M_0^E denoting Adversary's initial military power and $\kappa \geq 0$ the sensitivity to its share of military power, we parameterize

$$\tilde{\pi}_T(M_t) = a(M_t)\,\pi_T \quad (T \geq 2), \qquad \tilde{\pi}_1(M_t) = 1 - a(M_t)\,(1 - \pi_1),$$
$$a(M_t) = 1 - \kappa\,\frac{M_0^E}{M_t + M_0^E}.$$

The probability of reaching the attrition phase is therefore $a(M_t)(1 - \pi_1)$; conditional on reaching it, duration follows the fixed distribution $\pi_T/(1 - \pi_1)$ for $T \geq 2$.[16]

The allocation of the public good Re-index dates so that war begins at date zero. The parameters $\chi_s^{(T)}$ measure the effectiveness of military power in period s of a war lasting T periods; M_s^E denotes Adversary's military power. The postwar settlement changes the public good from its initial level G_0 according to

$$G_t = G_0 + \sum_{s=0}^{T-1} \chi_s^{(T)} \left(M_s - M_s^E\right) \quad \text{if } t \geq T, \qquad \text{and} \qquad G_t = G_0 \quad \text{otherwise.}$$

Adversary's policy and challenge decision Adversary's military power is constant in peace and follows an exogenous path $\{M_s^E\}_{s=0}^{T-1}$ in a war of length T. Its consumption is the same in peace and war.[17] Adversary attacks when $W_t^{A,C} + \eta_t > W_t^{A,P}$, where η_t is its political preference for war. It decides before the duration is known, so welfare from war averages welfare $W_t^{A,C,(T)}$ across durations:

$$W_t^{A,C} = \sum_{T=1}^{\bar{T}} \tilde{\pi}_T(M_t)\, W_t^{A,C,(T)}.$$

Sovereign's wartime and postwar policy As in the baseline model, Sovereign optimally chooses wartime military expenditure, trading off consumption against military gains in the settlement. After war, peace is permanent and the public good is fixed. The departure from the baseline is that postwar expenditure follows an exogenous rule: the government

[16] We represent this distinction in reduced form: the duration is drawn and revealed to both parties at the outbreak.

[17] An exogenous wartime consumption change would shift the deterrence constraint without affecting Sovereign's wartime choices. This shift is absorbed by calibrating Adversary's military power to US peacetime spending, leaving optimal peacetime policy unchanged.

spends a share θ of GDP on procurement, allocated across baskets with shares θ_j. This rule allows the model to capture a realistic postwar reduction in military procurement. Industrial policy is not needed during or after war. Appendices C.2 and C.1 give the wartime problem and postwar equilibrium.

Deterrence Equilibrium As in Definition **??**, a deterrence equilibrium is a competitive equilibrium (Definition 3) satisfying $\mathcal{D}_t(\mathbf{K}_t, \mathbf{M}_t) \geq \eta_t$ in every period. We derive $\mathcal{D}_t$ below.

The Simplified Deterrence Constraint Let $M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t)$ be Sovereign's optimal military power in period s of a war lasting T periods, starting from date-t stocks $(\mathbf{K}_t, \mathbf{M}_t)$. Since Adversary's consumption is unchanged by war, deterrence requires the expected discounted settlement to offset its preference for war (appendix C.2):

$$\mathcal{D}_t(\mathbf{K}_t, \mathbf{M}_t) = \sum_{T=1}^{\bar{T}} \tilde{\pi}_T(M_t) \frac{\beta^T}{1-\beta} \sum_{s=0}^{T-1} \chi_s^{(T)} \left(M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t) - M_s^E \right) \geq \eta_t, \qquad \forall t.$$

4.3 Calibration

We calibrate the annual model to the United States, with $N = 378$ sectors and $J = 8$ military baskets. The economic structure and military stocks are disciplined by sectoral accounts and defense procurement data; war durations and wartime spending discipline the conflict parameters. Appendix C.3 details the data construction and calibration procedure.

Economic structure Consumption and production weights reproduce spending shares in the 2017 BEA input-output accounts. Investment weights $\vartheta_{g,r}$ and sectoral depreciation rates δ_g come from Iovino et al. (2025), who extend the investment network of Vom Lehn and Winberry (2022) to the 2017 BEA detail level. The appendix explains how we choose and normalize the weights to match observed shares.

Table 1 reports the common economic parameters and their sources.

Military stocks and procurement The postwar spending share is $\theta = 3.37\%$ of GDP, the US average over 2015–2025 in SIPRI. Basket shares θ_j and within-basket weights $\psi_{j,g}$ match 2017 US defense procurement. We choose military-power weights ϕ_j so that the no-industrial-policy benchmark reproduces this spending composition.

Table 1: Economic parameters

Parameter	Value	Source
β, discount factor	0.96	Authors' calibration
ξ, aggregate Frisch elasticity	0.4	Reichling and Whalen (2012)
τ, sectoral labor supply elasticity	4.5	Vom Lehn and Winberry (2022)
φ, capital adjustment cost	0.5	Vom Lehn and Winberry (2022)
σ_Y, value added–intermediates elasticity	0.5	Baqaee and Farhi (2019)
σ_{VA}, capital–labor elasticity	0.5	Oberfield and Raval (2021)
σ_X, intermediate-input elasticity	0.001	Baqaee and Farhi (2019)

Peacetime depreciation rates come from BEA national-defense fixed assets, with ammunition depreciation measured by purchases relative to stocks in DoD inventory reports. Wartime rates add combat losses from the Russia–Ukraine war for land systems and US World War II experience for air and naval systems. Structures, other equipment, and intellectual property retain their peacetime rates; non-durables are consumed within the year, as is ammunition during war. Table 2 reports the resulting parameters. This calibration describes a high-intensity combined-arms war fought abroad; the next section considers alternatives.

Table 2: Military basket parameters. Depreciation rates are annual; all entries are rounded to three decimal places.

	Spending	Depreciation		Military-power
Basket	share, θ_j	Peace, δ_j	War, δ_j^w	weight, ϕ_j
Structures	0.030	0.020	0.020	0.026
Other equipment	0.232	0.116	0.116	0.184
Intellectual property	0.309	0.223	0.223	0.354
Non-durables	0.168	1.000	1.000	0.188
Land systems	0.007	0.155	0.383	0.007
Air systems	0.147	0.101	0.343	0.123
Naval systems	0.053	0.081	0.146	0.040
Ammunition	0.054	0.127	1.000	0.078

War duration Baseline probabilities π_T match annual duration frequencies of inter-state wars with at least 1,000 battle deaths in the Correlates of War data, 1816–2007. Wars lasting six years or more are pooled at $\bar{T} = 6$. Conditional on avoiding a quick defeat, these frequencies determine the duration of the industrial war of attrition. We set $\kappa =$

0.33 for the sensitivity to the initial balance of power, based on the relationship between war duration and the initiator's prewar share of the two sides' military personnel in the National Material Capabilities data.

Wartime spending We recover combat efficiency parameters $\chi_s^{(T)}$ from the procurement conditions that make the wartime spending path estimated by Marzian and Trebesch (2026) optimal. Starting from the no-industrial-policy steady state, we target spending increases above the prewar level of $(5.3, 15, 19, 21.5, 20.5)\%$ of steady-state GDP in the first five years. A war of length T matches the first $T-1$ targets, since final-period procurement cannot contribute to wartime military power. Table 9 in the appendix reports the fitted parameters. Outbreak efficiency does not affect procurement choices; we set $\chi_0 = 113.6$, the probability-weighted mean of subsequent efficiencies across war lengths.

Peacetime military expenditure We calibrate Adversary's political preference for war, η_t, so that military expenditure in the no-industrial-policy peacetime steady state matches the US average of 3.37% of GDP. Under our baseline symmetry assumption, Adversary's military power equals Sovereign's wartime path from this steady state for each war length. The deterrence constraint therefore binds at the observed expenditure level when $\eta_t = 0$, which is our calibrated value.

5 Quantitative results

In this section, we first briefly describe Sovereign's peacetime planning problem, and the formula for optimal subsidies that the government uses to decentralize the planner's allocation and achieve the optimal deterrence equilibrium. We then present the numerical results on the optimal deterrence equilibrium in our model.

5.1 Optimal Deterrence Equilibrium

Sovereign's peacetime planning problem generalizes that in Section **??**. Given initial stocks, the planner solves

$$\begin{aligned} &\max_{\{c_{g,t},\bar{L}_{g,t},x_{g,r,t},i_{g,r,t},I_{g,t},K_{g,t+1},m_{j,g,t},m_{j,t},M_{j,t+1}\}_{t=0}^{\infty}} \quad \sum_{t=0}^{\infty}\beta^t\left(\log C_t - \frac{\tilde{L}_t^{1+1/\xi}}{1+1/\xi} + G_0\right) \\ &\text{s.t. (i) the technology and market-clearing conditions (120), for all } t, \\ &\quad \text{(ii) the laws of motion of capital and military stocks (121), for all } t, \\ &\quad \text{(iii) the deterrence constraint } \mathcal{D}_t(\mathbf{K}_t,\mathbf{M}_t)\geq\eta_t, \text{ for all } t, \\ &\quad \text{(iv) the non-negativity constraints } I_{g,t}\geq 0 \text{ and } m_{j,t}\geq 0, \text{ for all } t. \end{aligned} \tag{33}$$

Proposition 2 likewise generalizes to this economy. Let $\gamma_t \geq 0$ denote the shadow value of relaxing the date-$(t+1)$ deterrence constraint, expressed in units of date-$(t+1)$ consumption.

Proposition 7. *An optimal deterrence equilibrium can be implemented with sectoral capital subsidy rates*

$$s_{g,t+1} = \frac{\gamma_t}{P^I_{g,t+1}}\frac{\partial \mathcal{D}_{t+1}}{\partial K_{g,t+1}}, \qquad g\in\mathcal{G}.$$

Proof. See Appendix C.4. □

The subsidy compensates firms for capital's marginal deterrence value, which they do not internalize. Dividing by the investment-basket price $P^I_{g,t+1}$ converts this value into a rate on capital's replacement value. Capital can affect deterrence through military production and through supplies of intermediate and investment goods to other sectors; the derivative captures these effects through the wartime equilibrium. When $\gamma_t > 0$, positive marginal deterrence effects warrant subsidies and negative effects warrant taxes.

5.2 Aggregate outcomes

Our model studies how countries can deter efficiently. We are interested in how countries can use industrial policy to lower the cost of sustaining peace. Table 3 shows how industrial policy achieves this in our model. Both steady states provide the same level of deterrence. By complementing peacetime military strength with industrial capacity, the government attains this level with a lower military procurement share of GDP. Relative

to the benchmark of deterrence through procurement alone, steady-state welfare rises by 2.35% in consumption-equivalent terms.[18]

Table 3: Aggregate outcomes in the steady state.

	optimal deterrence equilibrium	no industrial policy
deterrence, $\mathcal{D}$ (index)	1	1
welfare gain, consumption-equivalent (%)	2.35	-
consumption, relative	1.033	1
military procurement, % of GDP	2.46	3.37
real GDP, relative	1.042	1
aggregate capital, relative	1.091	1
labor supply, relative	1.011	1

Deterrence is indexed to one at its common binding level, $\mathcal{D} = \eta = 0$. Relative quantities are normalized to the no-industrial-policy benchmark; GDP and capital are Fisher quantity indices. Welfare gains are expressed as equivalent permanent increases in benchmark consumption.

The table also explains why steady-state welfare rises. The main driver is a 3.3% increase in consumption, supported by a lower military procurement burden and an increase in steady-state real GDP. Military procurement's share of GDP falls from 3.37% to 2.46%, while real GDP rises by 4.2%. The increase in output is supported by a 9.1% rise in the aggregate capital stock. As discussed in the next subsection, most sectors receive positive subsidies, which encourage investment and capital accumulation. A modest 1.1% increase in labor supply also contributes to production. This increase in labor offsets part of the consumption benefit, explaining why the welfare gain is smaller than the increase in consumption.

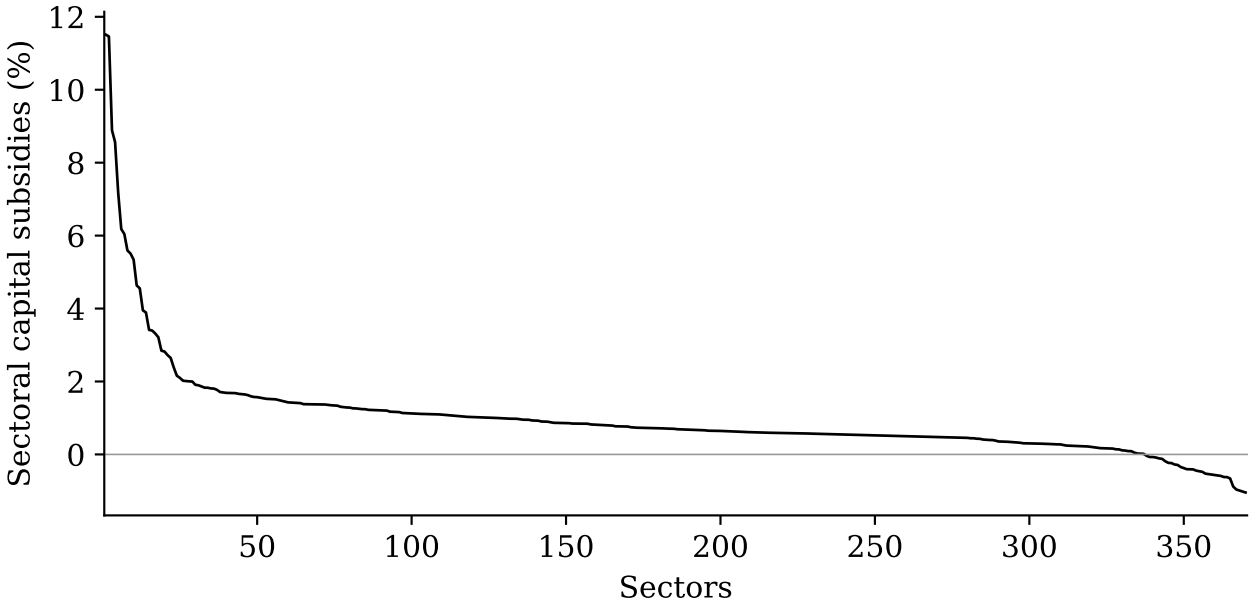
5.3 Industrial priorities in the baseline

We now use the quantitative model to study which industries a government should encourage and how much their subsidies should differ. We begin by examing the cross-sectoin of industries. Figure 5 provides a first insight: optimal subsidy rates decline rapidly as one moves down the sectoral ranking. This result is interesting because there a soft consensus among economist that the national security argument for industrial policy at best applies to a narrow range of industries. Figure 5 shows the model delivers this

[18]Eliminating the deterrence constraint altogether would raise welfare by 4.74% relative to the same benchmark. Industrial policy therefore eliminates approximately half of the steady-state welfare cost of deterrence.

quantitatively: optimal subsidies declines by 75% within the first twenty sectors falling from 11.33% to 2.83%

Figure 5: The cross-section of optimal capital subsidies.



Each bar is one of the 378 sectors, ordered by decreasing subsidy. The subsidy s_g is a rate on the replacement value of the sector's capital stock.

Next, we present the ranking of industrial priorities. Table 4 reports the twenty industries with the highest subsidy rates; the full ranking appears in Appendix C.7. Three broad categories dominate the ranking. First are producers of munitions and their components, including guided missiles, propulsion units, and ammunition. Second are manufacturers of heavy military equipment—ships, aircraft, and armored vehicles—and their input suppliers. Third are producers of advanced electronics, including search, detection, and navigation instruments, electrical and signal testing equipment, and communications equipment.

What determines whether a sector is an industrial priority? Our model highlights two aspects. First, military stocks differ in how quickly they are depleted. Military power requires a mix of equipment, with some scope for substitution. Tanks, for example, need shells, but the two stocks are depleted at different rates: a shell can be fired only once, whereas a tank can fire many shells. Sustaining a military force therefore requires replenishing munitions more frequently than tanks. Tanks, in turn, suffer battlefield losses and require more frequent replacement than military structures in our baseline calibration. These differences create particularly strong replacement demand for munitions and equipment with high wartime loss rates, raising the value of the capacity to produce them.

Table 4: The 20 sectors with the highest optimal capital subsidies.

rank	sector	s_g (%)
1	Guided missile and space vehicle manufacturing	11.33
2	Propulsion units and parts for space vehicles and guided missiles	10.91
3	Ship building and repairing	9.17
4	Search, detection, and navigation instruments manufacturing	9.12
5	Ammunition, arms, ordnance, and accessories manufacturing	8.55
6	Military armored vehicle, tank, and tank component manufacturing	7.24
7	Other aircraft parts and auxiliary equipment manufacturing	7.06
8	Electricity and signal testing instruments manufacturing	6.69
9	Electronic and precision equipment repair and maintenance	5.56
10	Architectural, engineering, and related services	5.06
11	Aircraft manufacturing	4.89
12	Broadcast and wireless communications equipment	4.48
13	Switchgear and switchboard apparatus manufacturing	4.00
14	Scientific research and development services	3.59
15	Motor vehicle electrical and electronic equipment manufacturing	3.47
16	Other electronic component manufacturing	3.42
17	Other computer related services, including facilities management	3.24
18	Air transportation	2.86
19	Aircraft engine and engine parts manufacturing	2.84
20	Cutlery and handtool manufacturing	2.83

The increase in replacement demand is especially pronounced for stocks whose loss rates rise sharply between peace and war.

Second, a sector can become an industrial priority through its exposure to military demand along the input-output and investment networks. Expanding capacity in component suppliers can lower the cost of military equipment and allow a larger wartime buildup. In Appendix C.8, we show numerically how this mechanism operates across sectors. In propulsion unit manufacturing, additional capacity supports direct military purchases and lowers the cost of inputs used in missile production. Suppliers of investment goods, such as research services, also help expand production capacity during war. Because the full model does not yield a closed-form ranking, we develop a simple analytical example in Appendix XX. We show that, all else equal, optimal subsidies are higher for sectors that derive more revenue from military demand relative to civilian demand, accounting for both direct and indirect sales.The ranking implied by this example correlates well with the full model's ranking in Table 4.

5.4 Industrial priorities and the nature of conflict

Optimal industrial policy depends not only on economic fundamentals, such as the input-output and investment networks, but also on the war a country expects to fight. Different conflicts require different mixes of military equipment, giving rise to different industrial priorities. When wars may be prolonged, economic policymaking and military analysis therefore need to be closely linked. To illustrate this point, we use the model to consider three stylized scenarios: a land war, an air war, and a naval war. In each scenario, we set the military-power weights of the other two platform categories to zero and solve again for the optimal deterrence equilibrium. Common military inputs, including ammunition, continue to contribute to military power in all three scenarios.

Table 5 reports baseline subsidy rates and the changes under each scenario for five industries whose subsidies vary substantially across scenarios. Shipbuilding receives a subsidy of 9.17% in the baseline, which rises to 10.05% in the naval scenario but falls to around 1% in the land and air scenarios. The subsidy to armored vehicles and tanks rises in the land scenario and falls by roughly four percentage points in the other two. Subsidies to aircraft and their components remain close to their baseline levels in the air scenario but fall substantially in the land and naval scenarios.

Table 5: Changes in optimal capital subsidies across conflict scenarios.

	Baseline	Change from baseline (pp)		
Sector	(%)	Land	Air	Naval
Shipbuilding	9.17	−8.13	−7.89	+0.88
Armored vehicles, tanks, and components	7.24	+0.60	−4.03	−3.67
Other aircraft parts and auxiliary equipment	7.06	−5.95	−0.38	−5.99
Aircraft manufacturing	4.89	−3.55	−0.13	−3.93
Aircraft engines and parts	2.84	−1.57	−0.11	−1.87

Baseline entries are subsidy rates s_g in percent; scenario entries are changes in percentage points. Sector names are abbreviated. Appendix **??** reports subsidy levels for all twenty industries in Table 4.

Other industries remain priorities across all three scenarios. As the fuller comparison in Appendix **??** shows, guided missile and propulsion unit manufacturing retain subsidy rates around 11%, while ammunition and search, detection, and navigation instruments also remain highly subsidized. These industries supply military inputs that remain important across the scenarios.

6 Conclusion

We develop a dynamic model of rational deterrence to study how procurement and industrial policy can jointly sustain peace at the lowest cost. When wars are short, procurement alone suffices. When wars last long enough for wartime production to affect military outcomes, industrial capacity itself contributes to deterrence. Private investors do not internalize this strategic benefit, creating a national security externality that justifies support for industrial investment and more scalable production technologies. We develop an Integrated Deterrence–Economy Model (IDEM) to operationalize this framework and show how it can generate quantitative rankings of industries to guide the targeting of industrial policy.

An important implication of our analysis is that economic policy for national security should be closely linked to military analysis. IDEM takes a step in this direction, but there is much more that could be done. Contemporaneous work by Grassi and Markó (2026) integrating combat models that generate Lanchester laws with general equilibrium models of production could further enrich IDEM's military block. Introducing geography would allow us to study how breakthroughs expose industrial capacity to capture and how the scope for retreat—strategic depth—in larger countries can prolong conflict. It would also allow a richer treatment of air power. Our current model is closest to its tactical role in supporting other armed forces on the front line, while a spatial model could also capture its strategic role in disrupting an adversary's economy.

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A Proofs

A.1 Derivation of (24) and (25)

Proof. We derive the planner optimality conditions in the peacetime problem. Suppress the argument $z_t = P$ and write the constraints as

$$c_t^S + i_t^S = l_{y,t}^S, \tag{34}$$
$$K_{t+1}^S = (1-\delta_K)K_t^S + i_t^S, \tag{35}$$
$$l_{y,t}^S + l_{m,t}^S = \bar{l}_t^S, \tag{36}$$
$$m_t^S = (K_t^S)^\alpha (l_{m,t}^S)^{1-\alpha}, \tag{37}$$
$$M_{t+1}^S = (1-\delta_M)M_t^S + m_t^S, \tag{38}$$
$$\mathcal{D}_t(K_t^S, M_t^S) \geq \eta_t^A. \tag{39}$$

Let $\lambda_{y,t}^S$ be the multiplier on (34), $\lambda_{K,t}^S$ on (35), $\lambda_{l,t}^S$ on (36), $\lambda_{m,t}^S$ on (37), $\lambda_{M,t}^S$ on (38), and γ_t^S on (39). The Lagrangian is

$$\begin{aligned}\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big(& c_t^S + \lambda_{y,t}^S (l_{y,t}^S - c_t^S - i_t^S) + \lambda_{K,t}^S \big((1-\delta_K)K_t^S + i_t^S - K_{t+1}^S\big) \\ & + \lambda_{l,t}^S (\bar{l}_t^S - l_{y,t}^S - l_{m,t}^S) + \lambda_{m,t}^S \big((K_t^S)^\alpha (l_{m,t}^S)^{1-\alpha} - m_t^S\big) \\ & + \lambda_{M,t}^S \big((1-\delta_M)M_t^S + m_t^S - M_{t+1}^S\big) + \gamma_t^S \big(\mathcal{D}_t(K_t^S, M_t^S) - \eta_t^A\big) \Big).\end{aligned}$$

Step 1: First-order conditions (within period). For each t, the FOCs w.r.t. $c_t^S, i_t^S, l_{y,t}^S, l_{m,t}^S, m_t^S$ are

$$[c_t^S]: \quad 1 - \lambda_{y,t}^S = 0, \tag{40}$$
$$[i_t^S]: \quad -\lambda_{y,t}^S + \lambda_{K,t}^S = 0, \tag{41}$$
$$[l_{y,t}^S]: \quad \lambda_{y,t}^S - \lambda_{l,t}^S = 0, \tag{42}$$
$$[l_{m,t}^S]: \quad -\lambda_{l,t}^S + \lambda_{m,t}^S (1-\alpha)(K_t^S)^\alpha (l_{m,t}^S)^{-\alpha} = 0, \tag{43}$$
$$[m_t^S]: \quad -\lambda_{m,t}^S + \lambda_{M,t}^S = 0. \tag{44}$$

Step 2: Envelope conditions (choice variables M^S_{t+1} and K^S_{t+1}). Differentiating $\mathcal{L}$ w.r.t. M^S_{t+1} yields

$$[M^S_{t+1}]: \qquad -\beta^t \lambda^S_{M,t} + \beta^{t+1}(1-\delta_M)\lambda^S_{M,t+1} + \beta^{t+1}\gamma^S_t \frac{\partial \mathcal{D}_{t+1}}{\partial M^S_{t+1}} = 0. \tag{45}$$

Differentiating $\mathcal{L}$ w.r.t. K^S_{t+1} yields

$$[K^S_{t+1}]: \qquad -\beta^t \lambda^S_{K,t} + \beta^{t+1}(1-\delta_K)\lambda^S_{K,t+1} + \beta^{t+1}\lambda^S_{m,t+1}\alpha(K^S_{t+1})^{\alpha-1}(l^S_{m,t+1})^{1-\alpha} + \beta^{t+1}\gamma^S_t \frac{\partial \mathcal{D}_{t+1}}{\partial K^S_{t+1}} = 0. \tag{46}$$

Step 3: Simplification to obtain (24) and (25). From (40)–(41) we have

$$\lambda^S_{y,t} = 1 \quad \text{and} \quad \lambda^S_{K,t} = \lambda^S_{y,t} = 1.$$

From (44) we have $\lambda^S_{M,t} = \lambda^S_{m,t}$ for all t.

Divide (45) by β^t and use $\lambda^S_{M,t} = \lambda^S_{m,t}$:

$$\lambda^S_{m,t} = \beta\Big(\gamma^S_t \frac{\partial \mathcal{D}_{t+1}}{\partial M^S_{t+1}} + (1-\delta_M)\lambda^S_{m,t+1}\Big),$$

which is (24).

Next divide (46) by β^t and use $\lambda^S_{K,t} = 1$:

$$1 = \beta\Big(\gamma^S_t \frac{\partial \mathcal{D}_{t+1}}{\partial K^S_{t+1}} + (1-\delta_K)\lambda^S_{K,t+1} + \lambda^S_{m,t+1}\alpha(K^S_{t+1})^{\alpha-1}(l^S_{m,t+1})^{1-\alpha}\Big).$$

Finally, using $\lambda^S_{K,t+1} = 1$ yields

$$1 = \beta\Big(\gamma^S_t \frac{\partial \mathcal{D}_{t+1}}{\partial K^S_{t+1}} + (1-\delta_K) + \lambda^S_{m,t+1}\alpha(K^S_{t+1})^{\alpha-1}(l^S_{m,t+1})^{1-\alpha}\Big),$$

which is (25). □

A.2 Proof of Proposition 1

Proof. Fix a possible challenge date $\bar{t}$. The deterrence constraint compares peace with the off-equilibrium continuation in which Adversary challenges at $\bar{t}$. From the simplified deterrence constraint, and suppressing terms that do not depend on Sovereign's state

variables, deterrence can be written as

$$\mathcal{D}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) = \frac{\chi\beta^{T-1}}{1-\beta} \sum_{t=\bar{t}}^{\bar{t}+T-1} M_{t,\bar{t}}^{S,C}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) + \text{constant}.$$

Thus the deterrent value of a state variable is its effect on the sequence of military stocks that Sovereign would deploy in that continuation, multiplied by the strategic value of those stocks for the allocation of the public good.

Short wars. Suppose first that $T = 1$. Then only the military stock available at the possible challenge date matters for the outcome:

$$\mathcal{D}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) = \frac{\chi}{1-\beta} M_{\bar{t}}^{S,P} + \text{constant}.$$

Military production in that continuation affects $M_{\bar{t}+1,\bar{t}}^{S,C}$, but this stock is reached only after a short war would have ended. It therefore has no effect on the allocation of the public good. Hence

$$\frac{\partial \mathcal{D}_{\bar{t}}}{\partial M_{\bar{t}}^{S,P}} = \frac{\chi}{1-\beta} > 0, \qquad \frac{\partial \mathcal{D}_{\bar{t}}}{\partial K_{\bar{t}}^{S,P}} = 0.$$

For the proposition, the relevant possible challenge date is $\bar{t} = t + 1$. This gives the first part of the proposition.

Long wars. Suppose now that $T = 2$. The relevant part of the deterrence function is

$$\mathcal{D}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) = \frac{\chi\beta}{1-\beta}\left(M_{\bar{t}}^{S,P} + M_{\bar{t}+1,\bar{t}}^{S,C}\right) + \text{constant}.$$

In this off-equilibrium continuation, the date-$\bar{t}$ stocks are predetermined, so

$$M_{\bar{t}+1,\bar{t}}^{S,C} = (1-\delta_M) M_{\bar{t}}^{S,P} + m_{\bar{t},\bar{t}}^{S,C}.$$

The wartime choice of military production depends on $K_{\bar{t}}^{S,P}$, but not on $M_{\bar{t}}^{S,P}$, because $M_{\bar{t}}^{S,P}$ enters the wartime objective additively and separably. Therefore

$$\frac{\partial \mathcal{D}_{\bar{t}}}{\partial M_{\bar{t}}^{S,P}} = \frac{\chi\beta}{1-\beta}\Big(1 + (1-\delta_M)\Big) > 0.$$

It remains to characterize the derivative with respect to the defense-industrial capital stock. In the first period of the off-equilibrium long-war continuation, capital is predeter-

mined and Sovereign chooses labor in the defense sector. Using the resource constraint to substitute out consumption, the relevant static problem is

$$\max_{0\leq l_m\leq \bar{l}^S}\left\{-l_m+\frac{\chi\beta}{1-\beta}A(K_{\bar{t}}^{S,P})^{\alpha}l_m^{1-\alpha}\right\}.$$

This problem captures the tradeoff in the conflict continuation: allocating more labor to military production lowers current consumption, but increases the military stock in the second period of the long-war continuation.

The non-negativity constraint does not bind. If the non-negativity constraint on numeraire labor does not bind, the choice of l_m is interior. The first-order condition is

$$\frac{\chi\beta}{1-\beta}A(1-\alpha)(K_{\bar{t}}^{S,P})^{\alpha}l_m^{-\alpha}=1,$$

so

$$l_{m,\bar{t},\bar{t}}^{S,C}(K_{\bar{t}}^{S,P})=\left(\frac{\chi\beta}{1-\beta}A(1-\alpha)\right)^{1/\alpha}K_{\bar{t}}^{S,P}.$$

Substituting this policy into the production function gives

$$m_{\bar{t},\bar{t}}^{S,C}(K_{\bar{t}}^{S,P})=A\left(\frac{\chi\beta}{1-\beta}A(1-\alpha)\right)^{\frac{1-\alpha}{\alpha}}K_{\bar{t}}^{S,P}.$$

It follows that

$$\frac{\partial\mathcal{D}_{\bar{t}}}{\partial K_{\bar{t}}^{S,P}}=\frac{\chi\beta}{1-\beta}\frac{\partial m_{\bar{t},\bar{t}}^{S,C}}{\partial K_{\bar{t}}^{S,P}}=\left(\frac{\chi\beta A}{1-\beta}\right)^{\frac{1}{\alpha}}(1-\alpha)^{\frac{1-\alpha}{\alpha}}>0.$$

The non-negativity constraint binds. If instead the non-negativity constraint on numeraire labor binds, then all labor is allocated to the defense sector: $l_m^{S,C}=\bar{l}^S$. In that case

$$m_{\bar{t},\bar{t}}^{S,C}=A(K_{\bar{t}}^{S,P})^{\alpha}(\bar{l}^S)^{1-\alpha},$$

and therefore

$$\frac{\partial\mathcal{D}_{\bar{t}}}{\partial K_{\bar{t}}^{S,P}}=\frac{\chi\beta}{1-\beta}A\alpha(K_{\bar{t}}^{S,P})^{\alpha-1}(\bar{l}^S)^{1-\alpha}>0.$$

The proposition evaluates the deterrent value of $(K_{t+1}^{S,P},M_{t+1}^{S,P})$ for the possible challenge date $t+1$. Setting $\bar{t}=t+1$ gives the stated expressions and completes the proof. □

A.3 Proof of Proposition 2

Proof. Throughout, the deterrence constraint is imposed one period ahead: at date t the government chooses allocations subject to

$$\mathcal{D}_{t+1}(K^S_{t+1}, M^S_{t+1}) \geq \eta^A_{t+1},$$

with multiplier γ^S_t.

Private-sector equilibrium conditions. Consider the defense-industry firm in country S. Taking as given prices $(p^S_{m,t}, p^S_{n,t}, w^S_t)$ and a per-unit subsidy $s^{S,P}_{K,t}$ on the capital stock K^S_t, it chooses $\{i^S_t, l^S_{m,t}\}_{t\geq 0}$ to maximize discounted profits subject to $m^S_t = (K^S_t)^\alpha (l^S_{m,t})^{1-\alpha}$ and $K^S_{t+1} = (1-\delta_K)K^S_t + i^S_t$. Let $\hat{\lambda}^S_{K,t}$ denote the firm's multiplier on the capital accumulation constraint. The firm's first-order conditions are

$$[l^S_{m,t}]: \qquad w^S_t = p^S_{m,t}(1-\alpha)(K^S_t)^\alpha (l^S_{m,t})^{-\alpha}, \tag{47}$$

$$[i^S_t]: \qquad p^S_{n,t} = \hat{\lambda}^S_{K,t}, \tag{48}$$

$$[K^S_{t+1}]: \qquad \hat{\lambda}^S_{K,t} = \frac{1}{1+r^S_t}\Big(s^{S,P}_{K,t+1} + p^S_{m,t+1}\alpha (K^S_{t+1})^{\alpha-1}(l^S_{m,t+1})^{1-\alpha} + (1-\delta_K)\hat{\lambda}^S_{K,t+1}\Big). \tag{49}$$

In equilibrium with linear utility and the linear numeraire technology, $p^S_{n,t} = w^S_t = 1$ for all t, and $\frac{1}{1+r^S_t} = \beta$. Using (48), this implies $\hat{\lambda}^S_{K,t} = 1$ for all t. Substituting these restrictions into (49) yields the firm Euler equation

$$1 = \beta\Big((1-\delta_K) + p^S_{m,t+1}\alpha (K^S_{t+1})^{\alpha-1}(l^S_{m,t+1})^{1-\alpha} + s^{S,P}_{K,t+1}\Big). \tag{50}$$

Decentralization We look for a decentralization of the planner allocation using a supporting price sequence for military goods satisfying

$$p^S_{m,t} = \lambda^S_{m,t} \qquad \forall t, \tag{51}$$

where $\lambda^S_{m,t}$ is the planner multiplier on (2). With (51), the firm Euler equation (50) becomes

$$1 = \beta\Big((1-\delta_K) + \lambda^S_{m,t+1}\alpha (K^S_{t+1})^{\alpha-1}(l^S_{m,t+1})^{1-\alpha} + s^{S,P}_{K,t+1}\Big). \tag{52}$$

Wedge relative to the planner and closing it. The planner Euler equation for K^S_{t+1} is (25). Comparing (52) with (25) shows that, under the supporting prices (51), the only difference between the private-sector Euler equation and the planner Euler equation is

the deterrence wedge $\gamma_t^S \frac{\partial \mathcal{D}_{t+1}}{\partial K_{t+1}^S}$. Setting the capital-stock subsidy to

$$s_{K,t+1}^{S,P} = \gamma_t^S \frac{\partial \mathcal{D}_{t+1}}{\partial K_{t+1}^S}$$

exactly closes this wedge, so the firm's optimal accumulation of K_{t+1}^S coincides with the planner's. Given the same initial conditions and with lump-sum taxes available to finance the subsidy, the remaining equilibrium conditions coincide by construction, and the resulting competitive equilibrium implements the planner allocation. This proves the proposition. □

A.4 Proof of Proposition 5

The two-technology deterrence constraint. Fix a possible challenge date $\bar{t}$. With two technologies, military output is $m_t^S = \sum_{g=1}^{2} (K_{g,t}^S)^{\alpha_K} (l_{g,t}^S)^{\alpha_{L,g}}$ and $M_{t+1}^S = (1-\delta_M) M_t^S + m_t^S$. Suppressing terms that do not depend on Sovereign's state variables, the deterrence constraint is

$$\mathcal{D}_{\bar{t}}(K_{1,\bar{t}}^{S,P}, K_{2,\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) = \frac{\chi \beta^{T-1}}{1-\beta} \sum_{t=\bar{t}}^{\bar{t}+T-1} M_{t,\bar{t}}^{S,C} + \text{constant}.$$

If wars are short, $T = 1$, only $M_{\bar{t}}^{S,P}$ enters this sum. Since technology-specific capital affects military production only after the possible challenge date, $\partial \mathcal{D}_{\bar{t}} / \partial K_{g,\bar{t}}^{S,P} = 0$ for each g. If wars are long, $T = 2$, then

$$\mathcal{D}_{\bar{t}} = \frac{\chi \beta}{1-\beta} \left(M_{\bar{t}}^{S,P} + M_{\bar{t}+1,\bar{t}}^{S,C} \right) + \text{constant}, \quad M_{\bar{t}+1,\bar{t}}^{S,C} = (1-\delta_M) M_{\bar{t}}^{S,P} + \sum_{g=1}^{2} (K_{g,\bar{t}}^{S,P})^{\alpha_K} (l_{g,\bar{t},\bar{t}}^{S,C})^{\alpha_{L,g}}.$$

Therefore technology-g capital contributes to deterrence through its effect on first-period conflict production. At an active interior allocation this follows from the labor FOC, and at an active constrained allocation it follows directly from the positive marginal product of $K_{g,\bar{t}}^{S,P}$. Hence $\partial \mathcal{D}_{\bar{t}} / \partial K_{g,\bar{t}}^{S,P} > 0$ for each active technology.

Firm and planner FOCs. Consider technology g. Under the supporting price $p_{m,t}^S = \lambda_{m,t}^S$, the firm's Euler equation for $K_{g,t+1}^S$ and the corresponding planner Euler equation

are

$$\text{firm:} \quad 1 = \beta\Big((1-\delta_K) + \lambda^S_{m,t+1}\alpha_K (K^S_{g,t+1})^{\alpha_K - 1}(l^S_{g,t+1})^{\alpha_{L,g}} + s^{S,P}_{K,g,t+1}\Big),$$

$$\text{planner:} \quad 1 = \beta\Big((1-\delta_K) + \lambda^S_{m,t+1}\alpha_K (K^S_{g,t+1})^{\alpha_K - 1}(l^S_{g,t+1})^{\alpha_{L,g}} + \gamma^S_t \frac{\partial \mathcal{D}_{t+1}}{\partial K^S_{g,t+1}}\Big).$$

The only wedge is the deterrence value of technology-g capital. Setting

$$s^{S,P}_{K,g,t+1} = \gamma^{S,P}_t \frac{\partial \mathcal{D}_{t+1}}{\partial K^{S,P}_{g,t+1}}$$

closes this wedge for each $g \in \{1,2\}$ and implements the planner allocation.

Short and long wars. The shape of the deterrence constraint gives the rest of the result. If wars are short, the derivative with respect to $K^{S,P}_{g,t+1}$ is zero, so $s^{S,P}_{K,g,t+1} = 0$. If wars are long and the deterrence constraint binds, then $\gamma^{S,P}_t > 0$ and the derivative is strictly positive for each active technology, so $s^{S,P}_{K,g,t+1} > 0$. This proves the proposition. □

A.5 Proof of Proposition 6

Setup. Consider the steady state of the two-technology economy when wars are long, and suppress country and peacetime superscripts. Technology g produces $m_g = A_g K_g^{\alpha_K} l_g^{\alpha_{L,g}}$, with $\alpha_K + \alpha_{L,g} < 1$. By Proposition 5, $s^P_{K,g} = \gamma^P \partial \mathcal{D}/\partial K_g$. Since the deterrence constraint binds, $\gamma^P > 0$, so the comparison of subsidies reduces to a comparison of deterrence derivatives.

Conflict versus peace. Let $B \equiv \chi\beta/(1-\beta)$ denote the marginal value of first-period conflict production in a long war, and let p^P_m be the peacetime price of military goods. The conflict and peacetime labor FOCs for technology g are

$$B A_g \alpha_{L,g} K_g^{\alpha_K} (l^C_g)^{\alpha_{L,g}-1} = 1, \qquad p^P_m A_g \alpha_{L,g} K_g^{\alpha_K} (l^P_g)^{\alpha_{L,g}-1} = 1.$$

Hence $l^C_g / l^P_g = (B/p^P_m)^{1/(1-\alpha_{L,g})}$. The assumption that military production rises during conflict implies $B/p^P_m > 1$.

The deterrence derivative. In a long war, technology-g capital affects deterrence by changing first-period conflict production. Differentiating the induced conflict production

function gives

$$\frac{\partial \mathcal{D}}{\partial K_g} = B \frac{\alpha_K}{1-\alpha_{L,g}} A_g K_g^{\alpha_K - 1} (l_g^C)^{\alpha_{L,g}}.$$

Using the peacetime capital Euler equation, $p_m^P \alpha_K A_g K_g^{\alpha_K - 1} (l_g^P)^{\alpha_{L,g}} = \beta^{-1} - (1-\delta_K) - s_{K,g}^P$, and substituting for l_g^C / l_g^P gives

$$\frac{\partial \mathcal{D}}{\partial K_g} = \Phi(\alpha_{L,g}) \left(\beta^{-1} - (1-\delta_K) - s_{K,g}^P \right), \qquad \Phi(a) \equiv \frac{1}{1-a} \left(\frac{B}{p_m^P} \right)^{\frac{1}{1-a}}.$$

Because $B / p_m^P > 1$, $\Phi(a)$ is strictly increasing in a.

Comparing subsidies. Combining $s_{K,g}^P = \gamma^P \partial \mathcal{D} / \partial K_g$ with the previous display gives

$$s_{K,g}^P = \gamma^P \Phi(\alpha_{L,g}) \left(\beta^{-1} - (1-\delta_K) - s_{K,g}^P \right), \quad \text{so} \quad s_{K,g}^P = \frac{\gamma^P \Phi(\alpha_{L,g})}{1 + \gamma^P \Phi(\alpha_{L,g})} \left(\beta^{-1} - (1-\delta_K) \right).$$

Since $\beta^{-1} - (1-\delta_K) > 0$, the right-hand side is strictly increasing in $\Phi(\alpha_{L,g})$, and Φ is strictly increasing in $\alpha_{L,g}$. Therefore $\alpha_{L,1} > \alpha_{L,2}$ implies $s_{K,1}^P > s_{K,2}^P$. Restoring time subscripts gives $s_{K,1,t+1}^{S,P} > s_{K,2,t+1}^{S,P}$. □

B Extensions

B.1 Positive Probability of War

In this appendix we study how Adversary can be deterred efficiently. We extend the baseline model to allow for shocks to adversary's willingness to start a war. We study how Sovereign can attain some desired set of war probabilities most efficiently. Effectively, we generalize our notion of a best deterrence equilibrium to a more general case.

B.1.1 Economic Block

States. First we introduce notation to keep track of the states of geopolitics. As in the baseline case war is game-ending in that it can only be started once. Our convention will be that state P_t means that peace is reached at date t. The state $C_{t,\bar{t}}$ means that war started at date $\bar{t}$ and the economy is in the corresponding conflict continuation at date $t \geq \bar{t}$. Let Q_t^P and $Q_{t,\bar{t}}^C$ denote the date-0 Arrow-Debreu prices of the numeraire in these states. Note that we use the language of Arrow-Debreu prices to refer to states of geopolitics rather than states of nature.

We let $\bar{P}_t^C$ be the conditional probability that war starts at date t, given that peace has survived until the date-t challenge. Let $\bar{P}_t^P \equiv 1 - \bar{P}_t^C$. The implied unconditional probabilities are

$$\Pi_{0,t}^P \equiv \Pi_{0,t-1}^P \bar{P}_t^P, \qquad \Pi_{0,t}^C \equiv \Pi_{0,t-1}^P \bar{P}_t^C.$$

where we use the convention that $\Pi_{0,-1}^P \equiv 1$. Thus $\Pi_{0,t}^P$ is the probability that the date-t peacetime allocation is reached, and $\Pi_{0,t}^C$ is the probability that war first starts at date t.

Household. The household in Sovereign chooses consumption in all states to maximize

$$\sum_{t=0}^{\infty} \beta^t \Pi_{0,t}^P \left(G_0^{S,P} + c_t^{S,P} \right) + \sum_{\bar{t}=0}^{\infty} \sum_{t=\bar{t}}^{\infty} \beta^t \Pi_{0,\bar{t}}^C \left(G_{t,\bar{t}}^{S,C} + c_{t,\bar{t}}^{S,C} \right)$$

subject to

$$\begin{aligned} &\sum_{t=0}^{\infty} Q_t^P c_t^{S,P} + \sum_{\bar{t}=0}^{\infty} \sum_{t=\bar{t}}^{\infty} Q_{t,\bar{t}}^C c_{t,\bar{t}}^{S,C} \\ &\leq \sum_{t=0}^{\infty} Q_t^P \left(w_t^{S,P} \bar{l}_t^S + \Pi_{F,t}^{S,P} - T_t^{S,P} \right) + \sum_{\bar{t}=0}^{\infty} \sum_{t=\bar{t}}^{\infty} Q_{t,\bar{t}}^C \left(w_{t,\bar{t}}^{S,C} \bar{l}_t^S + \Pi_{F,t,\bar{t}}^{S,C} - T_{t,\bar{t}}^{S,C} \right). \end{aligned} \tag{53}$$

Firms. The numeraire firm solves the same static problem as in the baseline in each state. The defense firm chooses state-contingent production, labor, investment, and capital to maximize

$$\begin{aligned} &\sum_{t=0}^{\infty} Q_t^P \left[p_{m,t}^{S,P} y_{m,t}^{S,P} + s_{K,t}^{S,P} K_t^{S,P} - w_t^{S,P} l_{m,t}^{S,P} - i_t^{S,P} \right] \\ &+ \sum_{\bar{t}=0}^{\infty} \sum_{t=\bar{t}}^{\infty} Q_{t,\bar{t}}^C \left[p_{m,t,\bar{t}}^{S,C} y_{m,t,\bar{t}}^{S,C} + s_{K,t,\bar{t}}^{S,C} K_{t,\bar{t}}^{S,C} - w_{t,\bar{t}}^{S,C} l_{m,t,\bar{t}}^{S,C} - i_{t,\bar{t}}^{S,C} \right] \end{aligned} \tag{54}$$

subject to

$$\begin{aligned} y_{m,t}^{S,P} &= A (K_t^{S,P})^{\alpha} (l_{m,t}^{S,P})^{1-\alpha}, \qquad K_{t+1}^{S,P} = (1 - \delta_K) K_t^{S,P} + i_t^{S,P}, \\ y_{m,t,\bar{t}}^{S,C} &= A (K_{t,\bar{t}}^{S,C})^{\alpha} (l_{m,t,\bar{t}}^{S,C})^{1-\alpha}, \qquad K_{t+1,\bar{t}}^{S,C} = (1 - \delta_K) K_{t,\bar{t}}^{S,C} + i_{t,\bar{t}}^{S,C}, \end{aligned}$$

and

$$K_{\bar{t},\bar{t}}^{S,C} = K_{\bar{t}}^{S,P}.$$

Competitive equilibrium. Given a probability sequence $\{\bar{P}_t^C\}_{t \geq 0}$, government policies, and initial stocks, a competitive equilibrium is a state-contingent allocation, prices, profits, taxes, and transfers such that households and firms solve the problems above, the

government budget constraint holds state by state, markets clear state by state, and conflict continuations inherit the peacetime stocks at the war-start date.

B.1.2 Challenge Decision and Probabilistic Deterrence

Adversary's conflict value. At a date t reached in peace, let $W_t^{A,C}(K_t^{S,P}, M_t^{S,P})$ denote Adversary's continuation value from starting a war, before adding the attack shock. It is identical to that given in the main body of the paper (21). When starting a war adversary receives additional utility (possibly negative) given by η_t which it draws from a distribution denoted by F_t. The draw is realized at the beginning of period before it makes an attack decision. We assume that F_t is continuous and strictly increasing over the relevant range, has a finite conditional mean, and that the target probabilities are interior, $\bar{P}_t^C \in (0,1)$.

Adversary's peace value. Let $\mathbf{K}_t^{S,P} \equiv \{K_\tau^{S,P}\}_{\tau=t}^{\infty}$ and $\mathbf{M}_t^{S,P} \equiv \{M_\tau^{S,P}\}_{\tau=t}^{\infty}$ denote Sovereign's peacetime stock sequences from date t onward. Let $X_t^{A,P} = c_t^{A,P} + G_0^{A,P}$ where $c_t^{A,P}$ is given by (18) denote Adversary's flow payoff in peace, excluding the attack shock. If Adversary does not attack at date t, its continuation value solves

$$W_t^{A,P}(\mathbf{K}_t^{S,P}, \mathbf{M}_t^{S,P}) = X_t^{A,P} + \beta \int \max\left\{ W_{t+1}^{A,P}(\mathbf{K}_{t+1}^{S}, \mathbf{M}_{t+1}^{S}), W_{t+1}^{A,C}(K_{t+1}^{S,P}, M_{t+1}^{S,P}) + \eta \right\} dF_{t+1}(\eta). \tag{55}$$

Challenge decision. At date t, Adversary attacks if

$$W_t^{A,C}(K_t^{S,P}, M_t^{S,P}) + \eta_t^A \geq W_t^{A,P}(\mathbf{K}_t^{S,P}, \mathbf{M}_t^{S,P}).$$

Thus the conditional probability of war is

$$P_t^C = \Pr\left(\eta_t^A \geq W_t^{A,P}(\mathbf{K}_t^{S,P}, \mathbf{M}_t^{S,P}) - W_t^{A,C}(K_t^{S,P}, M_t^{S,P}) \right). \tag{56}$$

Probabilistic deterrence equilibrium. Fix $\{\bar{P}_t^C, \bar{P}_t^P\}_{t\geq 0}$ with $\bar{P}_t^P + \bar{P}_t^C = 1$. A $\{\bar{P}_t^C\}_{t\geq 0}$-deterrence equilibrium is a competitive equilibrium in which Adversary's optimal challenge decision generates

$$P_t^C = \bar{P}_t^C, \qquad P_t^P = \bar{P}_t^P$$

at every date t reached in peace.

Restrictions imposed by equilibrium. The notion of a probabilistic deterrence equilibrium allows us to set up the planning problem analogously to the non-probabilistic deterrence of problem in the body of the paper. We first define deterrence as

$$\mathcal{D}_t^A(\mathbf{K}_t^{S,P}, \mathbf{M}_t^{S,P}) \equiv W_t^{A,P}(\mathbf{K}_t^{S,P}, \mathbf{M}_t^{S,P}) - W_t^{A,C}(K_t^{S,P}, M_t^{S,P}). \tag{57}$$

A probabilistic deterrence equilibrium is consistent with a particular sequence of deterrence values. Note that since $P_t^C = 1 - F_t(\mathcal{D}_t^A)$, a $\{\bar{P}_t^C\}_{t\geq 0}$-deterrence equilibrium requires

$$\mathcal{D}_t^A(\mathbf{K}_t^{S,P}, \mathbf{M}_t^{S,P}) = \bar{\eta}_t, \qquad \text{where} \qquad \bar{\eta}_t \equiv F_t^{-1}(1 - \bar{P}_t^C). \tag{58}$$

It also simplifies (55). Let $\Omega_t \equiv \mathbb{E}[\eta_t^A \mid \eta_t^A \geq \bar{\eta}_t]$. Then

$$W_t^{A,P}(\mathbf{K}_t^{S,P}, \mathbf{M}_t^{S,P}) = X_t^{A,P} + \beta \left[\bar{P}_{t+1}^P W_{t+1}^{A,P}(\mathbf{K}_{t+1}^S, \mathbf{M}_{t+1}^S) + \bar{P}_{t+1}^C \left(W_{t+1}^{A,C}(K_{t+1}^{S,P}, M_{t+1}^{S,P}) + \Omega_{t+1} \right) \right]. \tag{59}$$

B.1.3 Peacetime Planning Problem

We now turn to the full statement of the planning problem.

Wartime problem. If war starts at $\bar{t}$, Sovereign enters the conflict continuation with $K_{\bar{t},\bar{t}}^{S,C} = K_{\bar{t}}^{S,P}$ and $M_{\bar{t},\bar{t}}^{S,C} = M_{\bar{t}}^{S,P}$. Welfare during wartime is given by

$$\begin{aligned} W_{\bar{t}}^{S,C}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) = \max_{\{c_{t,\bar{t}}^{S,C}, l_{m,t,\bar{t}}^{S,C}, i_{t,\bar{t}}^{S,C}, m_{t,\bar{t}}^{S,C}, K_{t,\bar{t}}^{S,C}, M_{t,\bar{t}}^{S,C}\}_{t=\bar{t}}^{\infty}} & \sum_{t=\bar{t}}^{\infty} \beta^{t-\bar{t}} \left(G_{t,\bar{t}}^{S,C} + c_{t,\bar{t}}^{S,C} \right) \\ \text{s.t.} \quad & c_{t,\bar{t}}^{S,C} + i_{t,\bar{t}}^{S,C} + l_{m,t,\bar{t}}^{S,C} = \bar{l}_t^S, \\ & m_{t,\bar{t}}^{S,C} = A(K_{t,\bar{t}}^{S,C})^{\alpha}(l_{m,t,\bar{t}}^{S,C})^{1-\alpha}, \\ & K_{t+1,\bar{t}}^{S,C} = (1-\delta_K)K_{t,\bar{t}}^{S,C} + i_{t,\bar{t}}^{S,C}, \\ & M_{t+1,\bar{t}}^{S,C} = (1-\delta_M)M_{t,\bar{t}}^{S,C} + m_{t,\bar{t}}^{S,C}, \\ & \text{conflict technology (14), for all } t \geq \bar{t}. \end{aligned} \tag{60}$$

Peacetime planning problem. We can now state the full peacetime planning problem. For a fixed probability sequence $\{\bar{P}_t^C\}_{t\geq 0}$, the inner peacetime planning problem is

$$\begin{aligned}
\max_{\{c_t^{S,P}, l_{m,t}^{S,P}, i_t^{S,P}, m_t^{S,P}, K_t^{S,P}, M_t^{S,P}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \left[\Pi_{0,t}^P \left(G_0^{S,P} + c_t^{S,P}\right) + \Pi_{0,t}^C W_t^{S,C}\left(K_t^{S,P}, M_t^{S,P}\right)\right] \\
\text{s.t.} \quad & c_t^{S,P} + i_t^{S,P} + l_{m,t}^{S,P} = \bar{l}_t^S, \quad \forall t, \\
& m_t^{S,P} = A\left(K_t^{S,P}\right)^{\alpha}\left(l_{m,t}^{S,P}\right)^{1-\alpha}, \quad \forall t, \\
& K_{t+1}^{S,P} = (1-\delta_K)K_t^{S,P} + i_t^{S,P}, \quad \forall t, \\
& M_{t+1}^{S,P} = (1-\delta_M)M_t^{S,P} + m_t^{S,P}, \quad \forall t, \\
& \mathcal{D}_t^A\left(\mathbf{K}_t^{S,P}, \mathbf{M}_t^{S,P}\right) = \bar{\eta}_t, \quad \forall t.
\end{aligned} \tag{61}$$

The initial stocks $K_0^{S,P}$ and $M_0^{S,P}$ are given.

B.1.4 Planner FOCs

Let $\mu_t^{S,P}, \nu_t^{S,P}, \lambda_{K,t}^{S,P}, \lambda_{M,t}^{S,P}$ be the current-value multipliers on the resource constraint, military production, defense-industrial capital accumulation, and military-capital accumulation, respectively. Let $\xi_t^{S,P}$ be the multiplier on $\mathcal{D}_t^A(\mathbf{K}_t^{S,P}, \mathbf{M}_t^{S,P}) - \bar{\eta}_t = 0$. For an interior allocation,

$$\mu_t^{S,P} = \Pi_{0,t}^P, \quad \lambda_{K,t}^{S,P} = \mu_t^{S,P}, \quad \mu_t^{S,P} = \nu_t^{S,P} A(1-\alpha)(K_t^{S,P})^{\alpha}(l_{m,t}^{S,P})^{-\alpha}, \quad \lambda_{M,t}^{S,P} = \nu_t^{S,P}. \tag{62}$$

The present-value FOC for $K_{t+1}^{S,P}$ is

$$\beta^t \lambda_{K,t}^{S,P} = \beta^{t+1}\left[(1-\delta_K)\lambda_{K,t+1}^{S,P} + \nu_{t+1}^{S,P} A\alpha (K_{t+1}^{S,P})^{\alpha-1}(l_{m,t+1}^{S,P})^{1-\alpha} + \Pi_{0,t+1}^C \frac{\partial W_{t+1}^{S,C}}{\partial K_{t+1}^{S,P}}\right] \tag{63}$$

$$+ \sum_{s=0}^{t+1} \beta^s \xi_s^{S,P} \frac{\partial \mathcal{D}_s^A(\mathbf{K}_s^S, \mathbf{M}_s^S)}{\partial K_{t+1}^{S,P}}. \tag{64}$$

The FOC for $M_{t+1}^{S,P}$ is

$$\beta^t \lambda_{M,t}^{S,P} = \beta^{t+1}\left[(1-\delta_M)\lambda_{M,t+1}^{S,P} + \Pi_{0,t+1}^C \frac{\partial W_{t+1}^{S,C}}{\partial M_{t+1}^{S,P}}\right] + \sum_{s=0}^{t+1} \beta^s \xi_s^{S,P} \frac{\partial \mathcal{D}_s^A(\mathbf{K}_s^S, \mathbf{M}_s^S)}{\partial M_{t+1}^{S,P}}. \tag{65}$$

B.1.5 Decentralization

Household FOC. For an interior allocation, the household FOCs imply that Arrow–Debreu prices are proportional to expected marginal utilities. Normalizing the multiplier on the household's date-0 budget constraint to one,

$$Q_t^P = \beta^t \Pi_{0,t}^P, \qquad Q_{t,\bar{t}}^C = \beta^t \Pi_{0,\bar{t}}^C. \tag{66}$$

Planner FOC in prices. Use (62) and support the military-goods price in peace with

$$p_{m,t}^{S,P} = \frac{\nu_t^{S,P}}{\Pi_{0,t}^P}.$$

Then (64) becomes

$$Q_t^P = Q_{t+1}^P \left[(1-\delta_K) + p_{m,t+1}^{S,P} \alpha A (K_{t+1}^{S,P})^{\alpha-1} (l_{m,t+1}^{S,P})^{1-\alpha}\right] + Q_{t+1,t+1}^C \frac{\partial W_{t+1}^{S,C}}{\partial K_{t+1}^{S,P}} + \sum_{s=0}^{t+1} \beta^s \xi_s^{S,P} \frac{\partial \mathcal{D}_s^A(\mathbf{K}_s^S, \mathbf{M}_s^S)}{\partial K_{t+1}^{S,P}}. \tag{67}$$

Firm FOC. Let $p_{K,t+1,t+1}^{S,C}$ denote the value, in state $C_{t+1,t+1}$, of one installed unit of defense-industrial capital at the start of the conflict continuation. The defense firm's peacetime Euler equation is

$$Q_t^P = Q_{t+1}^P \left[(1-\delta_K) + p_{m,t+1}^{S,P} \alpha A (K_{t+1}^{S,P})^{\alpha-1} (l_{m,t+1}^{S,P})^{1-\alpha} + s_{K,t+1}^{S,P}\right] + Q_{t+1,t+1}^C p_{K,t+1,t+1}^{S,C}. \tag{68}$$

B.1.6 Optimal Subsidies

Wartime envelope. Let $V_{K,t,\bar{t}}^{S,C}$ denote the current-value marginal value of installed capital in the wartime problem. The envelope condition gives

$$\frac{\partial W_{\bar{t}}^{S,C}}{\partial K_{\bar{t}}^{S,P}} = V_{K,\bar{t},\bar{t}}^{S,C}.$$

The envelope recursion is

$$V_{K,t,\bar{t}}^{S,C} = \nu_{t,\bar{t}}^{S,C} \alpha A (K_{t,\bar{t}}^{S,C})^{\alpha-1} (l_{m,t,\bar{t}}^{S,C})^{1-\alpha} + \beta(1-\delta_K) V_{K,t+1,\bar{t}}^{S,C}.$$

Because there is no remaining deterrence externality once war has begun, set $s^{S,C}_{K,t,\bar{t}} = 0$ for all $t \geq \bar{t}$. The firm's conflict-state stock price then satisfies

$$p^{S,C}_{K,t,\bar{t}} = p^{S,C}_{m,t,\bar{t}} \alpha A (K^{S,C}_{t,\bar{t}})^{\alpha-1} (l^{S,C}_{m,t,\bar{t}})^{1-\alpha} + \frac{Q^C_{t+1,\bar{t}}}{Q^C_{t,\bar{t}}} (1-\delta_K) p^{S,C}_{K,t+1,\bar{t}}.$$

Using $p^{S,C}_{m,t,\bar{t}} = \nu^{S,C}_{t,\bar{t}}$ and $Q^C_{t+1,\bar{t}}/Q^C_{t,\bar{t}} = \beta$, the two recursions coincide. Hence

$$p^{S,C}_{K,\bar{t},\bar{t}} = \frac{\partial W^{S,C}_{\bar{t}}}{\partial K^{S,P}_{\bar{t}}}. \tag{69}$$

Capital subsidy. Compare (67) with (68). Using (69), the direct wartime-value terms cancel. The optimal peacetime capital subsidy is therefore

$$Q^P_{t+1} s^{S,P}_{K,t+1} = \sum_{s=0}^{t+1} \beta^s \xi_s \frac{\partial \mathcal{D}^A_s(\mathbf{K}^S_s, \mathbf{M}^S_s)}{\partial K^{S,P}_{t+1}}. \tag{70}$$

Using (66),

$$s^{S,P}_{K,t+1} = \frac{1}{\beta^{t+1} \Pi^P_{0,t+1}} \sum_{s=0}^{t+1} \beta^s \xi_s \frac{\partial \mathcal{D}^A_s(\mathbf{K}^S_s, \mathbf{M}^S_s)}{\partial K^{S,P}_{t+1}}. \tag{71}$$

Thus the subsidy corrects the probability-implementation externality. The direct value of capital in wartime is already priced by firms through the conflict-state stock price.

B.2 Classical Rational Deterrence with Limited Commitment

Geopolitical states. Fix a possible challenge date $\bar{t}$. The baseline model assumes that if Adversary challenges the status quo, Sovereign fights. This extension makes that response explicit. There are three possible geopolitical states at $\bar{t}$:

$$z_{\bar{t}} \in \{P, B, C\}.$$

The first is peace, denoted by P, in which Adversary does not challenge and the status quo allocation of the public good remains in place. The second is backing down, denoted by B, in which Adversary challenges and Sovereign does not contest the challenge. The letter B stands for backing down. The third is conflict, denoted by C, in which Adversary challenges and Sovereign fights.

The desired equilibrium outcome is still peace. To sustain it, two incentive constraints must hold. First, Adversary must prefer not to challenge, taking as given that a challenge

would be met by conflict. This is the deterrence constraint already studied in the baseline model. Second, if Adversary were to challenge, Sovereign must prefer fighting to backing down. This is the new commitment constraint.

Peace. Let $K_{\bar{t}}^{S,P}$ and $M_{\bar{t}}^{S,P}$ denote the peacetime defense-industrial and military stocks at the possible challenge date. If the state is P, Sovereign's continuation welfare is

$$W_{\bar{t}}^{S,P}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) = \frac{G_0^{S,P}}{1-\beta} + V_{\bar{t}}^{S,P}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}), \tag{72}$$

where $V_{\bar{t}}^{S,P}$ is the consumption value of the peacetime continuation.

Backing down. If the state is B, Sovereign keeps the peacetime economic allocation but loses a present value $\Delta_{\bar{t}}^{S} \geq 0$ of the public good:

$$W_{\bar{t}}^{S,B}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) = W_{\bar{t}}^{S,P}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) - \Delta_{\bar{t}}^{S}. \tag{73}$$

This captures the idea that backing down preserves peace but concedes part of the disputed public good.

Conflict. If the state is C, the off-equilibrium continuation is the same wartime continuation as in the baseline model. Sovereign starts the conflict with the peacetime stocks $(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P})$, chooses its wartime allocation optimally, and obtains

$$\begin{aligned} W_{\bar{t}}^{S,C}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) = \frac{G_0^{S,P}}{1-\beta} + V_{\bar{t}}^{S,C}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) \\ + \frac{\chi\beta^{T-1}}{1-\beta} \sum_{t=\bar{t}}^{\bar{t}+T-1} \left(M_{t,\bar{t}}^{S,C}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) - M_0^{A,P} \right), \end{aligned} \tag{74}$$

where $V_{\bar{t}}^{S,C}$ is the consumption value of the optimal wartime continuation. The term with the summation is the present value of the change in Sovereign's public good generated by the conflict technology.

Deterrence constraint. The deterrence constraint is unchanged. Since the conjectured response to a challenge is conflict, Adversary compares peace with the conflict continuation. Define

$$\mathcal{D}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) \equiv W_{\bar{t}}^{A,P} - W_{\bar{t}}^{A,C}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}). \tag{75}$$

Adversary is deterred when

$$\mathcal{D}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) \geq \eta_{\bar{t}}^{A}. \tag{76}$$

Commitment constraint. The new object is Sovereign's commitment constraint. Let $\bar{V}_{\bar{t}}^{S}$ denote any exogenous commitment value, such as resolve, reputation, domestic audience costs, or other political costs of backing down. Sovereign is willing to fight after a challenge when

$$W_{\bar{t}}^{S,C}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) + \bar{V}_{\bar{t}}^{S} \geq W_{\bar{t}}^{S,B}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}). \tag{77}$$

Equivalently, define

$$\mathcal{C}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) \equiv W_{\bar{t}}^{S,C}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) - W_{\bar{t}}^{S,P}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) + \Delta_{\bar{t}}^{S} + \bar{V}_{\bar{t}}^{S}. \tag{78}$$

The commitment constraint is

$$\mathcal{C}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) \geq 0. \tag{79}$$

The two constraints play different roles. The deterrence constraint is Adversary's no-challenge condition. The commitment constraint is Sovereign's no-back-down condition. A deterrence equilibrium with limited commitment requires both.

Planning problem. Sovereign chooses a peacetime allocation intended to implement the P outcome. The allocation must be feasible, must deter Adversary from challenging, and must make Sovereign's threatened response credible. The primal planning problem is therefore

$$\begin{aligned}
\max_{\{c_t^{S,P}, y_{n,t}^{S,P}, y_{m,t}^{S,P}, l_{n,t}^{S,P}, l_{m,t}^{S,P}, i_t^{S,P}, m_t^{S,P}, K_t^{S,P}, M_t^{S,P}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \left(G_0^{S,P} + c_t^{S,P} \right) \\
\text{s.t.} \quad & \text{technology and market-clearing constraints (2)–(8)}, \\
& \mathcal{D}_t(K_t^{S,P}, M_t^{S,P}) \geq \eta_t^{A}, \quad \forall t, \\
& \mathcal{C}_t(K_t^{S,P}, M_t^{S,P}) \geq 0, \quad \forall t, \\
& \text{non-negativity constraints, for all } t.
\end{aligned} \tag{80}$$

As in the baseline problem, conflict is off the equilibrium path. The planner nevertheless has to choose peacetime stocks that make the off-equilibrium threat credible. The reason is that Adversary's no-challenge condition is evaluated under the conjecture that a challenge would be met by conflict.

First-order conditions. Let $\gamma_t^{S,P} \geq 0$ denote the multiplier on the date $t+1$ deterrence constraint, and let $\kappa_t^{S,P} \geq 0$ denote the multiplier on the date $t+1$ commitment constraint. The one-period-ahead convention matches the baseline derivation: choices at date t determine the stocks available at the possible challenge date $t+1$. The optimality condition for military capital is

$$\lambda_{m,t}^{S,P} = \beta \left((1-\delta_M)\lambda_{m,t+1}^{S,P} + \gamma_t^{S,P} \frac{\partial \mathcal{D}_{t+1}}{\partial M_{t+1}^{S,P}} + \kappa_t^{S,P} \frac{\partial \mathcal{C}_{t+1}}{\partial M_{t+1}^{S,P}} \right). \tag{81}$$

The optimality condition for defense-industrial capital is

$$1 = \beta \left((1-\delta_K) + \lambda_{m,t+1}^{S,P} \alpha A \left(K_{t+1}^{S,P}\right)^{\alpha-1} \left(l_{m,t+1}^{S,P}\right)^{1-\alpha} + \gamma_t^{S,P} \frac{\partial \mathcal{D}_{t+1}}{\partial K_{t+1}^{S,P}} + \kappa_t^{S,P} \frac{\partial \mathcal{C}_{t+1}}{\partial K_{t+1}^{S,P}} \right). \tag{82}$$

Relative to the baseline model, each stock now has two strategic values. It can deter Adversary by lowering Adversary's payoff from conflict, and it can strengthen commitment by raising Sovereign's payoff from fighting relative to backing down.

Optimal subsidy. The private Euler equation for defense-industrial capital is unchanged because firms only price the peacetime use of capital:

$$1 = \beta \left((1-\delta_K) + p_{m,t+1}^{S,P} \alpha A \left(K_{t+1}^{S,P}\right)^{\alpha-1} \left(l_{m,t+1}^{S,P}\right)^{1-\alpha} + s_{K,t+1}^{S,P} \right).$$

Using the same supporting price as in the baseline, $p_{m,t+1}^{S,P} = \lambda_{m,t+1}^{S,P}$, comparison with the planner's Euler equation gives the implementing subsidy

$$s_{K,t+1}^{S,P} = \gamma_t^{S,P} \frac{\partial \mathcal{D}_{t+1}}{\partial K_{t+1}^{S,P}} + \kappa_t^{S,P} \frac{\partial \mathcal{C}_{t+1}}{\partial K_{t+1}^{S,P}}. \tag{83}$$

The first term is the baseline deterrence externality: firms do not internalize that capital raises the cost of war for Adversary. The second term is the commitment externality: firms also do not internalize that capital can make Sovereign more willing to fight after a challenge.

Capital prices and multipliers. The commitment derivative can be written in terms of state-contingent stock prices of capital. These are prices for an installed stock of capital at the beginning of a continuation, not flow rental prices. To see the connection, fix a continuation state $z \in \{P, C\}$ and let $\bar{K}$ denote the initial stock of defense-industrial capital

available at $\bar{t}$. Write the continuation problem with an explicit initial-stock constraint,

$$K_{\bar{t}}^{S,z} \leq \bar{K},$$

and let $\mu_{K,\bar{t}}^{S,z}$ be the multiplier on this constraint. With the convention that the Lagrangian contains

$$\mu_{K,\bar{t}}^{S,z}\left(\bar{K} - K_{\bar{t}}^{S,z}\right),$$

the envelope theorem gives

$$\frac{\partial W_{\bar{t}}^{S,z}}{\partial \bar{K}} = \mu_{K,\bar{t}}^{S,z}. \tag{84}$$

This is a statement about the primal problem. We now show that the same multiplier is the competitive stock price. Let $p_{K,\tau}^{S,z}$ denote the date-τ price, in units of the numeraire, of one installed unit of defense-industrial capital in continuation z. A firm that owns an installed unit of capital receives the marginal product revenue from military production, any per-unit stock payment used to decentralize the continuation, and the depreciated resale value of the capital stock. Therefore its no-arbitrage condition is

$$p_{K,\tau}^{S,z} = p_{m,\tau}^{S,z}\alpha A\left(K_{\tau}^{S,z}\right)^{\alpha-1}\left(l_{m,\tau}^{S,z}\right)^{1-\alpha} + s_{K,\tau}^{S,z} + \beta(1-\delta_K)p_{K,\tau+1}^{S,z}. \tag{85}$$

The corresponding planner multiplier satisfies the same recursion. Differentiating the continuation Lagrangian with respect to $K_{\tau}^{S,z}$ yields

$$\mu_{K,\tau}^{S,z} = \lambda_{m,\tau}^{S,z}\alpha A\left(K_{\tau}^{S,z}\right)^{\alpha-1}\left(l_{m,\tau}^{S,z}\right)^{1-\alpha} + s_{K,\tau}^{S,z} + \beta(1-\delta_K)\mu_{K,\tau+1}^{S,z}, \tag{86}$$

where $s_{K,\tau}^{S,z}$ denotes the non-market shadow-value term in the planner recursion, written in the units of the stock subsidy that decentralizes the continuation. In the peace continuation these are the future deterrence and commitment wedges, which are priced by future subsidies. In the conflict continuation the term is zero unless additional off-path wedges are imposed. Since the supporting price of military goods satisfies $p_{m,\tau}^{S,z} = \lambda_{m,\tau}^{S,z}$, equations (85) and the multiplier recursion coincide. Under the same transversality condition,

$$p_{K,\bar{t}}^{S,z} = \mu_{K,\bar{t}}^{S,z} = \frac{\partial W_{\bar{t}}^{S,z}}{\partial K_{\bar{t}}^{S,P}}. \tag{87}$$

Thus the price that appears here is the stock price in the continuation problem. In particular, $p_{K,\bar{t}}^{S,P}$ is not merely the current marginal product of capital in peace. It capitalizes future peacetime returns and, in the implemented deterrence equilibrium, future subsidy

payments that price future deterrence and commitment externalities.

Characterizing the commitment derivative. Since $\Delta_{\bar{t}}^{S}$ and $\bar{V}_{\bar{t}}^{S}$ are exogenous, the marginal commitment value of defense-industrial capital is

$$\frac{\partial \mathcal{C}_{\bar{t}}}{\partial K_{\bar{t}}^{S,P}} = \frac{\partial W_{\bar{t}}^{S,C}}{\partial K_{\bar{t}}^{S,P}} - \frac{\partial W_{\bar{t}}^{S,P}}{\partial K_{\bar{t}}^{S,P}}. \tag{88}$$

Using the price-multiplier equivalence above,

$$\frac{\partial \mathcal{C}_{\bar{t}}}{\partial K_{\bar{t}}^{S,P}} = p_{K,\bar{t}}^{S,C} - p_{K,\bar{t}}^{S,P}. \tag{89}$$

The envelope theorem is useful because no derivative of the optimal wartime policy rules appears. Capital strengthens commitment precisely when the installed stock is more valuable in the conflict continuation than in the peace or backing-down continuation. Substituting this into the subsidy formula gives

$$s_{K,t+1}^{S,P} = \gamma_t^{S,P} \frac{\partial \mathcal{D}_{t+1}}{\partial K_{t+1}^{S,P}} + \kappa_t^{S,P} \left(p_{K,t+1}^{S,C} - p_{K,t+1}^{S,P} \right). \tag{90}$$

This expression separates the two reasons for subsidizing defense-industrial capital. The deterrence term reflects the effect of capital on Adversary's incentive to challenge. The commitment term reflects the price gap between the conflict continuation and the peace/backing-down continuation.

In the simple long-war calculation used to characterize the deterrence derivative, the relevant wartime term is

$$\max_{l_m \in [0, \bar{l}^S]} \left\{ -l_m + \frac{\chi\beta}{1-\beta} A \left(K_{\bar{t}}^{S,P} \right)^{\alpha} l_m^{1-\alpha} \right\}.$$

When the labor constraint does not bind, the envelope theorem gives the marginal conflict value generated by this term as

$$\frac{\chi\beta}{1-\beta} A\alpha \left(K_{\bar{t}}^{S,P} \right)^{\alpha-1} \left(l_{m,\bar{t},\bar{t}}^{S,C} \right)^{1-\alpha}. \tag{91}$$

There is no term involving $\partial l_{m,\bar{t},\bar{t}}^{S,C} / \partial K_{\bar{t}}^{S,P}$ because labor is chosen optimally in the wartime continuation. The full commitment derivative subtracts the peacetime shadow value of the same capital stock. Thus the sign of $\partial \mathcal{C}_{\bar{t}} / \partial K_{\bar{t}}^{S,P}$ depends on whether defense-industrial

capital is more valuable for fighting the challenge than for the peacetime/backing-down continuation.

B.3 Endogenous War Duration

This appendix extends the baseline model by allowing the duration of a conflict to depend on the military balance at the possible challenge date. The point is to show that the sufficient statistic for industrial policy is unchanged, but that the optimal subsidy is lower when Adversary assigns positive probability to a short war, under a natural condition.

Duration. Fix a possible challenge date $\bar{t}$. In the baseline model, the duration of the off-equilibrium conflict continuation is exogenous. Here, the continuation lasts two periods with probability

$$\pi_{\bar{t}} = \Pr(T = 2 \mid M_{\bar{t}}^{S,P}, M_0^{A,P}) = \phi + \eta_s \left(M_{\bar{t}}^{S,P} - M_0^{A,P} \right), \tag{92}$$

where we focus on the interior region in which $\pi_{\bar{t}} \in (0, 1)$. We assume the duration shock is realized at the possible challenge date, so the conditional conflict continuation is solved as in the baseline model once the duration is known. One microfoundation for (92) is that the continuation lasts two periods if

$$\varepsilon_{\bar{t}} \leq M_{\bar{t}}^{S,P} - M_0^{A,P} + \frac{\phi}{\eta_s}, \qquad \varepsilon_{\bar{t}} \sim \text{Unif}\left[0, \frac{1}{\eta_s}\right].$$

The parameter ϕ is the probability of a long war when the two sides have equal initial military capital, and η_s measures how sensitive the probability of a long war is to Sovereign's initial military advantage.

Deterrence. Let $\mathcal{D}_{\bar{t}}^1$ denote deterrence if the conflict continuation lasts one period and $\mathcal{D}_{\bar{t}}^2$ denote deterrence if it lasts two periods. Suppressing terms that do not depend on Sovereign's state variables,

$$\mathcal{D}_{\bar{t}}^1 = \frac{\chi}{1-\beta} \left(M_{\bar{t}}^{S,P} - M_0^{A,P} \right), \qquad \mathcal{D}_{\bar{t}}^2 = \frac{\chi\beta}{1-\beta} \left[M_{\bar{t}}^{S,P} + M_{\bar{t}+1,\bar{t}}^{S,C} - 2M_0^{A,P} \right].$$

The deterrence constraint is therefore

$$\mathcal{D}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) = (1 - \pi_{\bar{t}})\, \mathcal{D}_{\bar{t}}^1 + \pi_{\bar{t}} \mathcal{D}_{\bar{t}}^2. \tag{93}$$

This nests the baseline model as the limiting cases $\pi_{\bar{t}} = 0$ and $\pi_{\bar{t}} = 1$.

Marginal deterrent values. We focus on the case in which the non-negativity constraint on numeraire labor does not bind in the long-war continuation. Conditional on the continuation lasting two periods, the first-period military production problem is the same as in the baseline long-war case, so

$$m^{S,C}_{\bar{t},\bar{t}}(K^{S,P}_{\bar{t}}) = A\left(\frac{\chi\beta}{1-\beta}A(1-\alpha)\right)^{\frac{1-\alpha}{\alpha}} K^{S,P}_{\bar{t}}.$$

Using $M^{S,C}_{\bar{t}+1,\bar{t}} = (1-\delta_M)M^{S,P}_{\bar{t}} + m^{S,C}_{\bar{t},\bar{t}}$, the marginal deterrent values are

$$\frac{\partial \mathcal{D}_{\bar{t}}}{\partial K^{S,P}_{\bar{t}}} = \pi_{\bar{t}}\left(\frac{\chi\beta A}{1-\beta}\right)^{\frac{1}{\alpha}}(1-\alpha)^{\frac{1-\alpha}{\alpha}}, \tag{94}$$

$$\frac{\partial \mathcal{D}_{\bar{t}}}{\partial M^{S,P}_{\bar{t}}} = (1-\pi_{\bar{t}})\frac{\chi}{1-\beta} + \pi_{\bar{t}}\frac{\chi\beta}{1-\beta}(1+(1-\delta_M)) + \eta_s\left(\mathcal{D}^2_{\bar{t}} - \mathcal{D}^1_{\bar{t}}\right). \tag{95}$$

The industrial-capital derivative is the long-war derivative scaled by the probability that the second period is reached. Military capital has an additional duration effect because it changes this probability.

Optimal subsidy. The sufficient statistic for industrial policy is unchanged. Proposition 2 applies to any differentiable deterrence constraint, so the optimal capital subsidy is still

$$s^{S,P}_{K,t+1} = \gamma^{S,P}_t \frac{\partial \mathcal{D}_{t+1}}{\partial K^{S,P}_{t+1}}.$$

Endogenous duration changes the derivative that enters this expression, not the expression itself.

Subsidies and the private steady state. To obtain a sharper comparative static, we follow the steady-state exercise in the main text and use the private first-order conditions to write the subsidy directly as a function of the defense-sector labor-to-capital ratio. Let

$$R^{ss} \equiv \frac{l^{ss}_m}{K^{ss}}.$$

In a steady-state competitive equilibrium, the firm's labor and capital first-order conditions imply

$$1 = p_m^P A(1-\alpha)(R^{ss})^{-\alpha}, \qquad \beta^{-1} - (1-\delta_K) - s_K^{ss} = p_m^P A\alpha(R^{ss})^{1-\alpha}.$$

Combining them gives

$$s_K^{ss} = \beta^{-1} - (1-\delta_K) - \frac{\alpha}{1-\alpha}R^{ss}. \tag{96}$$

In the $\beta \to 1$ steady-state case used in the main text, this becomes the particularly simple expression

$$s_K^{ss} = \delta_K - \frac{\alpha}{1-\alpha}R^{ss}.$$

In either case, the optimal subsidy is decreasing in R^{ss}.

The efficient capital mix. It remains to characterize how endogenous duration affects R^{ss}. Let all derivatives of $\mathcal{D}$ be evaluated at the steady state, and focus on an interior steady state with $\partial\mathcal{D}/\partial M^{S,P} > 0$. Since $m^{ss} = A(R^{ss})^{1-\alpha}K^{ss}$,

$$M^{ss} = \frac{A}{\delta_M}(R^{ss})^{1-\alpha}K^{ss}.$$

The planner chooses K^{ss} and R^{ss} to minimize the resource cost $(\delta_K + R^{ss})K^{ss}$ subject to deterrence. The first-order conditions are

$$\delta_K + R^{ss} = \gamma\left(\frac{\partial\mathcal{D}}{\partial K^{S,P}} + \frac{\partial\mathcal{D}}{\partial M^{S,P}}\frac{A}{\delta_M}(R^{ss})^{1-\alpha}\right), \qquad 1 = \gamma\frac{\partial\mathcal{D}}{\partial M^{S,P}}\frac{A}{\delta_M}(1-\alpha)(R^{ss})^{-\alpha}.$$

Eliminating γ gives

$$\frac{\delta_M}{A}\frac{\partial\mathcal{D}/\partial K^{S,P}}{\partial\mathcal{D}/\partial M^{S,P}}(R^{ss})^{\alpha} + \alpha R^{ss} = (1-\alpha)\delta_K. \tag{97}$$

The left-hand side is increasing in both R^{ss} and the ratio

$$\frac{\partial\mathcal{D}/\partial K^{S,P}}{\partial\mathcal{D}/\partial M^{S,P}}.$$

Therefore R^{ss} is decreasing in this ratio.

Comparative static. Let the superscript L denote the long-war-for-sure case and E the endogenous-duration case. From (94),

$$\left(\frac{\partial \mathcal{D}}{\partial K^{S,P}}\right)^{E} = \pi_{\bar{t}} \left(\frac{\partial \mathcal{D}}{\partial K^{S,P}}\right)^{L}.$$

For military capital, (95) implies

$$\left(\frac{\partial \mathcal{D}}{\partial M^{S,P}}\right)^{E} = \pi_{\bar{t}} \left(\frac{\partial \mathcal{D}}{\partial M^{S,P}}\right)^{L} + (1-\pi_{\bar{t}})\frac{\chi}{1-\beta} + \eta_s \left(\mathcal{D}^2_{\bar{t}} - \mathcal{D}^1_{\bar{t}}\right).$$

Thus

$$\frac{\left(\partial \mathcal{D}/\partial K^{S,P}\right)^{E}}{\left(\partial \mathcal{D}/\partial M^{S,P}\right)^{E}} < \frac{\left(\partial \mathcal{D}/\partial K^{S,P}\right)^{L}}{\left(\partial \mathcal{D}/\partial M^{S,P}\right)^{L}}$$

whenever

$$(1-\pi_{\bar{t}})\frac{\chi}{1-\beta} + \eta_s \left(\mathcal{D}^2_{\bar{t}} - \mathcal{D}^1_{\bar{t}}\right) > 0.$$

This condition is automatically satisfied if long wars are at least as good for deterrence as short wars, $\mathcal{D}^2_{\bar{t}} \geq \mathcal{D}^1_{\bar{t}}$. In that case, endogenous duration lowers the ratio of marginal deterrent values. Equation (97) then implies that R^{ss} is higher than in the long-war-for-sure case. Finally, (96) implies that the optimal capital subsidy is lower.

B.4 Strategic Interactions

This appendix extends the baseline model by allowing for a limited strategic interaction in which Adversary responds to Sovereign's choices of $K^{S,P}_{t+1}$ and $M^{S,P}_{t+1}$. We focus on a simple interaction in which Adversary chooses $M^{A,P}_{t+1}$ according to a linear rule. This allows us to study Sovereign's optimal policy when Adversary responds strategically, while staying close to the baseline model in which there is no war on the equilibrium path.[19]

B.4.1 Setup

Response rule. At date t, Sovereign chooses the stocks that will be available at the next possible challenge date, $K^{S,P}_{t+1}$ and $M^{S,P}_{t+1}$. After observing these stocks, Adversary sets its

[19] Fully endogenizing Adversary's decision would require changing the model in several ways because, in a deterrence equilibrium, Adversary never uses its military stockpile in war and hence has no direct reason to accumulate it. One would therefore have to introduce other motives for holding a stockpile of military equipment, such as a positive probability of war or the possibility that Sovereign or third parties attack Adversary.

next-period military stock according to

$$M_{t+1}^{A,P} = M_0^{A,P} + \varphi_K K_{t+1}^{S,P} + \varphi_M M_{t+1}^{S,P}, \tag{98}$$

where $\varphi_K, \varphi_M \geq 0$. The coefficient φ_K measures how much Adversary expands its military stock in response to Sovereign's defense-industrial capital, while φ_M measures the response to Sovereign's military capital. We assume the parameters are such that it is feasible for Adversary to follow this rule.

Wars last either one or two periods, $T \in \{1, 2\}$. During a conflict that starts at $\bar{t}$, Adversary's military stock is fixed at its stock at the possible challenge date:

$$M_{\tau,\bar{t}}^{A,C} = M_{\bar{t}}^{A,P} \qquad \text{for all } \tau \in \{\bar{t}, \bar{t}+1\} \text{ reached during the conflict.}$$

Response-adjusted deterrence. Let $\mathcal{D}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}; M_{\bar{t}}^{A,P})$ denote the baseline deterrence function when Adversary's military stock at the possible challenge date is treated as a parameter. With the response rule (98), the deterrence function is

$$\widetilde{\mathcal{D}}_{\bar{t}}(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}) \equiv \mathcal{D}_{\bar{t}}\left(K_{\bar{t}}^{S,P}, M_{\bar{t}}^{S,P}; M_0^{A,P} + \varphi_K K_{\bar{t}}^{S,P} + \varphi_M M_{\bar{t}}^{S,P}\right). \tag{99}$$

The deterrence constraint is therefore

$$\widetilde{\mathcal{D}}_t(K_t^{S,P}, M_t^{S,P}) \geq \eta_t^A. \tag{100}$$

The distinction from the baseline is that Sovereign now internalizes how its peacetime choices affect Adversary's military stock, and hence the tightness of the deterrence constraint.

Marginal deterrent values. The response rule changes the marginal deterrent values through a chain-rule term. Suppressing time subscripts and evaluating derivatives at the relevant possible challenge date,

$$\frac{\partial \widetilde{\mathcal{D}}}{\partial K^{S,P}} = \frac{\partial \mathcal{D}}{\partial K^{S,P}} + \varphi_K \frac{\partial \mathcal{D}}{\partial M^{A,P}}, \tag{101}$$

$$\frac{\partial \widetilde{\mathcal{D}}}{\partial M^{S,P}} = \frac{\partial \mathcal{D}}{\partial M^{S,P}} + \varphi_M \frac{\partial \mathcal{D}}{\partial M^{A,P}}. \tag{102}$$

Since a higher adversary military stock makes conflict more attractive to Adversary, $\partial \mathcal{D}/\partial M^{A,P} < 0$. Hence an increase in φ_K lowers the deterrent value of Sovereign's defense-industrial

capital, and an increase in φ_M lowers the deterrent value of Sovereign's military capital.

B.4.2 Optimal Policy

Planning problem. Sovereign chooses the peacetime allocation taking the response rule as given. Since the response rule is already embedded in $\widetilde{\mathcal{D}}_t$, Adversary's military stock does not need to be tracked as an additional constraint in the reduced-form primal problem. The planning problem is

$$\begin{aligned} \max_{\{c_t^{S,P}, y_{n,t}^{S,P}, y_{m,t}^{S,P}, l_{n,t}^{S,P}, l_{m,t}^{S,P}, i_t^{S,P}, m_t^{S,P}, K_t^{S,P}, M_t^{S,P}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \left(G_0^{S,P} + c_t^{S,P} \right) \\ \text{s.t.} \quad & \text{technology and market-clearing constraints (2)–(8)}, \\ & \widetilde{\mathcal{D}}_t(K_t^{S,P}, M_t^{S,P}) \geq \eta_t^A, \quad \forall t, \\ & \text{non-negativity constraints, for all } t. \end{aligned} \tag{103}$$

First-order conditions. Let $\gamma_t^{S,P} \geq 0$ denote the multiplier on the date $t+1$ deterrence constraint. The optimality conditions for military capital and defense-industrial capital are

$$\lambda_{m,t}^{S,P} = \beta \left((1-\delta_M) \lambda_{m,t+1}^{S,P} + \gamma_t^{S,P} \frac{\partial \widetilde{\mathcal{D}}_{t+1}}{\partial M_{t+1}^{S,P}} \right), \tag{104}$$

$$1 = \beta \left((1-\delta_K) + \lambda_{m,t+1}^{S,P} \alpha A \left(K_{t+1}^{S,P} \right)^{\alpha-1} \left(l_{m,t+1}^{S,P} \right)^{1-\alpha} + \gamma_t^{S,P} \frac{\partial \widetilde{\mathcal{D}}_{t+1}}{\partial K_{t+1}^{S,P}} \right). \tag{105}$$

The only difference from the baseline FOCs is that the planner uses the total derivatives in (101) and (102), rather than the partial derivatives holding Adversary's military stock fixed.

Optimal subsidy. The private Euler equation for defense-industrial capital is unchanged. Comparing it with (105), and using the same supporting price for military goods as in the baseline, gives the implementing subsidy

$$s_{K,t+1}^{S,P} = \gamma_t^{S,P} \frac{\partial \widetilde{\mathcal{D}}_{t+1}}{\partial K_{t+1}^{S,P}} = \gamma_t^{S,P} \left(\frac{\partial \mathcal{D}_{t+1}}{\partial K_{t+1}^{S,P}} + \varphi_K \frac{\partial \mathcal{D}_{t+1}}{\partial M_{t+1}^{A,P}} \right). \tag{106}$$

Thus the sufficient statistic is unchanged in form: the subsidy equals the shadow value of deterrence times the marginal contribution of capital to deterrence. The response rule

changes the relevant marginal contribution. Since $\partial \mathcal{D}/\partial M^{A,P} < 0$, a stronger adversary response to $K^{S,P}$ directly lowers the marginal deterrent value of defense-industrial capital.

B.4.3 Comparative Static: Result

Statement. We now ask how the optimal steady-state capital subsidy changes with the reaction coefficients φ_K and φ_M. Focus on long wars, $T = 2$, and on an interior region in which the response-adjusted marginal deterrent values of both $K^{S,P}$ and $M^{S,P}$ are positive. In this region, increasing the response to Sovereign's defense-industrial capital lowers the optimal capital subsidy, while increasing the response to Sovereign's military capital raises it:

$$\frac{\partial s^{S,P}_{K,ss}}{\partial \varphi_K} < 0, \qquad \frac{\partial s^{S,P}_{K,ss}}{\partial \varphi_M} > 0. \tag{107}$$

The intuition is simple. A higher φ_K means Adversary arms more in response to Sovereign's defense-industrial capital. This makes defense-industrial capital less effective at providing deterrence, so the planner tilts away from it and the capital subsidy falls. A higher φ_M means Adversary arms more in response to Sovereign's military capital. This makes military capital less effective at providing deterrence relative to the industrial base, so the planner tilts toward defense-industrial capital and the capital subsidy rises.

B.4.4 Comparative Static: Derivation

This section derives the comparative static in (107). Throughout we focus on the case where wars are long.

Deterrent values. In a long war, and suppressing terms that do not depend on Sovereign's state variables, the baseline deterrence function is

$$\mathcal{D}_{\bar{t}}(K^{S,P}_{\bar{t}}, M^{S,P}_{\bar{t}}; M^{A,P}_{\bar{t}}) = \frac{\chi\beta}{1-\beta}\left[M^{S,P}_{\bar{t}} + M^{S,C}_{\bar{t}+1,\bar{t}} - 2M^{A,P}_{\bar{t}}\right].$$

Thus

$$\frac{\partial \mathcal{D}}{\partial M^{A,P}} = -\frac{2\chi\beta}{1-\beta}.$$

Equations (101) and (102) become

$$\frac{\partial \widetilde{\mathcal{D}}}{\partial K^{S,P}} = \frac{\partial \mathcal{D}}{\partial K^{S,P}} - \frac{2\chi\beta}{1-\beta}\varphi_K, \tag{108}$$

$$\frac{\partial \widetilde{\mathcal{D}}}{\partial M^{S,P}} = \frac{\partial \mathcal{D}}{\partial M^{S,P}} - \frac{2\chi\beta}{1-\beta}\varphi_M. \tag{109}$$

Subsidies and the private steady state. We next derive an expression relating subsidies to the labor-to-capital ratio. The private-sector optimality conditions in the steady state imply

$$1 = p_m^P A(1-\alpha)(R_{ss})^{-\alpha}, \qquad \beta^{-1} - (1-\delta_K) - s_{K,ss}^{S,P} = p_m^P A\alpha(R_{ss})^{1-\alpha},$$

where $R_{ss} \equiv l_{m,ss}/K_{ss}$. Combining these conditions gives

$$s_{K,ss}^{S,P} = \beta^{-1} - (1-\delta_K) - \frac{\alpha}{1-\alpha}R_{ss}. \tag{110}$$

Therefore the optimal capital subsidy is decreasing in R_{ss}.

The efficient capital mix. Next we pin down the R_{ss} desired by the planner. Let all derivatives of $\widetilde{\mathcal{D}}$ be evaluated at the steady state, and assume $\partial\widetilde{\mathcal{D}}/\partial M^{S,P} > 0$. Since $m_{ss} = A(R_{ss})^{1-\alpha}K_{ss}$,

$$M_{ss} = \frac{A}{\delta_M}(R_{ss})^{1-\alpha}K_{ss}.$$

The planner chooses K_{ss} and R_{ss} to minimize the resource cost $(\delta_K + R_{ss})K_{ss}$ subject to the response-adjusted deterrence constraint. The FOCs are

$$\delta_K + R_{ss} = \gamma\left(\frac{\partial\widetilde{\mathcal{D}}}{\partial K^{S,P}} + \frac{\partial\widetilde{\mathcal{D}}}{\partial M^{S,P}}\frac{A}{\delta_M}(R_{ss})^{1-\alpha}\right), \qquad 1 = \gamma\frac{\partial\widetilde{\mathcal{D}}}{\partial M^{S,P}}\frac{A}{\delta_M}(1-\alpha)(R_{ss})^{-\alpha}.$$

Eliminating γ gives

$$\frac{\delta_M}{A}\frac{\partial\widetilde{\mathcal{D}}/\partial K^{S,P}}{\partial\widetilde{\mathcal{D}}/\partial M^{S,P}}(R_{ss})^{\alpha} + \alpha R_{ss} = (1-\alpha)\delta_K. \tag{111}$$

Thus R_{ss} is decreasing in the ratio

$$\frac{\partial\widetilde{\mathcal{D}}/\partial K^{S,P}}{\partial\widetilde{\mathcal{D}}/\partial M^{S,P}}.$$

Combining this with (110), the optimal capital subsidy is increasing in this ratio.

Signing the ratio. The response-adjusted ratio of marginal deterrent values is

$$\frac{\partial\widetilde{\mathcal{D}}/\partial K^{S,P}}{\partial\widetilde{\mathcal{D}}/\partial M^{S,P}} = \frac{\partial\mathcal{D}/\partial K^{S,P} + \varphi_K \partial\mathcal{D}/\partial M^{A,P}}{\partial\mathcal{D}/\partial M^{S,P} + \varphi_M \partial\mathcal{D}/\partial M^{A,P}}.$$

Because $\partial\mathcal{D}/\partial M^{A,P} < 0$, and because we focus on the interior region in which the numerator and denominator are positive,

$$\frac{\partial}{\partial\varphi_K}\left(\frac{\partial\widetilde{\mathcal{D}}/\partial K^{S,P}}{\partial\widetilde{\mathcal{D}}/\partial M^{S,P}}\right) = \frac{\partial\mathcal{D}/\partial M^{A,P}}{\partial\widetilde{\mathcal{D}}/\partial M^{S,P}} < 0,$$
$$\frac{\partial}{\partial\varphi_M}\left(\frac{\partial\widetilde{\mathcal{D}}/\partial K^{S,P}}{\partial\widetilde{\mathcal{D}}/\partial M^{S,P}}\right) = -\frac{\left(\partial\widetilde{\mathcal{D}}/\partial K^{S,P}\right)\left(\partial\mathcal{D}/\partial M^{A,P}\right)}{\left(\partial\widetilde{\mathcal{D}}/\partial M^{S,P}\right)^2} > 0.$$

Equation (111) then implies that increasing φ_K raises R_{ss}, while increasing φ_M lowers R_{ss}. Finally, (110) implies that increasing φ_K lowers the optimal capital subsidy, while increasing φ_M raises it.

C Quantitative model

C.1 Competitive equilibrium

Throughout, $\Lambda_t \equiv u'(C_t)$, $\rho \equiv 1/\beta - 1$, and $\rho_V = \frac{\sigma_{VA}-1}{\sigma_{VA}}$, $\rho_X = \frac{\sigma_X-1}{\sigma_X}$, $\rho_Y = \frac{\sigma_Y-1}{\sigma_Y}$.

Household's problem The household chooses the consumption of each good, the labor it supplies to each sector, and holdings B_{t+1} of a risk-free bond in zero net supply, taking as given the goods prices $p_{g,t}$, the sectoral wages $w_{g,t}$, the interest rate r_t, the dividends Π_t of the producers it owns, and the lump-sum tax T_t:

$$\max_{\{c_{g,t},L_{g,t},B_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty}\beta^t\left(u(C_t) - \frac{\tilde{L}_t^{1+1/\xi}}{1+1/\xi} + G_t\right) \qquad \text{s.t.} \qquad \sum_{g\in\mathcal{G}} p_{g,t}c_{g,t} + \frac{B_{t+1}}{1+r_t} = \sum_{g\in\mathcal{G}} w_{g,t}L_{g,t} + \Pi_t - T_t + B_t$$

with C_t and $\tilde{L}_t$ the aggregators of section 3, $B_0 = 0$, and the usual no-Ponzi condition. The price index of the consumption basket is the numeraire.

Producers' problem The producer of sector g owns its capital stock and chooses labor, intermediate inputs, investment goods, and next period's capital, taking as given prices, wages, the interest rate, and the subsidy rate $s_{g,t}$ on the replacement value $P_{g,t}^I K_{g,t}$ of its

capital:

$$\max_{\{L_{g,t},x_{g,r,t},i_{g,r,t},K_{g,t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \Big(\prod_{\tau<t} \frac{1}{1+r_\tau} \Big) \Big(p_{g,t} Y_{g,t} + s_{g,t} P_{g,t}^I K_{g,t} - w_{g,t} L_{g,t} - \sum_{r \in \mathcal{G}} p_{r,t} \big(x_{g,r,t} + i_{g,r,t} \big) \Big)$$

subject to the technology, the investment aggregator, the law of motion of capital, and $I_{g,t} \geq 0$, with $K_{g,0}$ given; the period profit is paid out as the dividend.

Government The government buys each basket at minimum cost, pays the subsidies, and balances its budget with the lump-sum tax:

$$m_t^j = \prod_{g \in \mathcal{G}_j} \big(m_{g,t}^j \big)^{\psi_g^j}, \qquad T_t = \sum_{j \in \mathcal{J}} \sum_{g \in \mathcal{G}_j} p_{g,t} m_{g,t}^j + \sum_{g \in \mathcal{G}} s_{g,t} P_{g,t}^I K_{g,t}, \qquad \mathrm{GDP}_t = C_t + \sum_{g \in \mathcal{G}} P_{g,t}^I I_{g,t} + \sum_{j \in \mathcal{J}} p_{m,t}^j m_t^j$$

where GDP_t is final demand at market prices, in the numeraire, and $p_{m,t}^j$ is the price index of basket j given below. In the postwar phase the spending rule of section 3 replaces the choice of the baskets and $s_{g,t} = 0$.

Definition 3 (Competitive equilibrium of the quantitative model)**.** *Given the government's purchases and subsidy rates $\{m_t^j, s_{g,t}\}_{t=0}^{\infty}$ and the initial stocks $(\mathbf{K}_0, \mathbf{M}_0)$, a competitive equilibrium is a sequence of quantities $\{c_{g,t}, L_{g,t}, x_{g,r,t}, i_{g,r,t}, I_{g,t}, K_{g,t+1}, m_{g,t}^j, M_{t+1}^j\}_{t=0}^{\infty}$, prices $\{p_{g,t}, w_{g,t}, r_t, P_{g,t}^X, P_{g,t}^I, p$ and taxes and dividends $\{T_t, \Pi_t\}_{t=0}^{\infty}$ such that the household and the producers solve their problems, the government buys each basket at minimum cost and its budget holds, and the goods and labor markets clear in every period.*

The rest of this appendix collects the conditions that characterize the equilibrium.

Household: optimality conditions The household's problem separates into a static and a dynamic part. Given total consumption C_t, cost minimization over the basket yields the Cobb–Douglas demands and the price index, which we take as the numeraire. The first-order conditions for the bond and for sectoral labor supply give the discount factor and the sectoral wages in terms of marginal utility. Together:

$$c_{g,t} = \epsilon_g \frac{C_t}{p_{g,t}}, \qquad P_{C,t} = \prod_{g \in \mathcal{G}} \Big(\frac{p_{g,t}}{\epsilon_g} \Big)^{\epsilon_g} = 1, \tag{112}$$

$$\frac{1}{1+r_t} = \beta \frac{\Lambda_{t+1}}{\Lambda_t}, \qquad \Lambda_t \, w_{g,t} = \tilde{L}_t^{\,1/\xi} \Big(\frac{L_{g,t}}{\tilde{L}_t} \Big)^{\frac{1}{\tau}}. \tag{113}$$

Producers: optimality conditions The producer's static problem is cost minimization over labor and intermediate inputs; its dynamic problem is the choice of investment and capital. Let $q_{g,t}$ denote the shadow value of installed capital. Marginal-product pricing of labor and of the intermediate composite, the CES demands for individual intermediate inputs, and the Cobb–Douglas demands for investment goods are

$$w_{g,t} = p_{g,t}\,\alpha_{V,g}\,\alpha_{L,g}\Big(\frac{Y_{g,t}}{V_{g,t}}\Big)^{\frac{1}{\sigma_Y}}\Big(\frac{V_{g,t}}{L_{g,t}}\Big)^{\frac{1}{\sigma_{VA}}}, \qquad P^X_{g,t} = p_{g,t}\,\alpha_{X,g}\Big(\frac{Y_{g,t}}{X_{g,t}}\Big)^{\frac{1}{\sigma_Y}}, \tag{114}$$

$$x_{g,r,t} = \alpha_{g,r}^{\sigma_X}\Big(\frac{p_{r,t}}{P^X_{g,t}}\Big)^{-\sigma_X} X_{g,t}, \qquad P^X_{g,t} = \Big[\sum_{r\in\mathcal{G}} \alpha_{g,r}^{\sigma_X}\, p_{r,t}^{1-\sigma_X}\Big]^{\frac{1}{1-\sigma_X}}, \tag{115}$$

$$i_{g,r,t} = \vartheta_{g,r}\,\frac{P^I_{g,t} I_{g,t}}{p_{r,t}}, \qquad P^I_{g,t} = \prod_{r\in\mathcal{G}}\Big(\frac{p_{r,t}}{\vartheta_{g,r}}\Big)^{\vartheta_{g,r}}. \tag{116}$$

Irreversibility makes the investment decision a complementarity problem. The first-order condition for investment and the Euler equation for capital are

$$\frac{P^I_{g,t}}{1-\varphi\big(I_{g,t}/K_{g,t}-\delta_g\big)} \;\geq\; q_{g,t}, \qquad I_{g,t}\geq 0, \qquad \text{with complementary slackness,} \tag{117}$$

$$q_{g,t} = \beta\,\frac{\Lambda_{t+1}}{\Lambda_t}\Big[(1-\delta_g)\,q_{g,t+1} + s_{g,t+1}P^I_{g,t+1} + p_{g,t+1}\,\mathrm{MPK}_{g,t+1} + q_{g,t+1}\,\frac{\varphi}{2}\Big(\Big(\frac{I_{g,t+1}}{K_{g,t+1}}\Big)^2 - \delta_g^2\Big)\Big], \tag{118}$$

$$\mathrm{MPK}_{g,t} = \alpha_{V,g}\,\alpha_{K,g}\Big(\frac{Y_{g,t}}{V_{g,t}}\Big)^{\frac{1}{\sigma_Y}}\Big(\frac{V_{g,t}}{K_{g,t}}\Big)^{\frac{1}{\sigma_{VA}}}.$$

The producer values its capital at its marginal revenue product plus the subsidy; it does not value the effect of its capital on deterrence.

Government and markets: conditions Cost minimization within each military basket gives its price index and the demands for its goods. With the government budget, goods market clearing, and the laws of motion,

$$p^j_{m,t} = \prod_{g\in\mathcal{G}_j}\Big(\frac{p_{g,t}}{\psi^j_g}\Big)^{\psi^j_g}, \qquad m^j_{g,t} = \psi^j_g\,\frac{p^j_{m,t}m^j_t}{p_{g,t}}, \qquad T_t = \sum_{j\in\mathcal{J}} p^j_{m,t}m^j_t + \sum_{g\in\mathcal{G}} s_{g,t}P^I_{g,t}K_{g,t}, \tag{119}$$

$$Y_{g,t} = c_{g,t} + \sum_{b\in\mathcal{G}} x_{b,g,t} + \sum_{b\in\mathcal{G}} i_{b,g,t} + \sum_{j\in\mathcal{J}} m^j_{g,t}, \tag{120}$$

$$K_{g,t+1} = (1-\delta_g)K_{g,t} + I_{g,t} - \frac{\varphi}{2}\Big(\frac{I_{g,t}}{K_{g,t}} - \delta_g\Big)^2 K_{g,t}, \qquad M^j_{t+1} = (1-\delta_{j,t})M^j_t + m^j_t. \tag{121}$$

Given the government's instruments $\{m_t^j, s_{g,t}\}$ and the initial stocks, the competitive equilibrium is the solution of (112)–(121) together with the technology.

C.2 Adversary's welfare, Sovereign's wartime problem, and the deterrence function

This appendix derives the deterrence function $\mathcal{D}_t$ of section 3 and its derivatives. Suppose Adversary challenges at date t. The war is re-indexed to start at $s = 0$ from the peacetime stocks $(\mathbf{K}_t, \mathbf{M}_t)$, which are predetermined at the outbreak; its length T is drawn with probability $\tilde{\pi}_T(M_t)$ and then known to both parties.

Adversary's welfare Adversary's consumption is the same in peace and in war, so its welfare differs across the two states only through the public good. In peace the public good stays at its initial level; in a war of length T it is reduced from period T on by the settlement $\sum_{s=0}^{T-1} \chi_s^{(T)}(M_s - M_s^E)$, the territory Sovereign gains. Discounting the settlement from its first period,

$$W_t^{A,P} - W_t^{A,C,(T)} = \frac{\beta^T}{1-\beta} \sum_{s=0}^{T-1} \chi_s^{(T)} \Big(M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t) - M_s^E \Big),$$

where $M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t)$ is Sovereign's military power in period s of the war, derived below, with $M_0^{(T)} = M_t$. Adversary evaluates the war before its length is known, $W_t^{A,C} = \sum_T \tilde{\pi}_T(M_t) W_t^{A,C,(T)}$, so $\mathcal{D}_t = W_t^{A,P} - W_t^{A,C}$ is the expression of section 3. Consumption and the initial level of the public good cancel: neither enters the deterrence function.

Sovereign's wartime problem Sovereign's public good rises by the settlement from period T on, so its wartime government maximizes

$$\sum_{s=0}^{T-1} \beta^s \Big(u(C_s) - \frac{\tilde{L}_s^{1+1/\xi}}{1+1/\xi} + G_0 \Big) + \frac{\beta^T}{1-\beta} \sum_{s=0}^{T-1} \chi_s^{(T)} \big(M_s - M_s^E \big) + \beta^T V^{\text{post}}(\mathbf{K}_T, \mathbf{M}_T)$$

subject to the technology, market clearing, the laws of motion with the wartime depreciation rates δ_j^w, $m_s^j \geq 0$ and $I_{g,s} \geq 0$, where V^{post} is the household's welfare in the postwar competitive equilibrium under the spending rule, in which the public good is fixed at its postwar level. Because the settlement depends on the military stocks only, and sectoral capital affects it only through production, the wartime Euler equation for capital carries no deterrence wedge: the government's within-period allocation coincides with the com-

petitive one, the multipliers on the technology and market-clearing constraints equal the competitive prices, and the conditions (112)–(121) hold with $s_{g,s} = 0$ – there is no industrial policy during the war – and with δ_j^w in the laws of motion for the military stocks. The only conditions specific to the wartime problem concern procurement. Let μ_s^j denote the shadow value, in period-s utility, of a unit of stock j at the beginning of period s. The first-order condition for procurement and the recursion for the shadow values are

$$\Lambda_s \, p_{m,s}^j \;\geq\; \beta \, \mu_{s+1}^j, \qquad m_s^j \geq 0, \qquad \text{with complementary slackness,} \tag{122}$$

$$\mu_s^j = \frac{\beta^{T-s}}{1-\beta} \, \chi_s^{(T)} \, \phi_j \, \frac{M_s}{M_s^j} + \beta \, (1 - \delta_j^w) \, \mu_{s+1}^j, \qquad s = 0, \ldots, T-1, \qquad \mu_T^j = 0. \tag{123}$$

A unit of stock j is worth its contribution to the settlement in the current period, through the marginal product $\phi_j M_s / M_s^j$ of the stock in military power, plus its depreciated value next period. The terminal condition follows from the postwar spending rule, under which the stocks carried out of the war have no value; hence nothing is procured in the last period of the war. The wartime capital Euler equation (118) at $s = T-1$ involves the postwar prices, marginal products, and shadow value of capital at $s = T$. We therefore solve the war together with its postwar continuation as a single perfect-foresight path, with the postwar spending rule replacing (122)–(123) after the war and the path closed at the postwar steady state after H postwar periods. The solution delivers $M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t)$.

Derivatives of the deterrence function The peacetime planner (appendix C.4) and the no-industrial-policy government (appendix C.5) use the derivatives of $\mathcal{D}_t$ with respect to the stocks at the outbreak. The stocks enter through the wartime paths and through the war-length probabilities, so for $x \in \{K_{g,t}, M_t^j\}$

$$\frac{\partial \mathcal{D}_t}{\partial x} = \sum_{T=1}^{\bar{T}} \tilde{\pi}_T(M_t) \, \frac{\beta^T}{1-\beta} \sum_{s=0}^{T-1} \chi_s^{(T)} \, \frac{\partial M_s^{(T)}}{\partial x} + \frac{\partial M_t}{\partial x} \sum_{T=1}^{\bar{T}} \frac{\partial \tilde{\pi}_T}{\partial M} \, \frac{\beta^T}{1-\beta} \sum_{s=0}^{T-1} \chi_s^{(T)} \left(M_s^{(T)} - M_s^E \right),$$

with $\partial M_t / \partial K_{g,t} = 0$, $\partial M_t / \partial M_t^j = \phi_j M_t / M_t^j$, $\partial \tilde{\pi}_T / \partial M = a'(M_t) \, \pi_T$ for $T \geq 2$, $\partial \tilde{\pi}_1 / \partial M = -a'(M_t)(1 - \pi_1)$, and $a'(M) = \kappa M_0^E / (M + M_0^E)^2$. The derivatives $\partial M_s^{(T)} / \partial x$ are total derivatives through the wartime equilibrium: the wartime path is a function of the initial stocks defined implicitly by the system above, and its derivatives are obtained by differentiating that system (in the computation, by an adjoint solve with the same sparse Jacobian used to solve it). Since $M_s^{(T)}$ is a choice variable of the wartime government rather than its value function, no envelope simplification applies.

C.3 Calibration

Sectors The sectoral data are the 2017 BEA input-output accounts at the detail level. We drop finance and insurance, owner-occupied housing, and government, which leaves the $N = 378$ private industries that are the sectors of the model.

Technology and preference weights The consumption weights ϵ_g are the shares of final uses other than investment and military procurement. The CES weights of the production nests are calibrated to the cost shares in the data: the intermediate spending shares of the use table, the value-added share of gross output (compensation plus operating surplus), and the labor share of value added (compensation). Away from the Cobb–Douglas case a CES weight is not a cost share – at any allocation the shares are $s_{K,g} = \alpha_{K,g}(K_g/V_g)^{\rho_V}$, $s_{V,g} = \alpha_{V,g}(V_g/Y_g)^{\rho_Y}$, $s_{g,r} = \alpha_{g,r}(x_{g,r}/X_g)^{\rho_X}$, and analogously for $\alpha_{L,g}$ and $\alpha_{X,g}$ – so the weights are chosen jointly with the postwar steady state: the weights of every nest sum to one, and they are set so that the steady-state cost shares equal the data shares, one condition per weight. The sum-to-one restriction is what makes the weights determinate; without it the share conditions fix only the ratios of the weights within a nest, the level of each nest's output being a choice of units (see also Carvalho et al. (2025), Appendix B.14). The steady state then reproduces every data share by construction for any elasticities, which are set from outside because shares at a single point do not identify them.

Investment network and depreciation The investment weights $\vartheta_{g,r}$ are the shares of each producing industry in the investment of industry g. They are built from the BEA detailed fixed-asset investment by industry and asset type, bridging each asset type to its producing industries by the method of Vom Lehn and Winberry (2022), in the detail-level implementation of Iovino et al. (2025). The purchasing side is observed for 66 fixed-asset industries, so the detailed sectors within one fixed-asset industry share the same capital composition. The sectoral depreciation rate δ_g is the depreciation rate of the sector's steady-state capital stock implied by its investment mix over asset types and the BEA asset depreciation rates.

Military baskets Military procurement is the purchases of goods and services by national defense in the use table: the intermediate purchases of the national-defense government sector and defense investment in structures, equipment, and intellectual property. We first remove from these purchases the overhead services, which keep installations and personnel running rather than entering the arsenal: maintenance and repair

of structures, facilities support services, employment services, accommodation, food service, dry-cleaning and laundry, janitorial and grounds services, waste management and environmental consulting, leased real estate, advertising, business support and management consulting, passenger transport and couriers, information services, and morale and recreation services. A stock of these services does not raise military power, so they are not part of any basket; they are dropped from the procurement bundle rather than reassigned, and account for 4.8% of defense procurement. The remaining 184 detailed industries are partitioned into the eight baskets: land systems (military armored vehicles), air systems (aircraft, aircraft parts and engines), naval systems (ship and boat building), ammunition (ammunition, arms and ordnance; guided missiles; missile and space propulsion), structures, intellectual property (research and development, engineering, software), non-durables (fuel, pharmaceuticals, transportation, and other goods and services consumed within the year, which BEA expenses), and the residual other equipment. We classify the durable materials and components that BEA expenses – textiles, apparel, leather, curtains and linens, fabrics, fibers and yarns, rubber and tires, hoses and belting, plastics, resins and foams, glass, clay and refractories, abrasives, office supplies, surgical and dental appliances, and other miscellaneous manufactures – with equipment rather than non-durables: uniforms, tires and tentage are held as stocks and wear out at an equipment-like rate rather than being consumed within the period. These twenty-three industries are 1.5% of procurement. Three further reclassifications place each good with the stock it belongs to: aircraft engines and engine parts join the air systems basket with airframes and other aircraft parts, boat building joins ship building in naval systems, and the building materials that had been counted as structures – structural plate work, cut stone, sawmills, millwork and other wood products – join equipment, leaving structures the construction commodities that BEA capitalizes as national-defense structures. The within-basket weights ψ_g^j and the basket shares θ_j are the shares of the remaining purchases, and the spending share $\theta = 3.37\%$ of GDP is the US average over 2015–2025 in the SIPRI data, unchanged by the reclassification: the overhead purchases leave the composition of procurement, not the level of military spending. The peacetime depreciation rates δ_j are the post-1970 means of current-cost depreciation over the lagged net stock of Federal national-defense fixed assets in the BEA Fixed Assets tables: structures, total equipment, and intellectual property for their baskets, and vehicles, aircraft, and ships for land, air, and naval systems. Non-durables are consumed within the year. For ammunition the rate is the FY2014–2019 mean of procurement over the DoD Class V inventory of conventional ammunition and missiles, 12.7%. The wartime rates δ_j^w add combat losses. For land systems we use the Russia–Ukraine war: visually confirmed annual losses (de-

stroyed, captured, and abandoned, from Oryx) over the stock at the start of the year (IISS Military Balance), 2022–2025, give attrition of 18% for Russia and 27% for Ukraine; added to the peacetime rate these give wartime rates of 34 and 43%, and we use their mean, $\delta^w = 38.3\%$. Confirmed losses undercount, Russia's uncommitted fleets dilute its rate, and Ukraine's wartime deliveries inflate its denominator. For air and naval systems we use the losses of the US armed forces in the Second World War, the only war in which the US fought at scale for several years. For aircraft, attrition is the Army Air Forces' losses overseas from all causes, combat and operational, over the aircraft on hand at the start of the year (AAF Statistical Digest, Tables 83 and 99), 24.2% per year on average over 1942–45; adding the peacetime rate gives $\delta^w = 34.3\%$. For ships, it is major combatants sunk or damaged beyond repair (battleships, carriers, cruisers, destroyers, destroyer escorts, and submarines; Naval History and Heritage Command) over the average number in commission during the year (NHHC ship force levels), 6.5% per year on average; adding the peacetime rate gives $\delta^w = 14.6\%$. Structures, other equipment, and intellectual property keep their peacetime rates; non-durables are consumed within the year, and so is ammunition in war. Table 6 collects the rates. Table 7 lists every basket and its component sectors, and table 8 the overhead services excluded from the baskets.

Table 6: Depreciation of the military baskets in peace and in war. Combat attrition is the increment $\delta_j^w - \delta_j$: for land systems the mean of the Russian and Ukrainian rates in the Russia–Ukraine war, for air and naval systems the loss rates of the US armed forces in the Second World War; structures, other equipment, and intellectual property are not exposed to it. Non-durables and, in war, ammunition are consumed within the year, so their rate is one rather than a peacetime rate plus attrition.

Basket	Peacetime, δ_j	Combat attrition	Wartime, δ_j^w
Structures	0.020	0	0.020
Other equipment	0.116	0	0.116
Intellectual property	0.223	0	0.223
Non-durables	1.000	—	1.000
Land systems	0.155	0.228	0.383
Air systems	0.101	0.242	0.343
Naval systems	0.081	0.065	0.146
Ammunition	0.127	—	1.000

Table 7: The military baskets and their component sectors. Sectors are ordered within each basket by defense procurement. 'mfg.' abbreviates manufacturing, 'svcs.' services, 'equip.' equipment, 'instr.' instruments, 'prod.' product and 'exc.' except.

code	sector	code	sector
Structures (10 sectors)			
2334A0	Other residential structures	2332C0	Transportation structures & highways & streets
2332D0	Other nonresidential structures	233412	Multifamily residential structures
2332A0	Office & commercial structures	233230	Manufacturing structures
233262	Educational & vocational structures	233240	Power & communication structures
233210	Health care structures	233411	Single-family residential structures
Other equipment (105 sectors)			
334511	Search, detection, & navigation instr. mfg.	326220	Rubber & plastics hoses & belting mfg.
334220	Broadcast & wireless communications equip.	339990	All other miscellaneous mfg.
336320	Motor vehicle electrical & electronic equip. mfg.	332430	Metal can, box, & other metal container (light gauge) mfg.
334515	Electricity & signal testing instr. mfg.	336111	Automobile mfg.
334111	Electronic computer mfg.	315000	Apparel mfg.
335313	Switchgear & switchboard apparatus mfg.	333994	Industrial process furnace & oven mfg.
336350	Motor vehicle transmission & power train parts mfg.	332720	Turned prod. & screw, nut, & bolt mfg.
332200	Cutlery & handtool mfg.	33399A	Other general purpose machinery mfg.
314900	Other textile prod. mills	333130	Mining & oil & gas field machinery mfg.
33441A	Other electronic component mfg.	336112	Light truck & utility vehicle mfg.
33291A	Valve & fittings other than plumbing	33329A	Other industrial machinery mfg.
326290	Other rubber prod. mfg.	335220	Major household appliance mfg.
335314	Relay & industrial control mfg.	33131B	Aluminum prod. mfg. from purchased aluminum
334210	Telephone apparatus mfg.	332320	Ornamental & architectural metal prod. mfg.
326210	Tire mfg.	3219A0	All other wood prod. mfg.
339113	Surgical appliance & supplies mfg.	334513	Industrial process variable instr. mfg.
336120	Heavy duty truck mfg.	336500	Railroad rolling stock mfg.
334510	Electromedical & electrotherapeutic apparatus mfg.	334610	Manufacturing & reproducing magnetic & optical media
332991	Ball & roller bearing mfg.	326110	Plastics packaging materials & unlaminated film & sheet mfg.
336390	Other motor vehicle parts mfg.	327999	Miscellaneous nonmetallic mineral prod.
333314	Optical instrument & lens mfg.	335991	Carbon & graphite prod. mfg.
33399B	Fluid power process machinery	331410	Nonferrous metal (exc. aluminum) smelting & refining
334118	Computer terminals & other computer peripheral equip. mfg.	332999	Other fabricated metal mfg.
335999	All other miscellaneous electrical equip. & component mfg.	334512	Automatic environmental control mfg.
336999	All other transportation equip. mfg.	3252A0	Synthetic rubber & artificial & synthetic fibers & filaments mfg.
333120	Construction machinery mfg.	333612	Speed changer, industrial high-speed drive, & gear mfg.
333914	Measuring, dispensing, & other pumping equip. mfg.	334514	Totalizing fluid meter & counting device mfg.
332500	Hardware mfg.	327200	Glass & glass prod. mfg.

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Table 7 (continued)

code	sector	code	sector
333318	Other commercial & service industry machinery mfg.	335930	Wiring device mfg.
33451A	Watch, clock, & other measuring & controlling device mfg.	333613	Mechanical power transmission equip. mfg.
336310	Motor vehicle gasoline engine & engine parts mfg.	333316	Photographic & photocopying equip. mfg.
333611	Turbine & turbine generator set units mfg.	331420	Copper rolling, drawing, extruding & alloying
334112	Computer storage device mfg.	333517	Machine tool mfg.
334517	Irradiation apparatus mfg.	335210	Small electrical appliance mfg.
333415	Air conditioning, refrigeration, & warm air heating equip. mfg.	336991	Motorcycle, bicycle, & parts mfg.
316000	Leather & allied prod. mfg.	327910	Abrasive prod. mfg.
332119	Metal crown, closure, & other metal stamping (exc. automotive)	313100	Fiber, yarn, & thread mills
3363A0	Motor vehicle steering, suspension component (exc. spring), & brake systems mfg.	339940	Office supplies (exc. paper) mfg.
339112	Surgical & medical instrument mfg.	321910	Millwork
335312	Motor & generator mfg.	333514	Special tool, die, jig, & fixture mfg.
314120	Curtain & linen mills	33721A	Office furniture & custom architectural woodwork & millwork mfg.
339114	Dental equip. & supplies mfg.	332996	Fabricated pipe & pipe fitting mfg.
333920	Material handling equip. mfg.	326190	Other plastics prod. mfg.
333618	Other engine equip. mfg.	321100	Sawmills & wood preservation
332310	Plate work & fabricated structural prod. mfg.	336360	Motor vehicle seating & interior trim mfg.
331313	Alumina refining & primary aluminum prod.ion	333413	Industrial & commercial fan & blower & air purification equip. mfg.
313200	Fabric mills	333991	Power-driven handtool mfg.
327991	Cut stone & stone prod. mfg.	333414	Heating equip. (exc. warm air furnaces) mfg.
327100	Clay prod. & refractory mfg.	335912	Primary battery mfg.
334290	Other communications equip. mfg.	325211	Plastics material & resin mfg.
335920	Communication & energy wire & cable mfg.	326150	Urethane & other foam prod. (exc. polystyrene) mfg.
336212	Truck trailer mfg.	326140	Polystyrene foam prod. mfg.
333912	Air & gas compressor mfg.		
Intellectual property (6 sectors)			
541700	Scientific research & development svcs.	541512	Computer systems design svcs.
541300	Architectural, engineering, & related svcs.	511200	Software publishers
541511	Custom computer programming svcs.	54151A	Other computer related svcs., including facilities management
Non-durables (54 sectors)			
324110	Petroleum refineries	311119	Other animal food mfg.
811200	Electronic & precision equip. repair & maintenance	482000	Rail transportation
325412	Pharmaceutical preparation mfg.	311910	Snack food mfg.
481000	Air transportation	322220	Paper bag & coated & treated paper mfg.
517210	Wireless telecommunications carriers (exc. satellite)	325413	In-vitro diagnostic substance mfg.
484000	Truck transportation	312110	Soft drink & ice mfg.
518200	Data processing, hosting, & related svcs.	322210	Paperboard container mfg.
31161A	Animal (exc. poultry) slaughtering, rendering, & processing	5111A0	Directory, mailing list, & other publishers

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Table 7 (continued)

code	sector	code	sector
221100	Electric power generation, transmission, & distribution	311230	Breakfast cereal mfg.
221300	Water, sewage & other systems	311420	Fruit & vegetable canning, pickling, & drying
483000	Water transportation	324190	Other petroleum & coal prod. mfg.
311615	Poultry processing	312140	Distilleries
3259A0	All other chemical prod. & preparation mfg.	311940	Seasoning & dressing mfg.
517110	Wired telecommunications carriers	325610	Soap & cleaning compound mfg.
515100	Radio & television broadcasting	112A00	Animal prod.ion, exc. cattle & poultry & eggs
811300	Commercial & industrial machinery & equip. repair & maintenance	325310	Fertilizer mfg.
221200	Natural gas distribution	322230	Stationery prod. mfg.
325180	Other basic inorganic chemical mfg.	325620	Toilet preparation mfg.
323110	Printing	3118A0	Cookie, cracker, pasta, & tortilla mfg.
311513	Cheese mfg.	325320	Pesticide & other agricultural chemical mfg.
493000	Warehousing & storage	322120	Paper mills
311700	Seafood prod. preparation & packaging	325190	Other basic organic chemical mfg.
311810	Bread & bakery prod. mfg.	532400	Commercial & industrial machinery & equip. rental & leasing
532100	Automotive equip. rental & leasing	322130	Paperboard mills
517A00	Satellite, telecommunications resellers, & all other telecommunications	311225	Fats & oils refining & blending
311300	Sugar & confectionery prod. mfg.	339920	Sporting & athletic goods mfg.
311990	All other food mfg.	322299	All other converted paper prod. mfg.
Land systems (1 sector)			
336992	Military armored vehicle, tank, & tank component mfg.		
Air systems (3 sectors)			
336411	Aircraft mfg.	336412	Aircraft engine & engine parts mfg.
336413	Other aircraft parts & auxiliary equip. mfg.		
Naval systems (2 sectors)			
336611	Ship building & repairing	336612	Boat building
Ammunition (3 sectors)			
336414	Guided missile & space vehicle mfg.	33641A	Propulsion units & parts for space vehicles & guided missiles
33299A	Ammunition, arms, ordnance, & accessories mfg.		

Table 8: Overhead services excluded from the military baskets, ordered by defense procurement. These purchases keep installations and personnel running rather than entering the arsenal, so no stock of them raises military power; they are dropped from the procurement bundle rather than reassigned.

code	sector	code	sector
230301	Nonresidential maintenance & repair	561700	Services to buildings & dwellings
561200	Facilities support svcs.	562000	Waste management & remediation svcs.
561300	Employment svcs.	5416A0	Environmental & other technical consulting svcs.

continued on the next page

Table 8 (continued)

code	sector	code	sector
230302	Residential maintenance & repair	722A00	All other food & drinking places
721000	Accommodation	485000	Transit & ground passenger transportation
541800	Advertising, public relations, & related svcs.	512200	Sound recording industries
5191A0	News syndicates, libraries, archives & all other information svcs.	561400	Business support svcs.
722211	Limited-service restaurants	48A000	Scenic & sightseeing transportation & support activities for transportation
722110	Full-service restaurants	492000	Couriers & messengers
531ORE	Other real estate	541610	Management consulting svcs.
812300	Dry-cleaning & laundry svcs.	713900	Other amusement & recreation industries
811400	Personal & household goods repair & maintenance		

Weights of the baskets in military power The weights ϕ_j are chosen so that a government that deters with procurement only, the no-industrial-policy benchmark of appendix C.5, holds exactly the observed arsenal. At the peacetime steady state without subsidies, the procurement conditions (131) of appendix C.5 read $p_m^j(\rho + \delta_j) = \zeta\, \partial\mathcal{D}/\partial M^j$ for every basket, where ζ is the multiplier on the deterrence constraint. We evaluate these conditions at the arsenal implied by the observed spending shares, $p_m^j m^j = \theta_j \theta\, \text{GDP}$ and $m^j = \delta_j M^j$, and solve for the ϕ_j that make all of them hold, with ζ the geometric mean of $p_m^j(\rho + \delta_j)/(\partial\mathcal{D}/\partial M^j)$ across baskets. Since the derivatives of the deterrence function depend on ϕ through the wartime problems, this is a fixed point over ϕ, the intensities $\chi^{(T)}$, and the derivatives; it is solved by a quasi-Newton iteration. At the solution the peacetime steady state without subsidies coincides with the postwar steady state, so the observed composition of military spending is optimal for a government that does not use industrial policy.

War length The probabilities π_T are the frequencies of war lengths in years among the inter-state wars with at least 1,000 battle deaths in the Correlates of War data, 1816–2007, with wars of six years or more collected in $T = 6$. The sensitivity κ is estimated on the same wars by a linear probability model of an indicator for a war lasting more than one year on the initiator's share of the two sides' military personnel in the year before the war, from the Correlates of War National Material Capabilities; the initiator is the side coded as such, and the other side comprises its opponents in the war's first year. On the 87 wars with personnel data the slope is -0.33 (robust standard error 0.15); the share of military expenditure gives -0.23 (0.17) on 62 wars, and the composite capabilities index -0.29 (0.15) on 89. Since Adversary opens a war with the same military power as Sovereign, its share at the peacetime steady state is one half, $a = 1 - \kappa/2 = 0.835$, and the probability

of a war longer than one year is 0.264.

Combat efficiency parameters The combat efficiency parameters enter the wartime problem only through the procurement conditions (122)–(123). Fixing the path of military spending therefore fixes the wartime allocation independently of the intensities, and the procurement conditions can be solved for the intensities that make the path optimal: from (123) and the interior condition (122),

$$\chi_{s+1}^{(T)} = \frac{1-\beta}{\beta^{T-s}} \frac{M_{s+1}^{j}}{\phi_j M_{s+1}} \Big[\Lambda_s \, p_{m,s}^{j} - \beta(1-\delta_j^w) \, \Lambda_{s+1} \, p_{m,s+1}^{j} \Big], \qquad s = 0, \ldots, T-2,$$

for any basket j that is purchased in both periods; the ratio conditions across baskets pin down the composition of procurement without the intensities, so every basket returns the same value. The target path is the military-spending response to war estimated by Marzian and Trebesch (2026), the deviation of military spending from its pre-war level as a share of GDP over the first five years of a war, $\sigma^* = (5.3, 15, 19, 21.5, 20.5)\%$. For a war of length T we require nominal military spending in year s of the war to exceed its steady-state level by σ_s^* of steady-state GDP for $s = 0, \ldots, T-2$. The last year of the war is not targeted: by (123) nothing is procured in it. The $T-1$ targets pin down $\chi_1^{(T)}, \ldots, \chi_{T-1}^{(T)}$, which are computed by appending them to the unknowns of the joint war–postwar system of appendix C.2 together with the $T-1$ spending targets, and the wars are started from the peacetime steady state without subsidies. Table 9 reports the result.

Table 9: Combat efficiency parameters inverted from the wartime spending path

T	$\chi_1^{(T)}$	$\chi_2^{(T)}$	$\chi_3^{(T)}$	$\chi_4^{(T)}$	$\chi_5^{(T)}$
2	123.39				
3	50.61	177.20			
4	45.68	76.95	189.99		
5	47.87	70.94	85.45	194.35	
6	50.62	73.91	77.68	93.83	189.14

Adversary's military power and the level of deterrence The intensity at the outbreak, χ_0, multiplies the predetermined gap $M_0 - M_0^E$ and enters no first-order condition, so the spending path does not identify it; and since Adversary's path M_s^E enters the deterrence function additively, its level and η_t are collinear. We normalize $\eta_t = 0$ and set Adver-

sary's path in a war of length T equal to Sovereign's own path in that war started from the peacetime steady state without subsidies, $M_s^E = M_s^{(T)}(\mathbf{K}^{ss}, \mathbf{M}^{ss})$, computed once per calibration round. Every term of the deterrence function then vanishes at that steady state, so the deterrence constraint binds there for any value of χ_0, which is set to the probability-weighted mean over $T \geq 2$ of the intensities in the later periods of a war, $\chi_0 = 113.6$. The peacetime steady state without subsidies – the steady state of the government that deters with procurement only, the no-industrial-policy benchmark – is thus the observed steady state by construction: with the calibrated ϕ_j, the observed arsenal satisfies the procurement conditions, and with the calibrated enemy path it satisfies the deterrence constraint with equality.

C.4 Sovereign's peacetime problem

This appendix derives the optimality conditions of the peacetime planning problem (33) and compares them with the competitive equilibrium of appendix C.1 to prove proposition 7.

The Lagrangian Attach to the date-t market-clearing condition of sector g the multiplier $\beta^t \Lambda_t p_{g,t}$, to the law of motion of capital $\beta^t \Lambda_t q_{g,t}$, to the law of motion of military stock j the multiplier $\beta^t \Lambda_t \varpi_t^j$, and to the date-$(t+1)$ deterrence constraint $\beta^{t+1} \Lambda_{t+1} \gamma_t$, where $\Lambda_t \equiv u'(C_t)$. Normalizing the multipliers by Λ_t in this way states them in units of the date-t consumption good. Substituting the aggregators for C_t, $\tilde{L}_t$, $V_{g,t}$, $X_{g,t}$, $Y_{g,t}$, $I_{g,t}$ and m_t^j, the Lagrangian is

$$
\begin{aligned}
\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ & u(C_t) - \frac{\tilde{L}_t^{\,1+1/\xi}}{1+1/\xi} + G_0 \\
& + \sum_{g \in \mathcal{G}} \Lambda_t \, p_{g,t} \Big[Y_{g,t} - c_{g,t} - \sum_b x_{b,g,t} - \sum_b i_{b,g,t} - \sum_j m_{g,t}^j \Big] \\
& + \sum_{g \in \mathcal{G}} \Lambda_t \, q_{g,t} \Big[(1-\delta_g) K_{g,t} + I_{g,t} - \frac{\varphi}{2} \Big(\frac{I_{g,t}}{K_{g,t}} - \delta_g \Big)^2 K_{g,t} - K_{g,t+1} \Big] \\
& + \sum_{j \in \mathcal{J}} \Lambda_t \, \varpi_t^j \Big[(1-\delta_j) M_t^j + m_t^j - M_{t+1}^j \Big] \\
& + \beta \, \Lambda_{t+1} \, \gamma_t \Big[\mathcal{D}_{t+1}(\mathbf{K}_{t+1}, \mathbf{M}_{t+1}) - \eta_{t+1} \Big] \Big\},
\end{aligned}
$$

with $I_{g,t} \geq 0$, $m_t^j \geq 0$ and $\gamma_t \geq 0$. Deterrence and the public good depend on the predetermined stocks only, so the constraints that tie the periods together are the two laws of

motion and the deterrence constraint; every remaining choice is a within-period one.

Static conditions Consumption of good g and sectoral labor satisfy

$$\frac{\partial C_t}{\partial c_{g,t}} = \epsilon_g \frac{C_t}{c_{g,t}} = p_{g,t}, \qquad \tilde{L}_t^{1/\xi} \Big(\frac{L_{g,t}}{\tilde{L}_t} \Big)^{\frac{1}{\tau}} = \Lambda_t \, p_{g,t} \frac{\partial Y_{g,t}}{\partial L_{g,t}}, \tag{124}$$

and the intermediate input of good r in sector g satisfies $p_{r,t} = p_{g,t} \, \partial Y_{g,t} / \partial x_{g,r,t}$. The first condition inverts to the Cobb–Douglas demand of (112); multiplying it by $c_{g,t}$ and summing over g, linear homogeneity of the aggregator gives $\sum_g p_{g,t} c_{g,t} = C_t$, so the consumption price index equals one and the multipliers $p_{g,t}$ are prices in the numeraire of appendix C.1. Defining $w_{g,t} \equiv p_{g,t} \partial Y_{g,t} / \partial L_{g,t}$, the second condition is the labor supply condition (113) and the definition of $w_{g,t}$ is the marginal-product pricing of (114); the condition for $x_{g,r,t}$ is the intermediate demand (115). The planner's within-period allocation therefore coincides with the competitive one, and its static multipliers are the competitive prices.

Investment The planner buys $i_{g,r,t}$ of good r to produce sector g's investment good. Since $\partial K_{g,t+1} / \partial I_{g,t} = 1 - \varphi\big(I_{g,t}/K_{g,t} - \delta_g\big)$ and $\partial I_{g,t} / \partial i_{g,r,t} = \vartheta_{g,r} I_{g,t} / i_{g,r,t}$, the first-order condition for $i_{g,r,t}$ is

$$p_{r,t} \, i_{g,r,t} = q_{g,t} \Big[1 - \varphi\Big(\frac{I_{g,t}}{K_{g,t}} - \delta_g \Big) \Big] \vartheta_{g,r} \, I_{g,t}, \tag{125}$$

which is the Cobb–Douglas demand of (116). Summing over r and using $\sum_r p_{r,t} i_{g,r,t} = P^I_{g,t} I_{g,t}$ gives the value of an extra unit of the investment good, so that irreversibility yields the complementarity condition

$$\frac{P^I_{g,t}}{1 - \varphi\big(I_{g,t}/K_{g,t} - \delta_g\big)} \geq q_{g,t}, \qquad I_{g,t} \geq 0, \qquad \text{with complementary slackness,} \tag{126}$$

which is (117): the planner installs capital up to the point where its shadow value equals the marginal cost of installing it.

Capital The stock $K_{g,t+1}$ raises date-$(t+1)$ output, relaxes the date-$(t+1)$ deterrence constraint, and carries over into $K_{g,t+2}$. Differentiating the adjustment cost,

$$\begin{aligned}\frac{\partial K_{g,t+2}}{\partial K_{g,t+1}} &= (1-\delta_g) + \varphi\Big(\frac{I_{g,t+1}}{K_{g,t+1}} - \delta_g\Big)\frac{I_{g,t+1}}{K_{g,t+1}} - \frac{\varphi}{2}\Big(\frac{I_{g,t+1}}{K_{g,t+1}} - \delta_g\Big)^2 \\ &= (1-\delta_g) + \frac{\varphi}{2}\Big(\Big(\frac{I_{g,t+1}}{K_{g,t+1}}\Big)^2 - \delta_g^2\Big),\end{aligned}$$

so the first-order condition for $K_{g,t+1}$ is

$$q_{g,t} = \beta\,\frac{\Lambda_{t+1}}{\Lambda_t}\Big[\gamma_t\,\frac{\partial \mathcal{D}_{t+1}}{\partial K_{g,t+1}} + p_{g,t+1}\,\mathrm{MPK}_{g,t+1} + (1-\delta_g)\,q_{g,t+1} + q_{g,t+1}\,\frac{\varphi}{2}\Big(\Big(\frac{I_{g,t+1}}{K_{g,t+1}}\Big)^2 - \delta_g^2\Big)\Big], \tag{127}$$

where γ_t is the multiplier on the date-$(t+1)$ deterrence constraint, expressed in units of the date-$(t+1)$ numeraire, and $q_{g,t}$ is the shadow value of installed capital. Only the date-$(t+1)$ deterrence constraint contains $K_{g,t+1}$, so a single multiplier γ_t appears. The derivative $\partial \mathcal{D}_{t+1}/\partial K_{g,t+1}$ is the total derivative of the deterrence function with respect to the state, through the wartime paths and the war-length probabilities; it is derived in appendix C.2.

Procurement and the military stocks Procurement of good g for basket j satisfies $p_{g,t} = \omega_t^j\,\partial m_t^j/\partial m_{g,t}^j = \omega_t^j \psi_g^j m_t^j/m_{g,t}^j$, which is the Cobb–Douglas demand of (119); summing over $g \in \mathcal{G}_j$ gives $\sum_g p_{g,t} m_{g,t}^j = \omega_t^j m_t^j$, so that $\omega_t^j = p_{m,t}^j$ whenever $m_t^j > 0$, and $p_{m,t}^j \geq \omega_t^j$ in general. The first-order condition for M_{t+1}^j is

$$\Lambda_t\,\omega_t^j = \beta\,\Lambda_{t+1}\Big[\gamma_t\,\frac{\partial \mathcal{D}_{t+1}}{\partial M_{t+1}^j} + (1-\delta_j)\,\omega_{t+1}^j\Big]. \tag{128}$$

For a stock purchased in both periods $\omega_t^j = p_{m,t}^j$ and $\omega_{t+1}^j = p_{m,t+1}^j$, so that (128) holds with $p_{m,t}^j$ in place of ω_t^j; a stock not purchased at t has $p_{m,t}^j > \omega_t^j$, and the same expression holds as a strict inequality. Military stocks enter neither production nor utility, so, unlike capital, they are valued for deterrence and carry-over alone. Finally $\gamma_t \geq 0$, $\mathcal{D}_{t+1} \geq \eta_{t+1}$, and $\gamma_t\big(\mathcal{D}_{t+1} - \eta_{t+1}\big) = 0$.

Proof. Proof. Proof of Proposition 7. Let the planner's allocation solve problem (33), with multipliers $p_{g,t}$, $q_{g,t}$, ω_t^j and γ_t. Let the government purchase the planner's $m_{g,t}^j$, pay the

capital subsidies

$$s_{g,t+1}\, P^I_{g,t+1} = \gamma_t \, \frac{\partial \mathcal{D}_{t+1}}{\partial K_{g,t+1}}, \qquad g \in \mathcal{G}, \tag{129}$$

and balance its budget with the lump-sum tax of (119). Take $p_{g,t}$ as the goods prices, $w_{g,t} = p_{g,t}\partial Y_{g,t}/\partial L_{g,t}$ as the wages, and $1/(1+r_t) = \beta\Lambda_{t+1}/\Lambda_t$ as the bond price.

Compare the two sets of conditions term by term. The static conditions (124) and the condition for $x_{g,r,t}$ are (112)–(115); the investment bundle (125) is (116); the investment condition (126) is (117); the procurement conditions and the laws of motion are (119)–(121). The two systems therefore differ in one condition only, the Euler equation for capital. The competitive Euler (118) values a unit of installed capital at

$$(1-\delta_g)\, q_{g,t+1} + s_{g,t+1} P^I_{g,t+1} + p_{g,t+1}\, \mathrm{MPK}_{g,t+1} + q_{g,t+1}\, \frac{\varphi}{2}\Big(\Big(\frac{I_{g,t+1}}{K_{g,t+1}}\Big)^2 - \delta_g^2\Big),$$

while the planner's Euler (127) values it at the same expression with $\gamma_t \partial \mathcal{D}_{t+1}/\partial K_{g,t+1}$ in place of $s_{g,t+1}P^I_{g,t+1}$. The firm does not internalize the effect of its capital on deterrence and the planner does; the difference between the two problems is exactly that deterrence wedge. Under (129) the two conditions coincide, so the planner's allocation, together with the price system above, satisfies every condition of the competitive equilibrium of appendix C.1; goods market clearing and the government budget then make the household's budget constraint hold by Walras' law. Solving (129) for the rate $s_{g,t+1}$ gives the statement of the proposition. Since $\gamma_t \geq 0$, the sign of the subsidy is the sign of $\partial \mathcal{D}_{t+1}/\partial K_{g,t+1}$, which the production network does not restrict. □

C.5 Sovereign's peacetime problem without industrial policy

This appendix derives the optimality conditions of the no-industrial-policy problem, in which the government deters with procurement alone, and characterizes its steady state, the state at which the model is calibrated (appendix C.3).

The government's problem The private allocation is the competitive equilibrium of appendix C.1 with $s_{g,t} = 0$: given the government's purchases, prices and quantities satisfy (112)–(121), and sectoral capital obeys the firms' Euler equation (118) without the subsidy term. The government takes this allocation as given. Since taxes are lump sum and, given the purchases, the private allocation is the optimum of the household's and the producers' problems, the envelope theorem makes $-\beta^t \Lambda_t p_{g,t}$ the exact welfare effect of a unit of good g purchased at date t: the induced changes in prices and quantities are of second

order. It chooses the purchases $m^j_{g,t}$, the baskets m^j_t, and the military stocks M^j_{t+1}, and finances them with the lump-sum tax $T_t = \sum_j \sum_{g\in\mathcal{G}_j} p_{g,t} m^j_{g,t}$. Since the consumption price index is the numeraire, the envelope theorem applied to the household's problem gives the welfare cost of the tax: a unit of the numeraire raised at date t lowers the household's lifetime utility by $\beta^t \Lambda_t$, with $\Lambda_t \equiv u'(C_t)$. Attaching, as in appendix C.4, the multiplier $\beta^t \Lambda_t \varpi^j_t$ to the law of motion of stock j and $\beta^{t+1}\Lambda_{t+1}\gamma_t$ to the date-$(t+1)$ deterrence constraint, the government's Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ u(C_t) - \frac{\tilde{L}_t^{1+1/\xi}}{1+1/\xi} + G_0 - \Lambda_t \sum_{j\in\mathcal{J}} \sum_{g\in\mathcal{G}_j} p_{g,t}\, m^j_{g,t} + \sum_{j\in\mathcal{J}} \Lambda_t\, \varpi^j_t \Big[(1-\delta_j) M^j_t + m^j_t - M^j_{t+1} \Big] + \beta\, \Lambda_{t+1}\, \gamma_t \Big[\mathcal{D}_{t+1}(\mathbf{K}_{t+1}, \mathbf{M}_{t+1}) - \eta_{t+1} \Big] \Big\},$$

with $m^j_t \geq 0$ and $\gamma_t \geq 0$, where the first line is the household's utility along the competitive allocation, differentiated with respect to the tax. Unlike the planner of appendix C.4, the government has no first-order condition for sectoral capital, and the capital stocks $\mathbf{K}_{t+1}$ in the deterrence function are those chosen by the firms.

Procurement and the military stocks Procurement of good g for basket j satisfies $p_{g,t} = \varpi^j_t\, \partial m^j_t / \partial m^j_{g,t} = \varpi^j_t \psi^j_g m^j_t / m^j_{g,t}$, the Cobb–Douglas demand of (119); summing over $g \in \mathcal{G}_j$ gives $\varpi^j_t = p^j_{m,t}$ whenever $m^j_t > 0$, and $p^j_{m,t} \geq \varpi^j_t$ in general. The first-order condition for M^j_{t+1} is

$$\Lambda_t\, \varpi^j_t = \beta\, \Lambda_{t+1} \Big[\gamma_t\, \frac{\partial \mathcal{D}_{t+1}}{\partial M^j_{t+1}} + (1-\delta_j)\, \varpi^j_{t+1} \Big], \tag{130}$$

together with $\gamma_t \geq 0$, $\mathcal{D}_{t+1} \geq \eta_{t+1}$, and $\gamma_t \big(\mathcal{D}_{t+1} - \eta_{t+1}\big) = 0$. Condition (130) is the planner's condition (128): the two problems set procurement by the same rule, and differ only in the capital stocks at which the rule is evaluated. The derivative $\partial \mathcal{D}_{t+1} / \partial M^j_{t+1}$ is taken at given $\mathbf{K}_{t+1}$. A government that internalized the response of private investment to its purchases would add to the first-order condition for $m^j_{g,t}$ the term $\sum_{k\geq 0} \beta^{k+1} \frac{\Lambda_{t+k+1}}{\Lambda_t} \gamma_{t+k} \sum_{g'\in\mathcal{G}} \frac{\partial \mathcal{D}_{t+k+1}}{\partial K_{g',t+k+1}} \frac{\partial K_{g'}}{\partial}$ the deterrence value of the capital that its purchases crowd in or out at every later date; the benchmark leaves this value uninternalized, in the same way as the firms of appendix C.1 leave the deterrence value of their own capital uninternalized. Proposition 7 corrects the second wedge; the benchmark measures the outcome when neither is corrected.

Steady state In a steady state prices, marginal utility, and the stocks are constant, procurement replaces depreciation, $m^j = \delta_j M^j$, and the deterrence constraint binds at a constant multiplier $\gamma_t = \zeta$. For a basket held in positive amount, $\varpi^j = p_m^j$ and (130) becomes $p_m^j = \beta\left[\zeta\, \partial\mathcal{D}/\partial M^j + (1-\delta_j)\, p_m^j\right]$, that is

$$p_m^j\,(\rho + \delta_j) = \zeta\,\frac{\partial \mathcal{D}}{\partial M^j}, \qquad j \in \mathcal{J}, \tag{131}$$

with $\rho \equiv 1/\beta - 1$: the user cost of a unit of each stock, in the numeraire, equals its marginal deterrence value. Dividing the conditions of two baskets eliminates ζ, so (131) pins down the composition of the arsenal, and the binding constraint $\mathcal{D}(\mathbf{K}, \mathbf{M}) = \eta$ pins down its level. The steady state is therefore the solution of (131), the binding deterrence constraint, and the competitive steady state without subsidies at the resulting procurement, with the derivatives of the deterrence function computed from the wartime system of appendix C.2. The calibration of appendix C.3 uses (131) in reverse: it fixes the arsenal at the one implied by the observed spending shares and solves the conditions for the weights ϕ_j, so that at the calibrated weights this steady state coincides with the postwar steady state of the spending rule.

C.6 Solution method

All objects are computed exactly at the sectoral level, without approximation around a steady state. The postwar steady state, at which the CES nests are normalized, is solved in share form: prices from the linear cost-push system implied by the cost shares, nominal outputs from the Leontief inverse of the combined intermediate and investment requirements, and the labor block by Newton's method. A war of length T from given stocks is solved together with its postwar continuation as a single perfect-foresight path of $T + H$ periods – the spending rule replacing the procurement conditions after the war and the path closed at the steady state after $H = 8$ postwar periods – as one system of $T + H$ blocks of $4N + 1 + 2J$ unknowns, by a damped Newton method with an exact sparse Jacobian assembled by automatic differentiation; the complementarity conditions for irreversible investment and non-negative procurement are written as smoothed Fischer–Burmeister equations. The combat efficiency parameters $\chi^{(T)}$ are inverted by appending the spending targets to the wartime system and solving for χ jointly with the path, warm-started along a continuation in χ. The derivatives of the deterrence function are obtained by an adjoint solve with the same Jacobian. The weights ϕ_j are the fixed point of a quasi-Newton iteration over the calibration conditions, each evaluation re-inverting $\chi^{(T)}$. The

optimum with industrial policy is the root of the peacetime first-order conditions in the sectoral capital stocks, the baskets, and the multiplier on the deterrence constraint, found by a Levenberg–Marquardt Newton method whose Jacobian combines implicit differentiation of the static equilibrium with second-order adjoints through the wartime equilibria; wartime corners that switch regime along the iteration are pinned and released automatically. The code is available from the authors.

C.7 The full ranking of optimal capital subsidies

Table 10 reports the optimal capital subsidy of every sector, a rate on the replacement value of the sector's capital stock. Sectors are ordered by decreasing subsidy, reading left to right and then down; the 20 highest are Table 4 in the main text. The optimum is computed with the wartime complementarity conditions (irreversible investment and non-negative procurement) smoothed by a Fischer–Burmeister perturbation of 10^{-5}. At the exact conditions the peacetime first-order condition of a sector whose wartime investment sits exactly at the irreversibility switch has no root; two sectors are in this position (industrial mold manufacturing, mining and oil and gas field machinery) and receive a subsidy of zero within rounding.

Table 10: Optimal capital subsidies, all 378 sectors, in percent of the replacement value of the sector's capital stock, ordered by decreasing subsidy. 'mfg.' abbreviates manufacturing and 'svcs.' services.

rank	sector	s_g	rank	sector	s_g
1	Guided missile & space vehicle mfg.	11.33	24	Telephone apparatus mfg.	2.42
2	Propulsion units & parts for space vehicles & guided missiles	10.91	25	Water transportation	2.41
3	Ship building & repairing	9.17	26	All other chemical product & preparation mfg.	2.40
4	Search, detection, & navigation instr. mfg.	9.12	27	Ball & roller bearing mfg.	2.34
5	Ammunition, arms, ordnance, & accessories mfg.	8.55	28	Relay & industrial control mfg.	2.26
6	Military armored vehicle, tank, & tank component mfg.	7.24	29	Watch, clock, & other measuring & controlling device mfg.	2.19
7	Other aircraft parts & auxiliary equip. mfg.	7.06	30	Carbon & graphite product mfg.	2.18
8	Electricity & signal testing instr. mfg.	6.69	31	Semiconductor & related device mfg.	2.13
9	Electronic & precision equip. repair & maintenance	5.56	32	Other basic inorganic chemical mfg.	2.11
10	Architectural, engineering, & related svcs.	5.06	33	Computer systems design svcs.	2.11
11	Aircraft mfg.	4.89	34	Other rubber product mfg.	2.09
12	Broadcast & wireless communications equip.	4.48	35	Manufacturing & reproducing magnetic & optical media	2.09
13	Switchgear & switchboard apparatus mfg.	4.00	36	Pharmaceutical preparation mfg.	2.09
14	Scientific research & development svcs.	3.59	37	Water, sewage & other systems	2.00
15	Motor vehicle electrical & electronic equip. mfg.	3.47	38	Environmental & other technical consulting svcs.	1.98
16	Other electronic component mfg.	3.42	39	Petrochemical mfg.	1.95

continued on the next page

Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
17	Other computer related svcs., including facilities management	3.24	40	All other miscellaneous professional, scientific, & technical svcs.	1.89
18	Air transportation	2.86	41	Other basic organic chemical mfg.	1.86
19	Aircraft engine & engine parts mfg.	2.84	42	Communication & energy wire & cable mfg.	1.83
20	Cutlery & handtool mfg.	2.83	43	All other transportation equip. mfg.	1.80
21	Biological product (except diagnostic) mfg.	2.61	44	Industrial gas mfg.	1.79
22	Other textile product mills	2.57	45	Computer terminals & other computer peripheral equip. mfg.	1.78
23	Printed circuit assembly (electronic assembly) mfg.	2.48	46	Fluid power process machinery	1.77

continued on the next page

Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
47	Medicinal & botanical mfg.	1.73	81	Scenic & sightseeing transportation & support activities for transportation	1.35
48	Management consulting svcs.	1.73	82	Radio & television broadcasting	1.35
49	Leather & allied product mfg.	1.69	83	Wired telecommunications carriers	1.29
50	Other communications equip. mfg.	1.63	84	Wiring device mfg.	1.29
51	Plastics material & resin mfg.	1.62	85	Photographic svcs.	1.29
52	Synthetic rubber & artificial & synthetic fibers & filaments mfg.	1.61	86	Other petroleum & coal prod. mfg.	1.28
53	Wireless telecommunications carriers (except satellite)	1.61	87	Copper rolling, drawing, extruding & alloying	1.28
54	Pesticide & other agricultural chemical mfg.	1.60	88	Periodical Publishers	1.28
55	Computer storage device mfg.	1.60	89	Paint & coating mfg.	1.28
56	Petroleum refineries	1.60	90	Other motor vehicle parts mfg.	1.27
57	Specialized design svcs.	1.59	91	Toilet preparation mfg.	1.25
58	Printing ink mfg.	1.57	92	Alumina refining & primary aluminum production	1.25
59	In-vitro diagnostic substance mfg.	1.56	93	Adhesive mfg.	1.24
60	Automatic environmental control mfg.	1.54	94	Oil & gas extraction	1.22
61	Valve & fittings other than plumbing	1.54	95	Turbine & turbine generator set units mfg.	1.22
62	Motor & generator mfg.	1.53	96	Industrial process variable instr. mfg.	1.22
63	Office supplies (except paper) mfg.	1.53	97	Optical instrument & lens mfg.	1.22
64	Custom computer programming svcs.	1.52	98	Motor vehicle transmission & power train parts mfg.	1.21
65	Synthetic dye & pigment mfg.	1.52	99	Commercial & industrial machinery & equip. repair & maintenance	1.21
66	Data processing, hosting, & related svcs.	1.50	100	All other miscellaneous mfg.	1.21
67	Software publishers	1.49	101	Nonferrous metal (except copper & aluminum) rolling, drawing, extruding & alloying	1.20
68	Curtain & linen mills	1.48	102	Newspaper publishers	1.19
69	Soap & cleaning compound mfg.	1.48	103	Mechanical power transmission equip. mfg.	1.17
70	Fertilizer mfg.	1.48	104	Hardware mfg.	1.17
71	Advertising, public relations, & related svcs.	1.47	105	News syndicates, libraries, archives & all other information svcs.	1.17
72	Apparel mfg.	1.45	106	Primary battery mfg.	1.16
73	Surgical appliance & supplies mfg.	1.45	107	Major household appliance mfg.	1.15
74	Directory, mailing list, & other publishers	1.44	108	Metal crown, closure, & other metal stamping (except automotive)	1.15
75	Satellite, telecommunications resellers, & all other telecommunications	1.43	109	Coating, engraving, heat treating & allied activities	1.14
76	All other forging, stamping, & sintering	1.39	110	Employment svcs.	1.14
77	Seafood product preparation & packaging	1.38	111	Nonferrous metal (except aluminum) smelting & refining	1.13
78	Accounting, tax preparation, bookkeeping, & payroll svcs.	1.37	112	Speed changer, industrial high-speed drive, & gear mfg.	1.12
79	Totalizing fluid meter & counting device mfg.	1.36	113	Doll, toy, & game mfg.	1.12
80	Tire mfg.	1.36	114	Jewelry & silverware mfg.	1.11

continued on the next page

Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
115	Other animal food mfg.	1.10	149	Ophthalmic goods mfg.	0.93
116	Textile & fabric finishing & fabric coating mills	1.09	150	Custom roll forming	0.93
117	Spring & wire product mfg.	1.09	151	Aluminum product mfg. from purchased aluminum	0.92
118	Analytical laboratory instrument mfg.	1.09	152	Electric lamp bulb & part mfg.	0.92
119	Office administrative svcs.	1.09	153	Facilities support svcs.	0.92
120	Animal (except poultry) slaughtering, rendering, & processing	1.09	154	Concrete pipe, brick, & block mfg.	0.92
121	Printing	1.08	155	Metal can, box, & other metal container (light gauge) mfg.	0.91
122	Cheese mfg.	1.08	156	Polystyrene foam product mfg.	0.90
123	Poultry processing	1.07	157	Motor home mfg.	0.90
124	Business support svcs.	1.06	158	Beef cattle ranching & farming, including feedlots & dual-purpose ranching & farming	0.89
125	Animal production, except cattle & poultry & eggs	1.06	159	Abrasive product mfg.	0.89
126	Electronic computer mfg.	1.05	160	Paperboard mills	0.89
127	Storage battery mfg.	1.04	161	Fiber, yarn, & thread mills	0.89
128	Machine shops	1.04	162	Cut stone & stone product mfg.	0.88
129	Veterinary svcs.	1.04	163	Other furniture related product mfg.	0.88
130	Glass & glass product mfg.	1.03	164	Personal & household goods repair & maintenance	0.88
131	Plastics pipe, pipe fitting, & unlaminated profile shape mfg.	1.02	165	Truck transportation	0.87
132	Management of companies & enterprises	1.00	166	Plastics packaging materials & unlaminated film & sheet mfg.	0.86
133	All other miscellaneous electrical equip. & component mfg.	1.00	167	Ferrous metal foundries	0.86
134	Laminated plastics plate, sheet (except packaging), & shape mfg.	1.00	168	Travel trailer & camper mfg.	0.86
135	Fabric mills	0.99	169	Legal svcs.	0.86
136	Other support svcs.	0.99	170	Travel arrangement & reservation svcs.	0.85
137	Dental equip. & supplies mfg.	0.98	171	Paperboard container mfg.	0.85
138	Plate work & fabricated structural product mfg.	0.97	172	Support activities for printing	0.85
139	Investigation & security svcs.	0.97	173	Other support activities for mining	0.85
140	Distilleries	0.96	174	Ground or treated mineral & earth mfg.	0.85
141	Other plastics product mfg.	0.96	175	Other fabricated metal mfg.	0.84
142	Turned product & screw, nut, & bolt mfg.	0.95	176	Flavoring syrup & concentrate mfg.	0.84
143	Paper bag & coated & treated paper mfg.	0.95	177	Fabricated pipe & pipe fitting mfg.	0.84
144	Motor vehicle steering, suspension component (except spring), & brake systems mfg.	0.94	178	Poultry & egg production	0.84
145	Fishing, hunting & trapping	0.94	179	Stationery product mfg.	0.84
146	Lime & gypsum product mfg.	0.94	180	Book publishers	0.84
147	Dental laboratories	0.94	181	Plastics bottle mfg.	0.83
148	Internet publishing & broadcasting & Web search portals	0.93	182	Automotive equip. rental & leasing	0.82

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Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
183	Asphalt paving mixture & block mfg.	0.81	217	Snack food mfg.	0.70
184	Couriers & messengers	0.80	218	Other personal svcs.	0.70
185	Fats & oils refining & blending	0.80	219	Tobacco mfg.	0.69
186	Steel product mfg. from purchased steel	0.79	220	Breweries	0.67
187	Irradiation apparatus mfg.	0.79	221	Civic, social, professional, & similar organizations	0.67
188	Small electrical appliance mfg.	0.79	222	Electromedical & electrotherapeutic apparatus mfg.	0.67
189	Paper mills	0.79	223	Power, distribution, & specialty transformer mfg.	0.66
190	Iron & steel mills & ferroalloy mfg.	0.78	224	Seasoning & dressing mfg.	0.66
191	Nonferrous metal foundries	0.77	225	Fruit & vegetable canning, pickling, & drying	0.65
192	Ornamental & architectural metal prod. mfg.	0.77	226	Natural gas distribution	0.65
193	Pulp mills	0.77	227	General & consumer goods rental	0.64
194	Bread & bakery product mfg.	0.76	228	Automotive repair & maintenance (including car washes)	0.64
195	All other food mfg.	0.76	229	Boat building	0.64
196	Sugar & confectionery product mfg.	0.76	230	Services to buildings & dwellings	0.64
197	Copper, nickel, lead, & zinc mining	0.76	231	Plumbing fixture fitting & trim mfg.	0.64
198	Soybean & other oilseed processing	0.75	232	Death care svcs.	0.63
199	Breakfast cereal mfg.	0.75	233	Miscellaneous nonmetallic mineral prod.	0.63
200	Mineral wool mfg.	0.75	234	Nonstore retailers	0.62
201	Commercial & industrial machinery & equip. rental & leasing	0.75	235	Dry, condensed, & evaporated dairy product mfg.	0.62
202	Wineries	0.75	236	Religious organizations	0.61
203	Electric power generation, transmission, & distribution	0.75	237	All other food & drinking places	0.61
204	Dry-cleaning & laundry svcs.	0.74	238	Building material & garden equip. & supplies dealers	0.61
205	Lighting fixture mfg.	0.74	239	Coffee & tea mfg.	0.61
206	Lessors of nonfinancial intangible assets	0.74	240	Support activities for agriculture & forestry	0.61
207	All other converted paper product mfg.	0.74	241	Health & personal care stores	0.61
208	Ready-mix concrete mfg.	0.73	242	Motorcycle, bicycle, & parts mfg.	0.61
209	Cement mfg.	0.73	243	Ice cream & frozen dessert mfg.	0.60
210	Wet corn milling	0.73	244	Gasoline stations	0.60
211	Flour milling & malt mfg.	0.72	245	Personal care svcs.	0.60
212	Urethane & other foam product (except polystyrene) mfg.	0.72	246	Other crop farming	0.60
213	Grain farming	0.72	247	All other retail	0.60
214	Clay product & refractory mfg.	0.72	248	Warehousing & storage	0.60
215	Other engine equip. mfg.	0.71	249	Grantmaking, giving, & social advocacy organizations	0.60
216	Soft drink & ice mfg.	0.71	250	Clothing & clothing accessories stores	0.60

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Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
251	Dairy cattle & milk production	0.60	285	Spectator sports	0.48
252	Cookie, cracker, pasta, & tortilla mfg.	0.60	286	Full-service restaurants	0.46
253	General merchandise stores	0.60	287	Museums, historical sites, zoos, & parks	0.45
254	Fluid milk & butter mfg.	0.59	288	Promoters of performing arts & sports & agents for public figures	0.43
255	Iron, gold, silver, & other metal ore mining	0.59	289	All other wood product mfg.	0.43
256	Sanitary paper product mfg.	0.59	290	Photographic & photocopying equip. mfg.	0.42
257	Frozen food mfg.	0.59	291	Independent artists, writers, & performers	0.42
258	Motor vehicle gasoline engine & engine parts mfg.	0.59	292	Limited-service restaurants	0.41
259	Wholesale electronic markets & agents & brokers	0.58	293	Coal mining	0.41
260	Offices of dentists	0.58	294	Rubber & plastics hoses & belting mfg.	0.41
261	Dog & cat food mfg.	0.57	295	Fruit & tree nut farming	0.41
262	Other household nonupholstered furniture	0.57	296	Accommodation	0.40
263	Heating equip. (except warm air furnaces) mfg.	0.56	297	Vegetable & melon farming	0.40
264	Offices of other health practitioners	0.56	298	Upholstered household furniture mfg.	0.39
265	Petroleum & petroleum prod.	0.54	299	Other real estate	0.39
266	Home health care svcs.	0.54	300	Heavy duty truck mfg.	0.38
267	Outpatient care centers	0.53	301	Rail transportation	0.38
268	Offices of physicians	0.53	302	Cutting & machine tool accessory, rolling mill, & other metalworking machinery mfg.	0.37
269	Other nondurable goods merchant wholesalers	0.53	303	Other concrete product mfg.	0.37
270	Waste management & remediation svcs.	0.53	304	Forestry & logging	0.35
271	Grocery & related product wholesalers	0.52	305	Hospitals	0.34
272	Oilseed farming	0.52	306	Motor vehicle & parts dealers	0.34
273	Drugs & druggists' sundries	0.51	307	Millwork	0.32
274	Medical & diagnostic laboratories	0.51	308	Elementary & secondary schools	0.31
275	Cable & other subscription programming	0.50	309	Gambling industries (except casino hotels)	0.31
276	Other ambulatory health care svcs.	0.50	310	Carpet & rug mills	0.31
277	Household appliances & electrical & electronic goods	0.50	311	Junior colleges, colleges, universities, & professional schools	0.31
278	Asphalt shingle & coating materials mfg.	0.49	312	Transit & ground passenger transportation	0.30
279	Motor vehicle & motor vehicle parts & supplies	0.49	313	Other amusement & recreation industries	0.30
280	Other durable goods merchant wholesalers	0.49	314	Other nonmetallic mineral mining & quarrying	0.30
281	Food & beverage stores	0.49	315	Amusement parks & arcades	0.30
282	Professional & commercial equip. & supplies	0.49	316	Nonresidential maintenance & repair	0.30
283	Machinery, equip., & supplies	0.48	317	Industrial & commercial fan & blower & air purification equip. mfg.	0.30
284	Greenhouse, nursery, & floriculture production	0.48	318	Performing arts companies	0.28

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Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
319	Wood kitchen cabinet & countertop mfg.	0.27	349	Veneer, plywood, & engineered wood product mfg.	-0.17
320	Other educational svcs.	0.26	350	Other nonresidential structures	-0.19
321	Child day care svcs.	0.25	351	Industrial process furnace & oven mfg.	-0.20
322	Individual & family svcs.	0.25	352	Manufacturing structures	-0.22
323	Community food, housing, & other relief svcs., including rehabilitation svcs.	0.23	353	Metal tank (heavy gauge) mfg.	-0.27
324	Surgical & medical instrument mfg.	0.19	354	Special tool, die, jig, & fixture mfg.	-0.33
325	Multifamily residential structures	0.18	355	Semiconductor machinery mfg.	-0.35
326	Residential maintenance & repair	0.17	356	Material handling equip. mfg.	-0.38
327	Other residential structures	0.16	357	Power boiler & heat exchanger mfg.	-0.39
328	Single-family residential structures	0.16	358	Machine tool mfg.	-0.43
329	Measuring, dispensing, & other pumping equip. mfg.	0.16	359	Other general purpose machinery mfg.	-0.45
330	Residential mental health, substance abuse, & other residential care facilities	0.15	360	Motor vehicle metal stamping	-0.47
331	Pipeline transportation	0.14	361	Other industrial machinery mfg.	-0.48
332	Sawmills & wood preservation	0.14	362	Showcase, partition, shelving, & locker mfg.	-0.49
333	Nursing & community care facilities	0.13	363	Automobile mfg.	-0.55
334	Sporting & athletic goods mfg.	0.06	364	Power & communication structures	-0.58
335	Drilling oil & gas wells	0.05	365	Motor vehicle body mfg.	-0.59
336	Construction machinery mfg.	0.02	366	Office & commercial structures	-0.60
337	Transportation structures & highways & streets	0.02	367	Health care structures	-0.60
338	Sound recording industries	-0.02	368	Office furniture & custom architectural woodwork & millwork mfg.	-0.61
339	Farm machinery & equip. mfg.	-0.04	369	Light truck & utility vehicle mfg.	-0.62
340	Nonupholstered wood household furniture mfg.	-0.04	370	Motor vehicle seating & interior trim mfg.	-0.67
341	Industrial mold mfg.	-0.05	371	Truck trailer mfg.	-0.67
342	Stone mining & quarrying	-0.09	372	Lawn & garden equip. mfg.	-0.69
343	Air conditioning, refrigeration, & warm air heating equip. mfg.	-0.10	373	Air & gas compressor mfg.	-0.70
344	Motion picture & video industries	-0.11	374	Institutional furniture mfg.	-0.71
345	Power-driven handtool mfg.	-0.12	375	Packaging machinery mfg.	-0.72
346	Other commercial & service industry machinery mfg.	-0.14	376	Sign mfg.	-0.79
347	Mining & oil & gas field machinery mfg.	-0.14	377	Railroad rolling stock mfg.	-1.00
348	Educational & vocational structures	-0.15	378	Audio & video equip. mfg.	-1.18

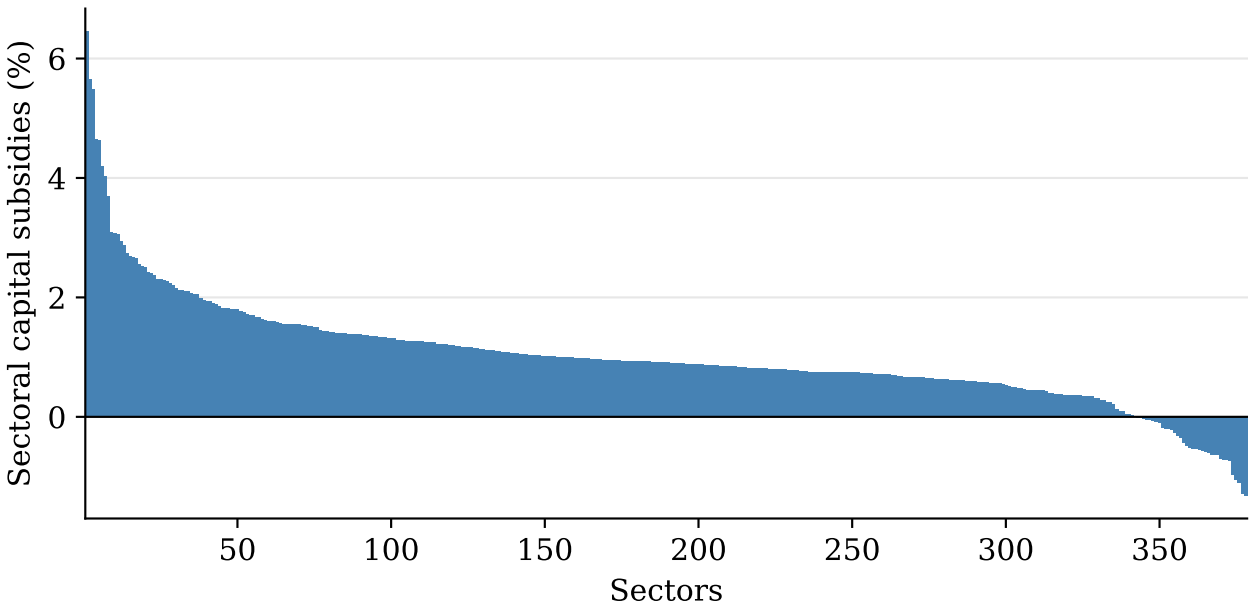
C.8 Additional results

Table 11: Aggregate outcomes in the steady state of the frictionless economy.

	optimal deterrence equilibrium	no industrial policy
military procurement, % of GDP	2.20	3.37
real GDP, relative	1.075	1
consumption, relative	1.052	1
aggregate capital, relative	1.178	1
labor supply, relative	1.007	1
welfare gain, consumption-equivalent (%)	4.52	-

Constructed as table 3, for the frictionless economy.

Figure 6: The cross-section of optimal capital subsidies in the frictionless economy.



Each bar is one of the 378 sectors, ordered by decreasing subsidy.

Table 12: The 20 sectors with the highest optimal capital subsidies in the frictionless economy.

rank	sector	s_g (%)
1	Guided missile and space vehicle manufacturing	6.45
2	Search, detection, and navigation instruments manufacturing	5.65
3	Propulsion units and parts for space vehicles and guided missiles	5.49
4	Ship building and repairing	4.64
5	Military armored vehicle, tank, and tank component manufacturing	4.63
6	Ammunition, arms, ordnance, and accessories manufacturing	4.19
7	Electricity and signal testing instruments manufacturing	4.03
8	Scientific research and development services	3.69
9	Broadcast and wireless communications equipment	3.08
10	Architectural, engineering, and related services	3.06
11	Air transportation	3.06
12	Other electronic component manufacturing	2.93
13	Aircraft manufacturing	2.87
14	Other aircraft parts and auxiliary equipment manufacturing	2.75
15	Switchgear and switchboard apparatus manufacturing	2.69
16	Other computer related services, including facilities management	2.66
17	Motor vehicle electrical and electronic equipment manufacturing	2.66
18	Water transportation	2.55
19	Aircraft engine and engine parts manufacturing	2.52
20	Watch, clock, and other measuring and controlling device manufacturing	2.50

Cobb–Douglas production nests, no capital adjustment cost, perfectly substitutable sectoral labour.

Table 13: The change in military basket prices under a 10% increase in one sector's capital stock at the outbreak of a war of length $T = 6$.

sector (+10% K)	str	eq	int	nd	land	air	naval	ammo
Guided missile & space vehicle mfg.	-0.000	+0.001	+0.000	+0.000	+0.001	+0.001	+0.001	-0.018
Propulsion units & parts for space vehicles & guided missiles	-0.001	+0.003	+0.001	+0.001	+0.003	+0.003	+0.003	-0.060
Ship building & repairing	-0.004	+0.016	+0.006	+0.006	+0.015	+0.015	-0.482	+0.010
Search, detection, & navigation instr. mfg.	-0.025	-0.585	+0.033	+0.052	+0.102	+0.093	-0.070	+0.055
Ammunition, arms, ordnance, & accessories mfg.	-0.003	+0.012	+0.006	+0.007	+0.012	+0.017	+0.012	-0.302
Military armored vehicle, tank, & tank component mfg.	-0.002	+0.006	-0.003	+0.005	-1.619	+0.006	+0.008	+0.007
Other aircraft parts & auxiliary equip. mfg.	-0.008	+0.039	+0.011	+0.020	+0.033	-0.364	+0.035	+0.025
Electricity & signal testing instr. mfg.	-0.003	-0.078	+0.002	+0.007	+0.010	+0.014	+0.009	+0.009
Electronic & precision equip. repair & maintenance	-0.011	+0.021	+0.010	-0.199	+0.026	+0.032	+0.022	+0.018
Architectural, engineering, & related svcs.	-0.030	+0.155	-0.472	+0.105	+0.208	+0.233	+0.141	+0.148
Aircraft mfg.	-0.010	+0.161	+0.046	+0.025	+0.065	-1.243	+0.128	+0.104
Broadcast & wireless communications equip.	-0.002	-0.320	+0.023	-0.001	+0.067	+0.020	+0.081	-0.018
Switchgear & switchboard apparatus mfg.	-0.004	-0.027	+0.001	+0.003	+0.004	+0.006	+0.005	+0.005
Scientific research & development svcs.	-0.020	-0.059	-0.315	+0.017	+0.106	-0.038	+0.086	+0.084
Motor vehicle electrical & electronic equip. mfg.	-0.002	-0.036	+0.002	+0.005	+0.007	+0.009	+0.006	+0.007
Other electronic component mfg.	-0.000	-0.030	-0.003	+0.001	+0.009	+0.007	+0.007	+0.008
Other computer related svcs., including facilities management	+0.002	+0.009	-0.019	+0.006	+0.012	+0.012	+0.010	+0.009
Air transportation	+0.015	+0.062	+0.045	-0.206	+0.076	-0.107	+0.073	+0.062
Aircraft engine & engine parts mfg.	+0.002	+0.015	+0.006	+0.008	+0.010	-0.088	+0.014	+0.012
Cutlery & handtool mfg.	-0.004	-0.021	+0.003	+0.003	-0.006	+0.005	+0.004	+0.001

Mean over the five war years in which the government procures. In red, the largest fall in each row.

Table 14: Output, price, and the composition of the rise in sales of the shocked sector.

	% change		composition of the rise in sales (%)			
sector (+10% K)	output	price	cons.	inv.	interm.	military
Guided missile & space vehicle mfg.	+0.02	-0.02	1.7	1.1	0.8	96.4
Propulsion units & parts for space vehicles & guided missiles	+0.13	-0.42	11.5	0.0	43.2	45.2
Ship building & repairing	+0.79	-0.41	0.0	3.0	0.2	96.9
Search, detection, & navigation instr. mfg.	+2.17	-1.67	0.0	7.7	3.7	88.6
Ammunition, arms, ordnance, & accessories mfg.	+1.02	-1.18	9.6	0.0	6.3	84.1
Military armored vehicle, tank, & tank component mfg.	+1.12	-1.39	9.7	0.0	2.5	87.7
Other aircraft parts & auxiliary equip. mfg.	+0.87	-1.75	1.0	0.0	31.8	67.2
Electricity & signal testing instr. mfg.	+1.34	-2.10	0.0	15.8	8.2	75.9
Electronic & precision equip. repair & maintenance	+0.78	-1.41	0.0	0.0	0.4	99.6
Architectural, engineering, & related svcs.	+1.17	-1.25	6.1	9.5	4.1	80.3
Aircraft mfg.	+2.18	-1.46	6.9	18.4	1.6	73.1
Broadcast & wireless communications equip.	+1.64	-1.72	0.0	32.9	10.7	56.5
Switchgear & switchboard apparatus mfg.	+0.92	-1.37	0.0	22.6	1.2	76.2
Scientific research & development svcs.	+1.06	-0.62	0.0	50.7	1.4	47.9
Motor vehicle electrical & electronic equip. mfg.	+0.28	-0.75	0.0	0.0	1.7	98.3
Other electronic component mfg.	+0.13	-1.07	0.0	0.0	18.9	81.1
Other computer related svcs., including facilities management	+0.31	-1.08	17.9	0.0	1.9	80.2
Air transportation	+1.68	-2.41	61.4	0.0	0.9	37.8
Aircraft engine & engine parts mfg.	+1.04	-2.09	64.0	0.5	21.0	14.6
Cutlery & handtool mfg.	+0.50	-1.39	0.0	2.2	1.0	96.8

Percent changes of the war totals. The last four columns split the rise in sales and sum to one hundred.